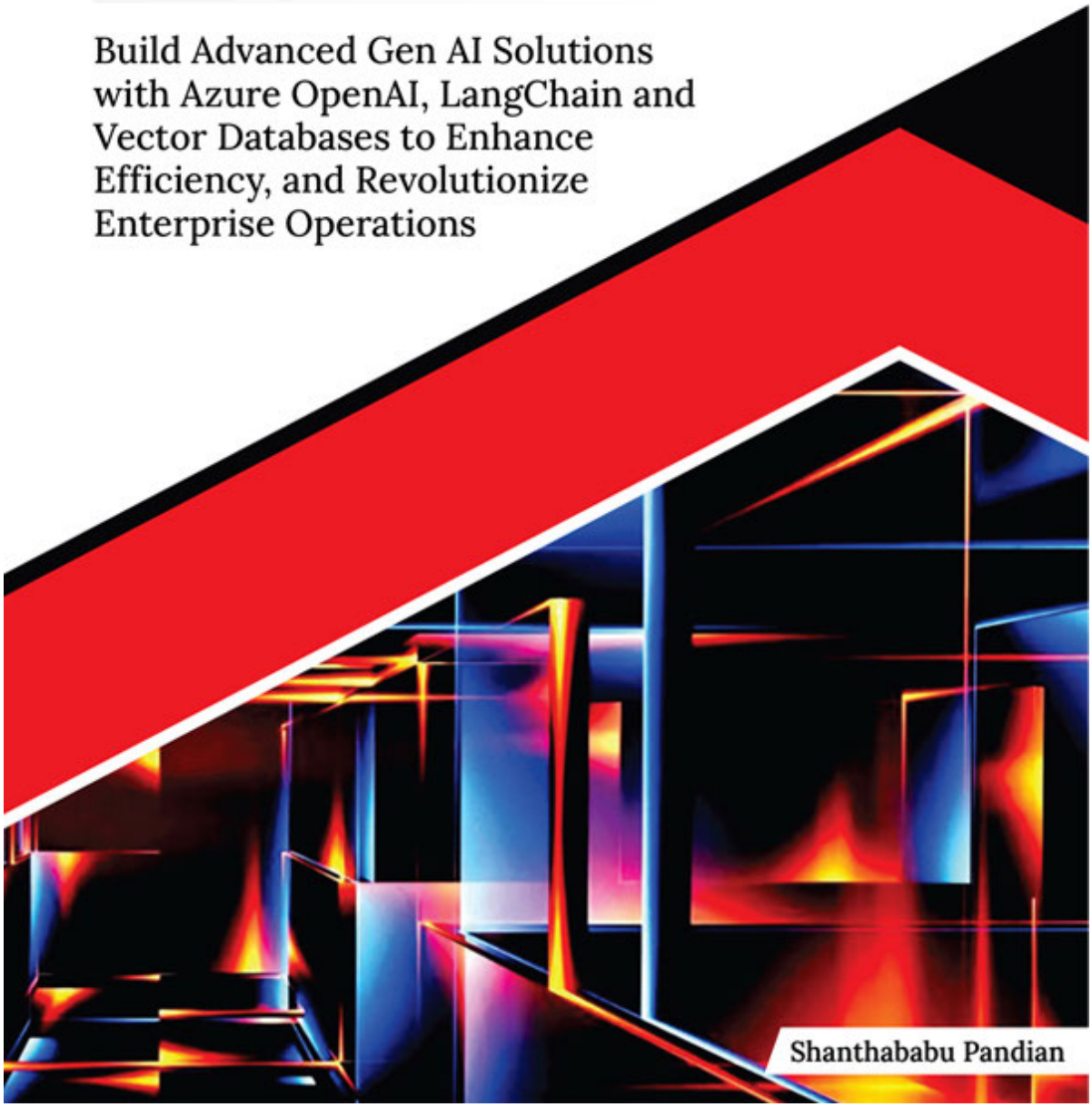
ULTIMATE

Azure AI Services for Gen AI Solutions

Build Advanced Gen AI Solutions with Azure OpenAI, LangChain and Vector Databases to Enhance Efficiency, and Revolutionize Enterprise Operations



Shanthababu Pandian

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Shanthababu Pandian

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Dedicated To

My Beloved Grandma:

Late Smt. Sornammal Subramanian

About the Author

With over 23 years of IT experience, Shanthababu Pandian is a seasoned tech executive specializing in data science, generative AI, machine learning, and artificial intelligence. He holds three master's degrees (M.Tech, MBA, and M.S.), a bachelor's degree in Electronics and Communication Engineering, a postgraduate degree in AI & ML from the University of Texas, and a certification in Data Science from IIT Guwahati. He is currently pursuing a PhD in AI at Anna University, Chennai.

Shanthababu has extensive experience in data engineering, analytics, and developing AI-driven solutions. He has successfully delivered projects for clients in the healthcare, financial, and retail sectors across the US and the UK. He is a master at implementing governance frameworks into place, directing agile, cross-functional teams, and creating AI-powered products.

He is a well-known thought leader who has presented more than 50 national and international lectures and authored more than 100 books and articles. He has also reviewed more than 75 technical publications in the areas of programming, data, artificial intelligence, and the cloud. He continues to motivate

the next generation of AI and data specialists by serving as a mentor and counselor to IT institutions.

About the Technical Reviewers

S.Ponmalar is an academician with over 25 years of field and domain experience in Information and Communication Engineering, specializing in areas such as optical communication, Networks, Machine Learning, and Computer vision. She has obtained her Doctoral degree in Information and Communcation Engineering from Anna University and is a recognized supervisor, guiding many PG and PhD scholars. She has published over 30 research papers in many reputed Journals and Conferences. Her interest in upgrading with recent trends and technology has obligated her to read and review articles and books from various publishers.

Tanay Agrawal is a seasoned Deep Learning Researcher with over eight years of experience in AI and machine learning, having made significant contributions across multiple startups. He began his journey by working on an AutoML platform at MateLabs, enabling efficient and accessible machine learning model creation. Tanay then joined Curl HG, where he played a pivotal role in building SARA, an advanced Information Extraction Engine for energy trading sector. Following the company's acquisition, he transitioned to Xplorazzi as an AI

Researcher, focusing on retail technology, including automated checkout kiosks and retail surveillance solutions.

As the author of the book *Hyperparameter Optimization in Machine* Tanay has shared his expertise through talks at esteemed conferences such as PyData London, PyData Delhi, and PyCon India.

In his entrepreneurial journey, Tanay co-founded TryonLabs.ai, where he developed a diffusion-based virtual try-on technology for garments. This technology, once open-sourced, has since become a foundational model that is widely adopted in the industry. Under his leadership, TryonLabs delivered AI-driven solutions to over 120 D2C fashion brands globally.

Currently, Tanay serves as the CTO of Videoit, a company he co-founded to transform the e-commerce experience for D2C brands. Videoit is building a suite of AI applications to bridge the gap between ad clicks and conversions. Tanay continues to innovate at the intersection of AI, retail, and consumer technology, driving an impact on a global scale.

Preface

The rapid evolution of **Generative AI (Gen AI) is** transforming industries, redefining how we interact with technology, and opening new possibilities for innovation. This book is a comprehensive guide tailored for learners at all levels—whether you are a beginner eager to explore Gen AI, a professional seeking to upskill, or an expert looking to enhance existing ML systems. By focusing on **Azure OpenAI Services, Azure AI** and practical implementation, this resource equips readers with the tools and knowledge to navigate the dynamic landscape of AI-driven solutions.

The content bridges theoretical understanding and real-world application, diving deep into **Large Language Models (LLMs), vector databases, embedding** and **prompt** Special emphasis has been placed on the Azure ecosystem, covering essential components such as **Azure Storage, Search Services, OpenAI** and **Prompt** Each chapter is designed to provide hands-on insights into building scalable, cost-effective, and responsible AI solutions using **Python programming** and **Azure** Whether you are developing AI models, optimizing data systems, or implementing cutting-edge applications, this book ensures a seamless learning experience.

This book aims to inspire confidence and creativity in applying Gen AI concepts across industries by including real-world examples, detailed workflows, and actionable strategies. You will embark on the fascinating world of Generative AI and Azure, unlocking the potential to transform ideas into impactful solutions.

Chapter 1. Introduction to Generative AI: This chapter explores the evolution of AI with a focus on transformer architectures, distinguishing Generative AI (Gen AI) from classical AI and ML. It examines the core characteristics of Gen AI, including its ability to transform text, images, and videos, and explains why generative models are referred to as Large Language Models (LLMs). The chapter discusses the theories behind these models, such as neural networks and training methodologies, highlighting how they produce human-like content. Practical examples, including code snippets, are provided to deepen understanding. The chapter concludes by addressing the ethical considerations and security challenges of Gen AI, emphasizing its importance and applications across industries.

Chapter 2. Exploring LLMs and Its Capabilities: This chapter explores the foundations of Large Language Models (LLMs),

focusing on their key components and functionality. It delves into embedding layers, transformer blocks, multi-head self-attention mechanisms, feed-forward neural networks, positional encoding, and decoders. The chapter also explains output layers, loss functions, attention masks, and fine-tuning layers, unravelling how these elements work together. Finally, it highlights the critical role of fine-tuning in optimizing LLMs for specific tasks.

Chapter 3. Vector Database and Embedding Techniques: This chapter explores the mechanisms of vector databases, including indexing, querying, and so on. It provides an analogy to simplify their concepts, and delves into underlying theories, similarity measures, and working mechanisms. The chapter also covers categories of vector databases, the concept of serverless vector databases, and popular options such as Pinecone, Faiss, and Chroma. Practical steps with code examples are included for working with Pinecone and Faiss, providing hands-on insights into their functionality.

Chapter 4. Prompt Engineering and Its Significance: This chapter delves into prompt engineering, its significance in Generative AI, and the components and categories of prompts. It explores techniques such as N-shot prompting, Chain-of-Thought (CoT) prompting, Directional Stimulus Prompting

(DSP), React prompting, and Multimodal CoT prompting. The chapter covers how to craft, evaluate, refine, and scale prompts for optimal model performance. Practical examples include Q&A, text summarization, chatbot responses, and document classification. Additionally, it provides the best practices and guidelines for crafting effective prompts to build strategic AI models.

Chapter 5. Azure Storage for Azure OpenAI Implementations: This chapter explores Azure Storage and Integration strategies and Azure Search Services for enhancing Azure OpenAI implementations. It discusses how Azure Blob Storage and Azure Data Lake Storage (ADLS) support unstructured data storage, enabling seamless access for AI applications' training, inference, and processing tasks. The chapter also examines scalable, secure, and high-availability storage solutions, along with architectural designs and best practices for Azure OpenAI integration. Strategies to improve data retrieval, storage efficiency, and search functionality are also covered.

Chapter 6. Azure AI Search Services for Azure OpenAI Implementations: This chapter explores Azure AI Search Services, highlighting their significance and core components, including indexes, indexers, and data sources. It provides a detailed guide on implementing Azure Search and building

solutions using these services. The chapter examines mechanisms for indexing and organizing data stored in Azure Storage, emphasizing scalability and performance for handling large data volumes and user queries. Best practices for ensuring fast, responsive search experiences in AI applications are also covered.

Chapter 7. Getting Started with Generative AI Using Azure OpenAI Services: This chapter provides an in-depth overview of Azure OpenAI Services and their practical applications. It explores Azure OpenAI Studio, including the Playground, Chat, and Assistant features, and guides you through setting up prompts, configuration parameters, and data integration. The chapter covers the OpenAI deployment process, building and fine-tuning models, and leveraging endpoints for custom solutions. It also delves into Azure OpenAI integration with LangChain, content moderation, version handling, and deployment scalability. Additionally, you will learn to build AI apps using public and private data, deploy web applications, and integrate AI solutions with MS Teams and other tools.

Chapter 8. Advanced Azure AI Studio–I: This chapter introduces Azure AI Studio as a transformative platform for Generative AI development, empowering developers to build, test, and deploy AI solutions confidently. It explores models, capabilities,

responsible AI practices, and prompt orchestration through features such as prompt flow. The chapter highlights how Azure AI Studio enables the creation of custom AI products, integration of pre-built services, and tools for prompt evaluation. It also addresses content safety, privacy, security, and compliance, equipping developers with ethical and responsible AI practices to navigate the complexities of Generative AI effectively.

Chapter 9. Advanced Azure AI Studio–II: This chapter explores Azure AI Speech, Language, and Vision Studios, showcasing their capabilities for easily creating custom projects and models. It covers both no-code and code-based approaches using client libraries and REST APIs. You will learn to utilize Speech Studio for text-to-speech solutions and Language Studio for processing textual data. The chapter also delves into Azure AI Vision, offering innovative features such as image analysis, text extraction via OCR, and facial recognition. Focusing on responsible AI practices, it demonstrates how to integrate vision capabilities into applications without requiring machine learning expertise.

Chapter 10. Generative AI Use Cases for Industries–I: This chapter explores the diverse use cases of Generative AI across industries and its transformative impact on business development, user benefits, and quality of life. It highlights AI's

role in healthcare, including medical report analysis, drug discovery, personalized medicine, and patient care. Applications in finance, such as fraud detection, financial forecasting, and customer service, are examined, along with its use in entertainment for generating music, art, and virtual influencers. The chapter also delves into Industry 4.0 applications in manufacturing, precision agriculture, and food processing. Finally, it discusses ethical considerations, security challenges, and the broader implications of Generative AI in these fields.

Chapter 11. Gen AI Implementation Use Case with Azure OpenAI–II: This chapter explores how Azure OpenAI solutions drive business growth and innovation across industries. It highlights applications in telehealth, analyzing call center data to improve patient experiences, and manufacturing, where it automates processes and predicts equipment failures. In banking and financial services, Azure OpenAI enhances advisor support, while in education, it streamlines workflows and supports learning implementations. The chapter showcases how these solutions elevate operational efficiency and provide strategic advantages for businesses.

Downloading the code

bundles and colored images

Please follow the link or scan the QR code to download the ***Code Bundles and Images*** of the book:

https://github.com/ava-orange-education/Ultimate-Azure-AI-Services-for-Gen-AI-Solutions

The code bundles and images of the book are also hosted on ***https://rebrand.ly/e17824***

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CHAPTER 1

Introduction to Generative AI

Introduction

This chapter will discuss Generative AI and the evolution of Artificial Intelligence and transformer architecture backgrounds. It will then discuss how Gen AI differs from classical AI and ML. The core of Generative AI and its characteristics transformed text, images, and video processing using Gen AI. We should remember that not all Gen AI models are Large Language Models (LLMs), but all LLMs are a type of Gen AI. Moreover, LLMs specifically refine how we generate and interact with the given textual data. The theories and rules behind these models and how they relate to neural networks, training approaches, and text generation are equivalent to human-like content. Why do industries require Gen AI, and what are the significant benefits along with Gen AI ethics and high-level security aspects of this technology?

Structure

In this chapter, we will discuss the following topics:

Introduction to Generative AI

Evolution of Gen AI

Generative AI and its Characteristics

Gen AI Transformer Architecture

How Gen AI Differs from Classical AI and ML

How Industries Get Benefits out of Generative AI

Gen AI Ethics and High-Level Security

Introduction to Generative AI

In recent years, Generative AI has become the buzzword in this busy world. Every industry and every commoner talks about AI in all forms. In this book, Generative AI, or, shortly, Gen AI, will be discussed at a very detailed level, and experiments will be conducted to build the Gen AI application using various tools and technologies.

Definition of Gen AI – A simple one-line answer is as follows:

"The application will generate new information for you from your input."

We cannot treat all this information as simple and take it lightly. It is compelling and insightful and provides solutions for various problem statements and decision-making purposes for numerous industries' specific challenges in multiple aspects. This is the main reason all industries are focusing on building their own Gen AI tools and integrating multiple AI tools to accomplish their needs and necessities.

It does not matter what input format you use; simultaneously, you can generate your desired output format. We will discuss the input and output type and format in detail in the upcoming chapters and how technologies are helping to achieve this. *Figure 1.1* shows the simple block diagram of the Gen AI System.

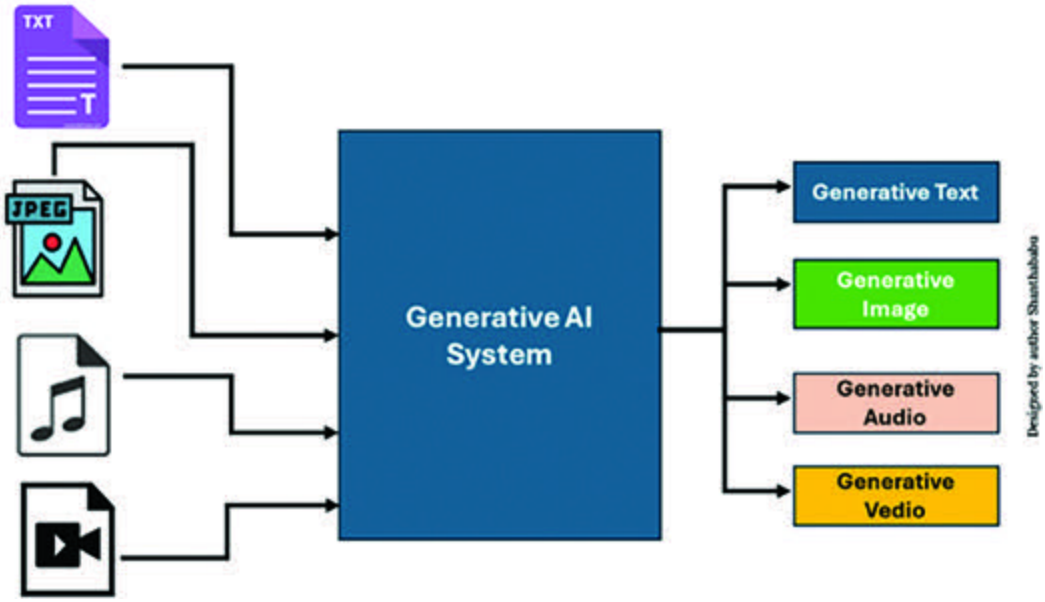

Figure 1.1: *Simple Block Diagram of Gen AI*

Technically, we can define Gen AI as a branch of that involves using specific algorithms and larger models to create new and original content from input. This category can generate content that is not replicated or copied from the original input data

and shares the relevant characteristics. Undoubtedly, the end product is entirely new and relevant to business aspects.

Characteristics of Gen AI Involving LLMs:

The characteristics of Gen AI involving LLMs are as follows:

Gen AI has non-deterministic outputs and creates new and original content based on the business point of view.

This involves a complex training mechanism to generate synthetic data for various business perspectives.

It consists of the massive number of Neural Networks involved internally to generate accurate data for analysis.

It adds more creative elements to generate precise content to attain the expectations.

Ultimately, it helps to generate additional training data to support unbalanced situations.

Benefits of LLMs:

We can build very creative and innovative products for business solutions.

We can enhance the customer experience journey from past collections of data to strengthen customer bonding.

We can construct high-time and cost-effective products from vast quantities of data.

We can customize the products to be highly scalable to achieve the results.

We can ease the process of data analysis with less effort by using tools and technologies.

We can automate the task behind the screen with practical solutions and reduce manual interventions.

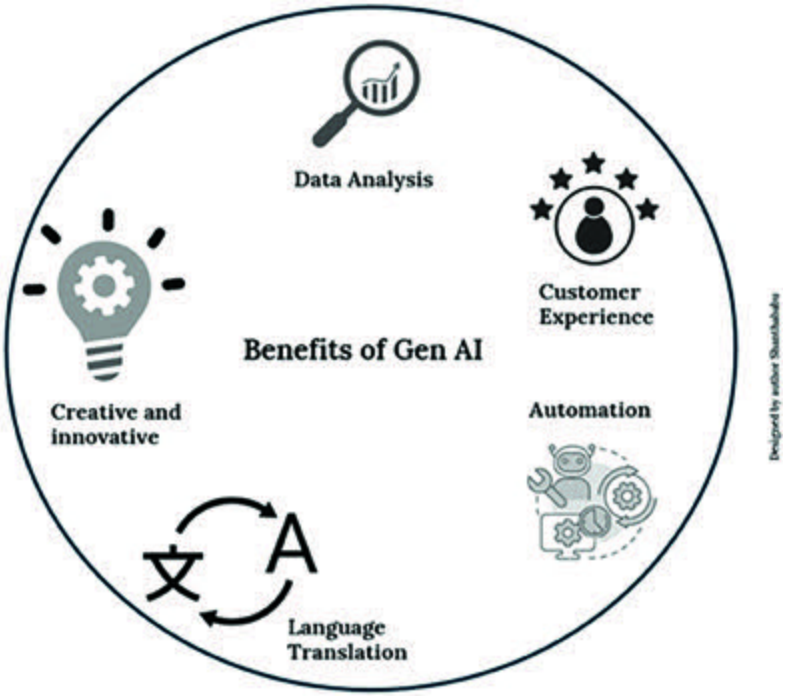

Figure 1.2: *Benefits of Gen AI*

Evolution of Gen AI

The popularity of Gen AI has been around for decades, but its actual breakthrough in the industry is subsequent to the introduction of ChatGPT in the market. The main objective is to generate the content, whether text, image, audio, or video. Gen AI is an exceptionally revolutionary technology that brings abundant benefits to businesses irrespective of any domain.

The Evolution of Gen AI started in the 1950s and has had a long journey and significant milestones.

1952—Love Letter: British computer scientist Christopher Strachey built a software program to generate Love Letter text. It was considered one of the earliest examples of creativity and natural language generated by TEXT.

1965-66—First Chatbot: During 1965-66, Joseph Weizenbaum, a computer scientist at the MIT Artificial Intelligence Laboratory, built the first chatbot application. It was called ELIZA. It was designed to simulate a discussion with a Rogerian psychotherapist using the concepts behind pattern-matching approaches and scripted-based responses.

This chatbot is a pioneer for pre-programmed responses based on those inputs. It paves the way for natural language processing, human-computer interaction, future chatbots, and conversational agent developments.

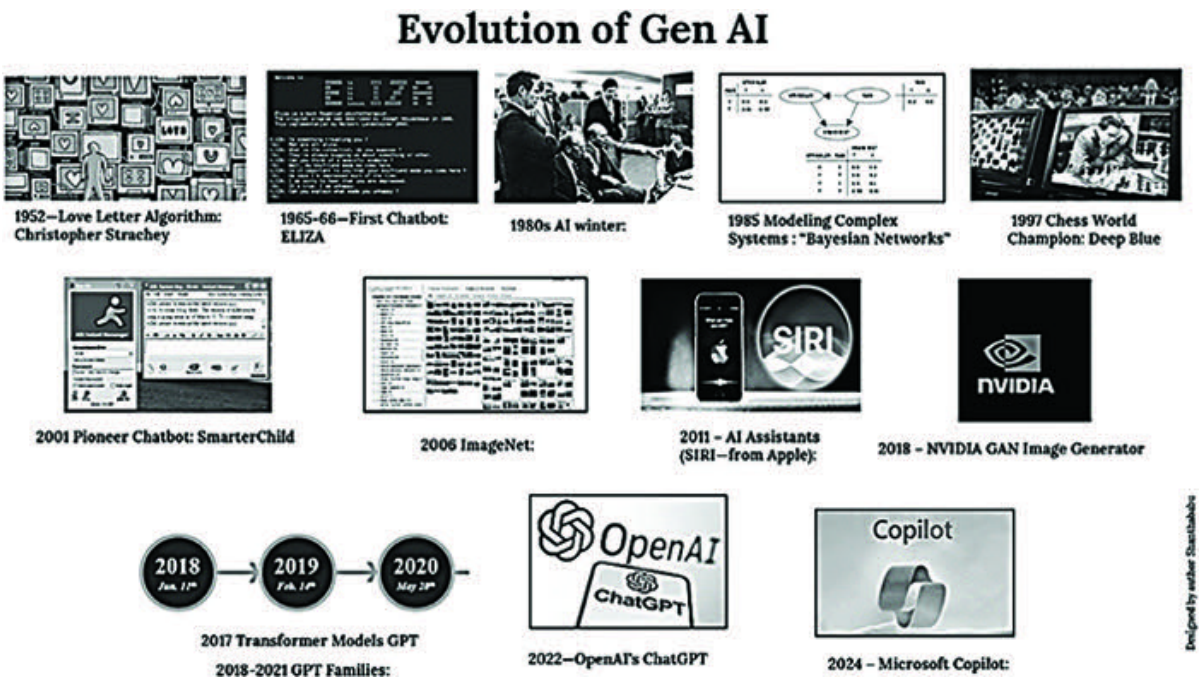

Figure 1.3: *Evolution of Gen AI*

1980s AI Winter: In AI history, there was a slowdown in this technology during the 1980s. During this period, action was taken to reduce funding and interest in AI research and development, and typically, in other fields, hype, disappointment, and critique

occurred in the AI domain, resulting in funding cuts. However, work continued in the area of neural networks.

1985 Modeling Complex Systems: During this period, Judea Pearl, a computational scientist, and great philosopher, introduced typical mathematical-based ***"Bayesian Networks"*** and seeded for the complex system modeling in the Generative AI era.

1997 Chess World Champion: The famous IBM chess computer "Deep Blue" defeated the chess world champion Garry Kasparov. This is the best evidence of AI's decision-making strategy and the event that unlocks the potential of Generative AI.

2001 Pioneer Chatbot: In 2001, Active Buddy developed the chatbot version of the Generative AI application and launched it on AOL's (American Online) Instant Messaging (AIM) platform. One of the pioneering CHATBOTS extensively used by most consumers, the product was designed to provide information regarding entertainment and services to major AIM users.

2006 Dr. Fei-Fei Li at Stanford University invented the "ImageNet" product, which utilizes a larger volume of visual databases to train the computer vision algorithm for object detection and recognition, which involves a more significant contribution of Deep Learning and AI.

2011 – AI Assistants (SIRI—from In 2011, Apple introduced the AI Assistants (SIRI). It revolutionized voice-based assistance and was the first product that came into public hands as a highly convenient and practical device for houses. It supports various aspects of assistance, such as Boss meeting reminders, voice recognition communication, and integrating and controlling other devices based on the command.

2017 Transformer Models: Open AI organization released a Generative Pre-Trained Transformer (GPT-1) language model that helps to generate the text based on the given input prompt.

2018 – NVIDIA GAN Image Generator: The breakthrough of NVIDIA's AI image generator is a landmark in the AI art creation environment. It introduces the StyleGAN architecture, which can generate highly creative, realistic, customizable images based on needs. Due of its compact size, training time, and excellent operation, all these abilities are set to transform the way we use Generative AI.

2018-2021 GPT Families: During this period, the extension of transmission models (GTP-3 and 4) expanded further along with the massive parameters inputs and generated creative human-like

text and images at different levels and incomparable height of the technology expansion in AI space.

2022—OpenAI's ChatGPT: After the introduction of ChatGPT, Gen AI utilization increased enormously since its capabilities include unlivable content creation, precise answers, and more interaction with humans. Nowadays, everyone has an account in ChatGPT.

2024 – Microsoft Copilot: As we know, Microsoft is always unique in technology adoption, no exception in this Gen AI; they introduced the Copilot by integrating ChatGPT with Microsoft 365, which is a significant milestone and brings excellent features for many business use cases. This is a major release from Microsoft's Azure OpenAI Studio with unique features to build highly intelligent tools and technologies for many industries' innovative solutions. Indeed, we are going to cover most of them in this book.

Note:

Evolution of Gen AI

1952—Love Letter: Christopher Strachey, a British Computer Scientist.

1965-66—First Chatbot: During 1965-66, Joseph Weizenbaum, a computer scientist, built the first chatbot application.

1980s- AI winter: In AI history, there was a slowdown in AI technology.

1985- Modeling Complex Systems: During this period, Judea Pearl, a computational scientist, and great philosopher, introduced typical mathematical-based systems.

1997- Chess World Champion: The famous IBM chess computer "Deep Blue" defeated the chess world champion Garry Kasparov.

2001- Pioneer Chatbot: In 2001, Active Buddy developed the "SmarterChild" chatbot version on AOL's (American Online) Instant Messaging (AIM) platform.

2006- ImageNet: Dr. Fei-Fei Li at Stanford University invented the "ImageNet" product, a computer vision algorithm for object detection and recognition.

2011– AI Assistants (SIRI—from Apple): In 2011, Apple introduced the AI Assistants (SIRI).

2017-22- Transformer Models: GPT Families and OpenAI's ChatGPT.

2024– Microsoft Copilot: As we know, Microsoft is always unique in technology adoption.

Gen AI - Transformer Architecture

Transformer Architecture Overview:

One of the Generative AI models is the **Transformer-Based** which primarily adopts neural network techniques and natural language processing methodologies, such as text generation, sentiment analysis, question-and-answer, text summarization, and language translation.

In simple terms, the transformer architecture works as a Food BALL team in that the players have the connection and thought process of passing the ball from one side of the court to the target to hit the ball inside the goalpost. This is a very high collaborative effort incurred process.

Since these transformer models are built from the deep neural network-based mathematical model, their structure and function replicate the human brain system, which consists of multiple layers of neurons. So, each neuron can receive various input data, perform a powerful mathematical computation internally with the data, and process the influential output.

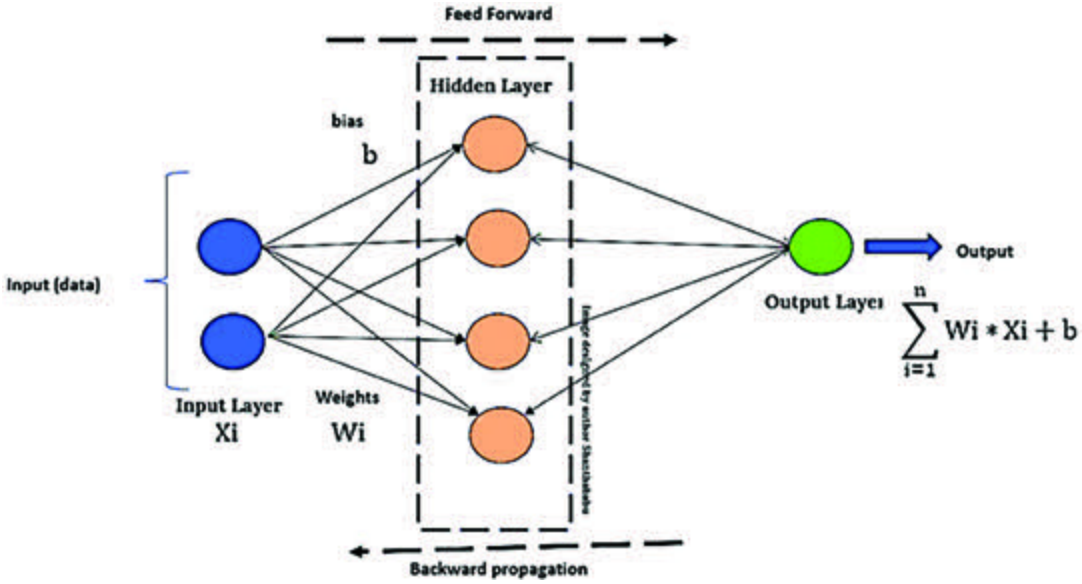

Figure 1.4: *Neural Network*

The transformer model is highly capable of generating high-quality text in various contexts based on the demand and requirements. This model uses a **SELF-ATTENTION MECHANISM,** a distinctive feature that captures long-range dependencies and is more effective than traditional Deep Learning and NLP models.

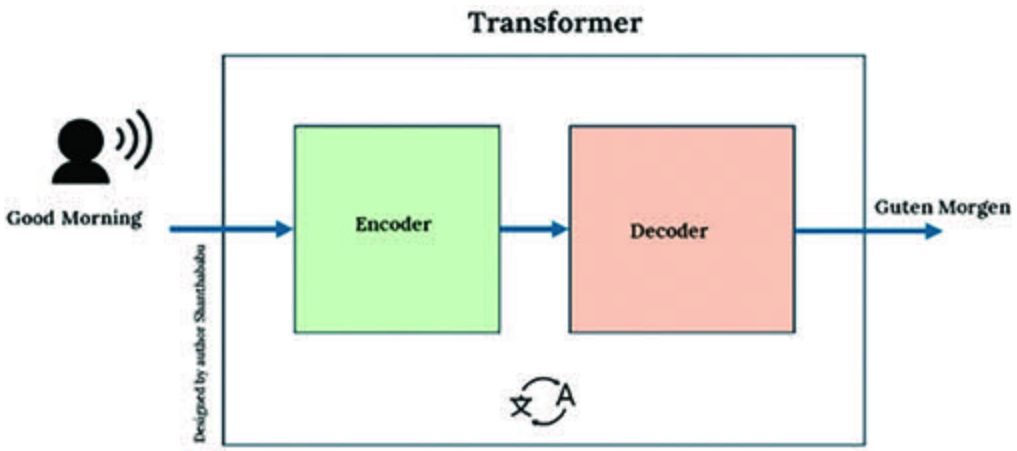

Figure 1.5: *Simple Language Translation Transformer Representation*

Components of Transformer Architecture

Let us discuss the components of Transformer Architecture:

At a very high level, this architecture uses two major pilers: **Encoder** and **Decoder.**

ENCODER: It takes the input feed, extracts context, and then the meanings from the data, which it passes onto the decoder.

DECODER: It is responsible for converting the input into several sequences from the encoder's feed. These embeddings are processed by the SELF-ATTENTION technique. The decoder's architecture is similar to the encoder and produces the generative output.

SELF-ATTENTION: Self-attention is an important process stage. In this stage, it captures the significance of each word and the relationships between them. This is typically how humans experience understanding what verbs, nouns, adjectives, and nouns are when they speak and write and the thought process of how each word relates to every other word in a sentence.

The rest of the components are Input and Output.

INPUT: This is just an input for the system.

OUTPUT: The output is the sequence generated by the transformer.

Let us think big. *Figure 1.6* shows the Transformer Architecture (Universal Picture). There is nothing new we cannot add here.

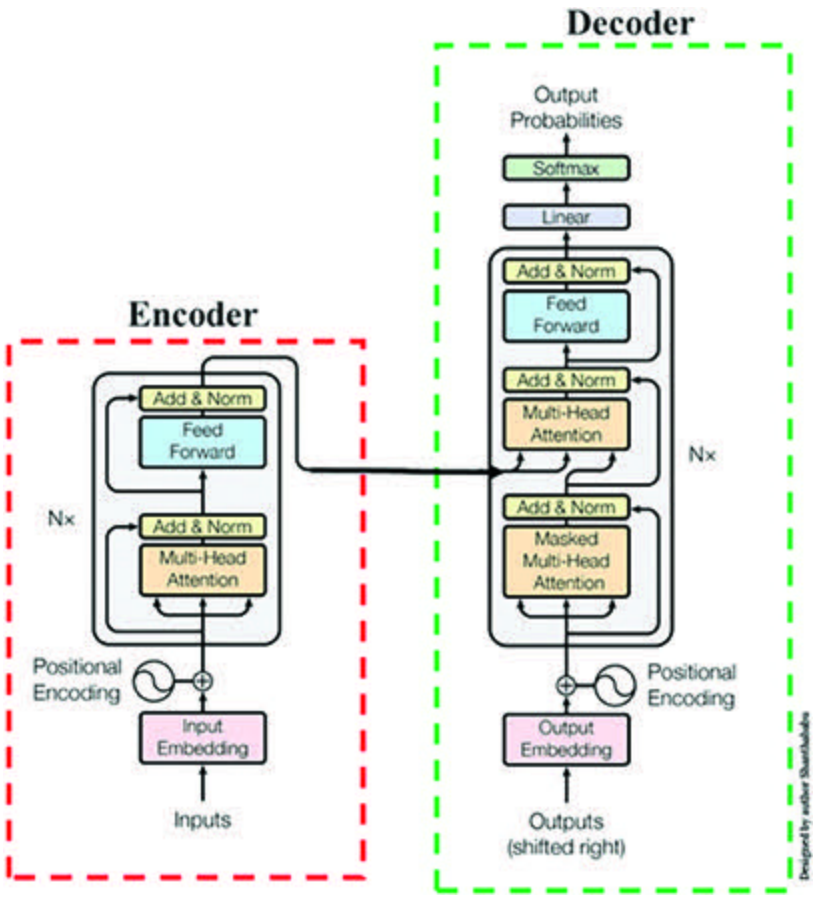

Figure 1.6: *Transformer Architecture (Universal Picture)*

As we understood from the earlier explanation, the follows an **ENCODER-DECODER** structure and does not depend on the model, such as the Recurrence and Convolutions, to generate an output.

The left part is **ENCODER,** and the right part is **DECODER.**

ENCODER

The Encoder consists of N identical layers. Each layer contains a MULTI-HEAD ATTENTION LAYER and a FEED FORWARD NEURAL NETWORK as a fully connected set of blocks. The output of each layer is used as an input to the subsequent layer with the same dimensionality. The block count depends on the value of x. If x=8, there are N8 blocks and layers inside the encoder at the left part of the encoder.

As per the mechanism, the **ENCODER** maps a set of inputs as sequences, and continuous inputs are representations. They undergo the Nx block and layer with x times the "Addition and Normalization" process, which is then fed into a DECODER; refer to *Figure*

Before getting into the encoder, let us understand what **Input Embedding** and **Positional Encoding** are.

Even the AI system cannot understand the word(s) directly, so we have to convert them into numbers prior to feeding them into the system; the mechanism of this conversion should be implemented in sensible ways to build an input sequence into the system.

This process is called the first component in the encoder's block. You will see it in *Figure* in which the input is transformed into sensible numbers, constructed in multidimensional vector space, and fed into Multi-Head Attention (MHA) later than positioning them with the help of the positional encoding process. Let us discuss the following steps:

Step 1: We have to organize the input sentence into a comma-separated list of the words.

Step 2: Add two additional tokens to indicate the start and end of the sentence.

Step 3: Follow the vocabulary list, have to map each vocabulary with a unique number; this is the so-called indexing.

Step 4: As per the standards, we have to normalize all these words as **One Hot Encoding** to build the high-dimensional vector base. The process of converting each word into these high-dimensional vectors is called **Input Embedding** or **Token**

Step 5: The Transformer works differently from LSTM, where the input is fed into the system parallelly. This means the order of the sentence might be lost. Technically, adding additional

information of the same dimension as the input vector to represent the sequence of the words in the sentence, which is called **Positional**

So, finally, the combination of input embedding and positional encoding vectors will be fed into Multi-Head Attention (MHA).

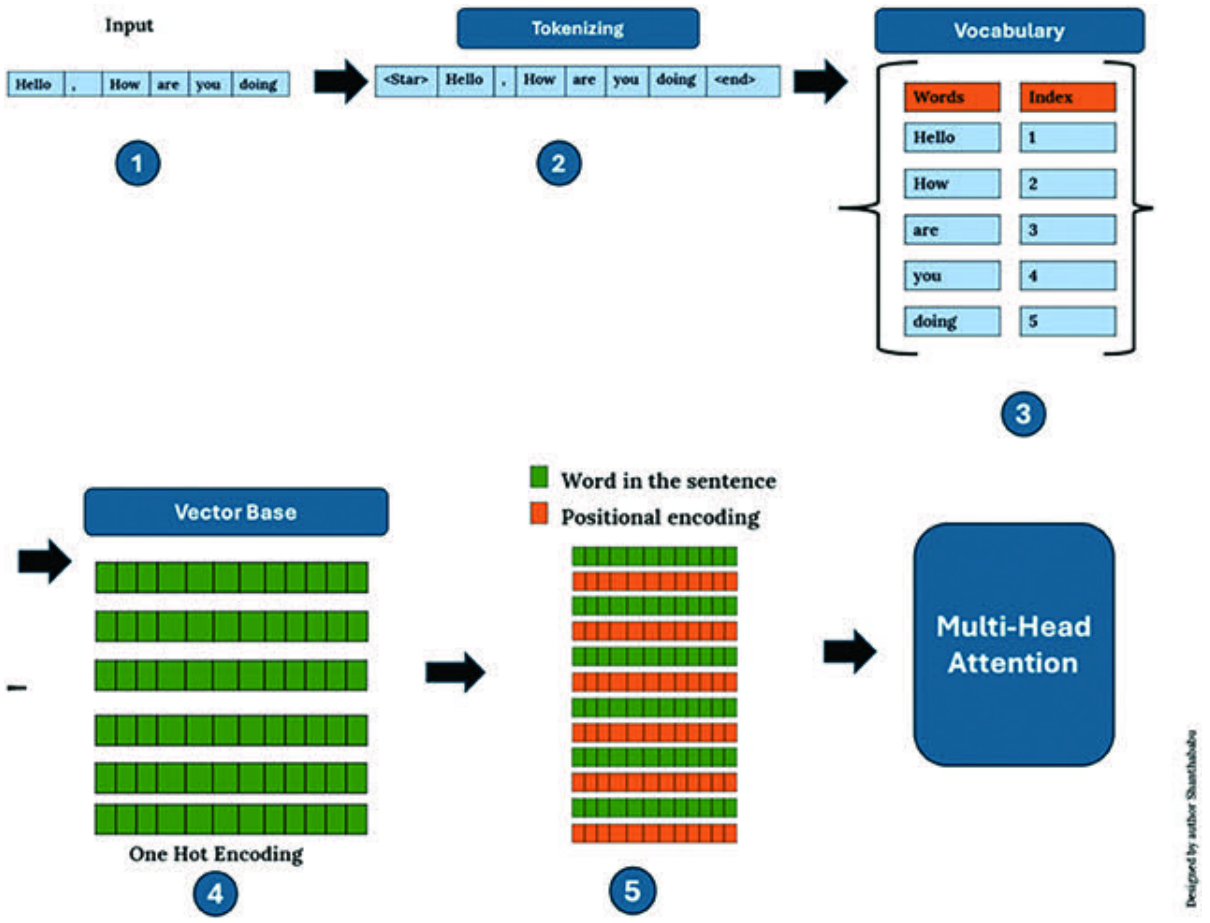

Figure 1.7: *Input and Positional Embedding in Transformer Architecture*

The Positional Encoding Layer is a Matrix, because: The positional encoding scheme in transformer architecture has each position versus index mapped to a vector. Therefore, the output of the positional encoding layer is a matrix.

Let us do a little deep dive into the **Single and Multi-Head Attention layers.**

As we can see, the attention layers are in the encoder and decoder, which are part of the transformer architecture.

Stage 1: Tokenization and Embeddings: This is the pre-stage in the attrition layer building; at this stage, the given input text has been tokenized, and the embedding of each word in the input query, as shown in *Figure*

Stage 2: Vectorization: In this stage of the attention layer, which consists of three neural networks, each word produces three output vectors, such as Queries, Keys, and Values. This process takes place for each word in the given input query.

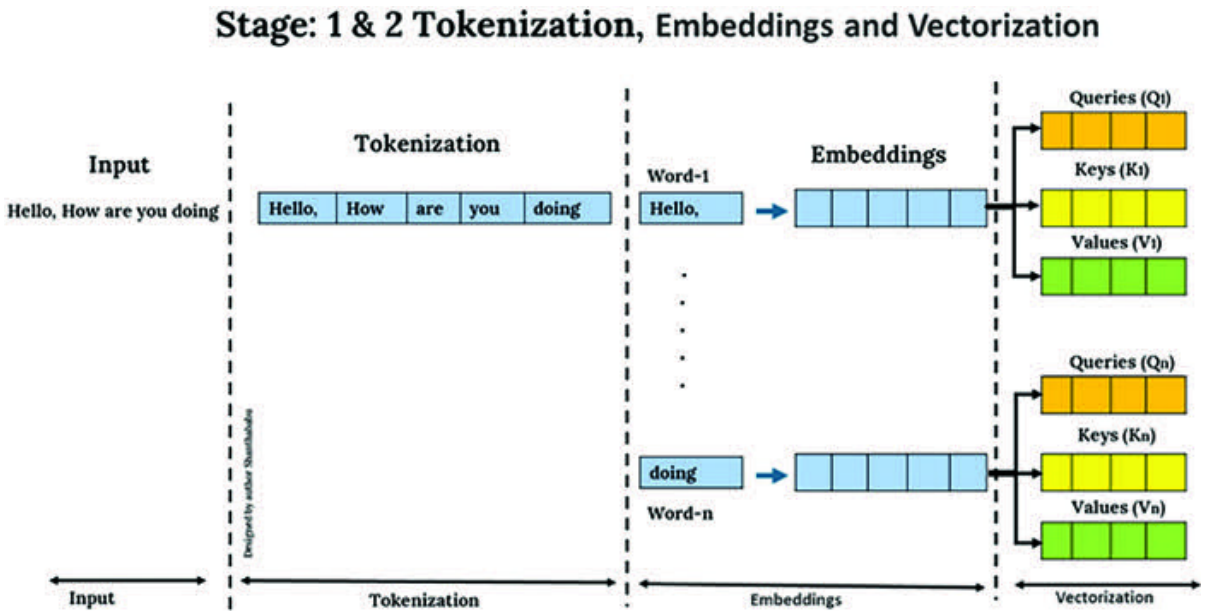

Figure 1.8: *Stage 1 and 2 Attention - Tokenization, Embeddings, and Vectorization*

Stage 3: Calculating the Word's In this stage, the score of each word is calculated by multiplying Q and K. The process starts with the dot product of the first word's Vector Query (Q) and all the Vectors of K of the input.

This process will confirm the similarity measure of each product of Q and K.

The exercises will start the dot product between the first word of the query (Q1) and all keys (K1...Kn) in the input sentence, which are used to produce the scores.

If the Q and K vectors are aligned and have meaningful values, their dot product will produce a significant value as an outcome.

If the Q and K vectors are close to orthogonal, then the result of the dot-product will approach zero.

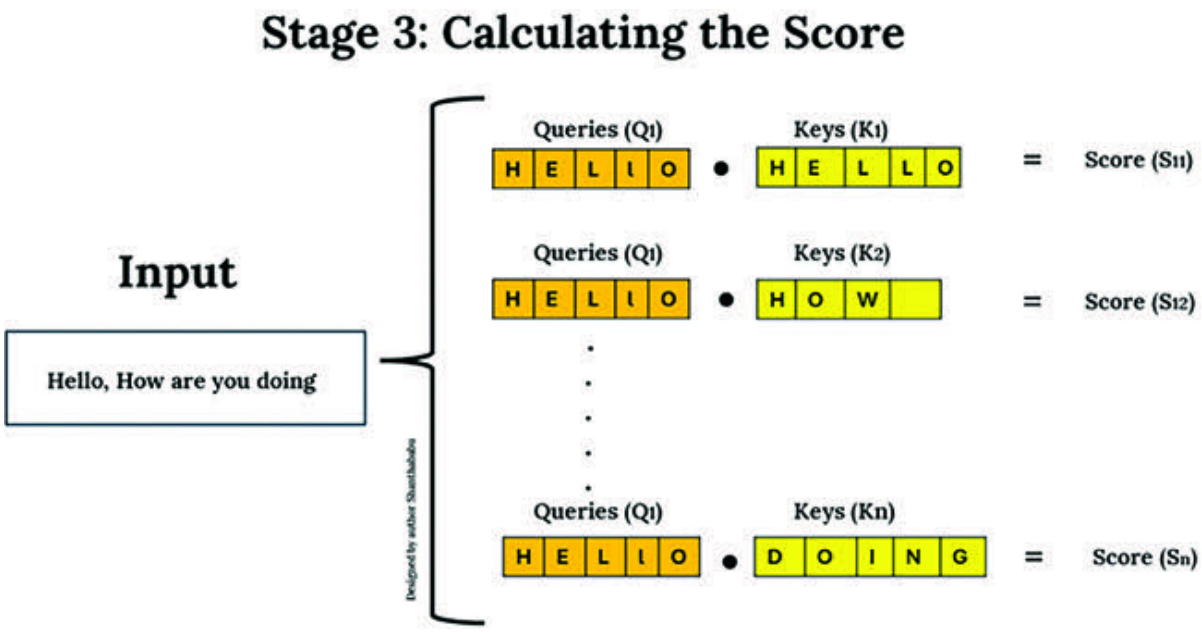

Figure 1.9: *Stage 3: Attention – Calculating the Score*

Stage 4: Calculating Scaled Value In this stage, the Values (V) are scaled up using the scores derived from Stage 3. The resulting scaled value vectors are summarized as a final step, building the attention vector for each word from the input.

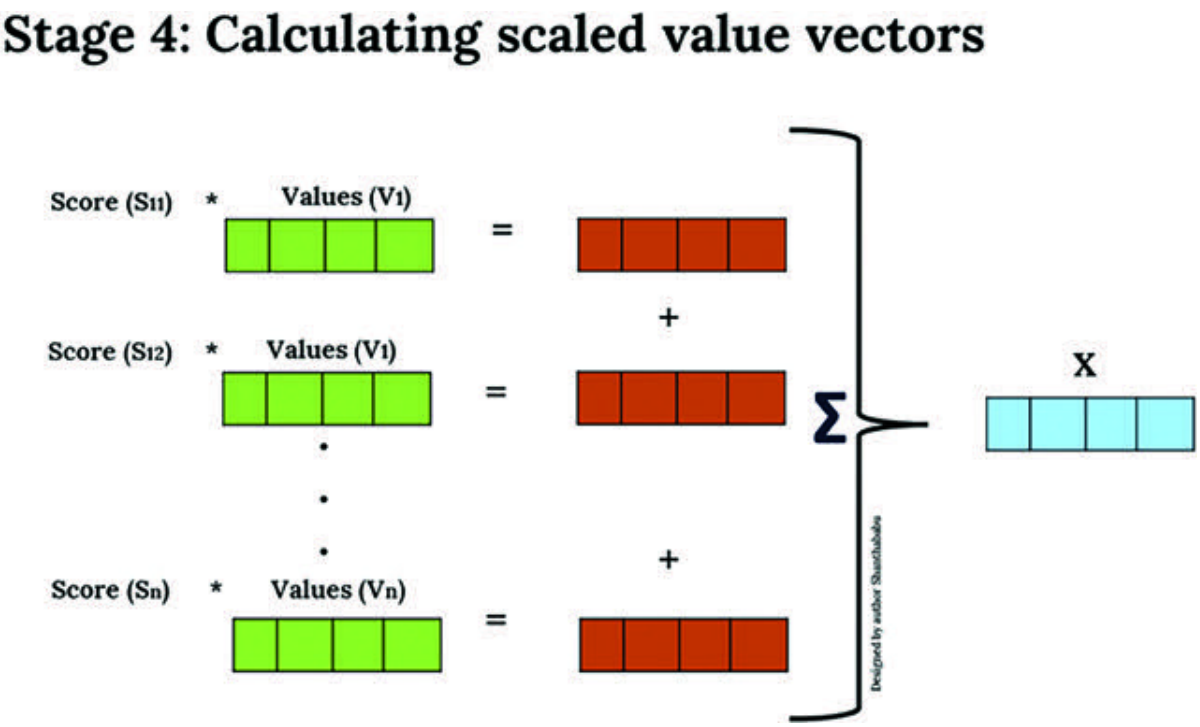

Figure 1.10: *Stage 4: Attention – Calculating Scaled Value Vectors*

This X vector is the attention vector for Query 1. We have to do this exercise for every query for the given input and build Xn vectors; the resulting vectors are the attention layer's outputs.

Each resulting vector, Xn, depends on the Query (Qn) for that word and also depends on the Keys (Kn) and Values (Vn) of all the words in the input text. The overall outcome makes the attention powerful in the form of sequential tasks and able to embed it into the context of the entire input representation.

So far, we have discussed the Single-Head Attention (SHA) layer. Let us get in touch with the Multi-Head Attention (MHA) layer. This is an expansion of **SHA** that allows faster computation.

As we know, in Single-Head Attention, the embedding vectors for each word are 512 dimensional. In Multi-Head Attention, the **Query, Key,** and **Value** vectors are equally split into **8** which means **Q** and **K** have **64-dimensional** Finally, this will be multiplied by **V (8 heads),** and the attention layer will be applied to each head using the same method we explored in SHA. Then, each head is concatenated together to form (64*6=512) 512-dimensional attention vectors as the resulting attention vectors, as shown in *Figure*

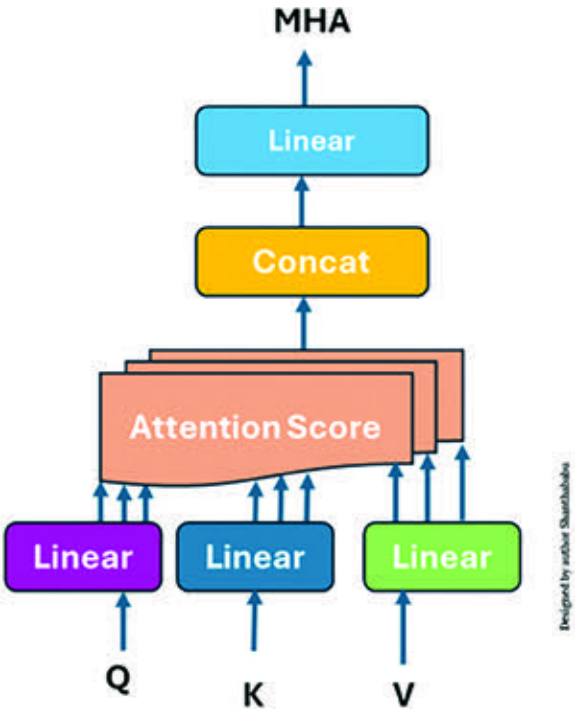

Figure 1.11: *Attention – MHA*

The encoder represents encodings and produces embeddings from each query using the attention mechanism; the decoder works in the other route and generates the output text. Let us discuss this quickly.

DECODER

The **DECODER part** receives the encoder's output succeeding a set of processes we discussed. Functionally, the decoder is similar to the encoder. It also consists of a stack of Nx layers x=6; the architecture has six matching layers, each composed of three sets of sublayers. The sublayers are classified as follows for better understanding:

Decoder Self-Attention

Encoder-Decoder Attention

Position-wise Feed-Forward Networks

These sublayers employ a residual connection around them, followed by layer normalization.

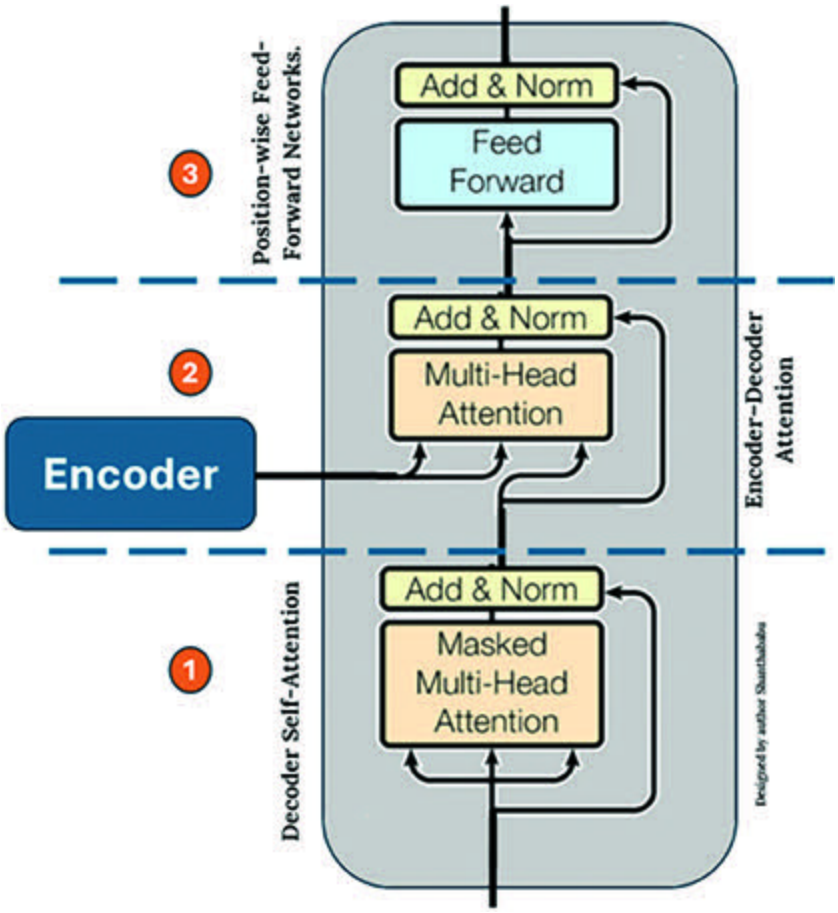

Figure 1.12: *DECODER and its Layers*

Decoder Self-Attention (First Sublayer):

This decoder self-attention is the first sublayer that receives the preceding output of the decoder, enriches the output with the positional information, and implements Multi-Head Self-Attention (MHA/MHSA) over it by introducing suppressing the matrix mask over the values produced by the matrices scaled multiplication.

As we discussed earlier, the encoder is designed to handle all words in the input sequence despite their position in the input order. Still, the decoder is designed to attend only to the previous words. This is the main difference between an encoder and a decoder.

Hence, the prediction for a word at position depends on the words previous to the input sequence. Please refer to *Figure 1.12* – indicated by 1.

Encoder-Decoder Attention (Second Sublayer):

This encoder-decoder attention is the second sublayer that receives multi-head mechanism input/queries from the decoder side from the previous decoder's output (the first sublayer's output) along with the keys(K) and values(V) from the encoder's output (the left side of the architecture diagram). This layer is also MHA. In this layer, the decoder attends to all the words in the input sequence concerning the MHSA mechanism, similar to layer one in the encoder.

Please refer to *Figure 1.12* – indicated by 2.

Position-wise Feed-Forward Networks (Third Sublayer):

This position-wise feed-forward network layer is the third and final layer in the decoder. It is similar to the one implemented in the second sublayer of the encoder. Along with the three sub-layers, the decoder has residual connections, followed by a normalization layer. Please refer to *Figure 1.12* – indicated by 3.

Linear and Softmax Layer:

This layer sits outside the encoder and decoder blocks and is the final layer in this architecture. The output vector from the decoder undergoes a linear transformation in this layer. There, the output changes the dimension of the vector size 512 into the size of the vocabulary. The Softmax layer further converts the vector into ~10,000 probabilities. Basically, the Softmax function is used to create a probability distribution that reveals the likelihood of each token in the sequence.

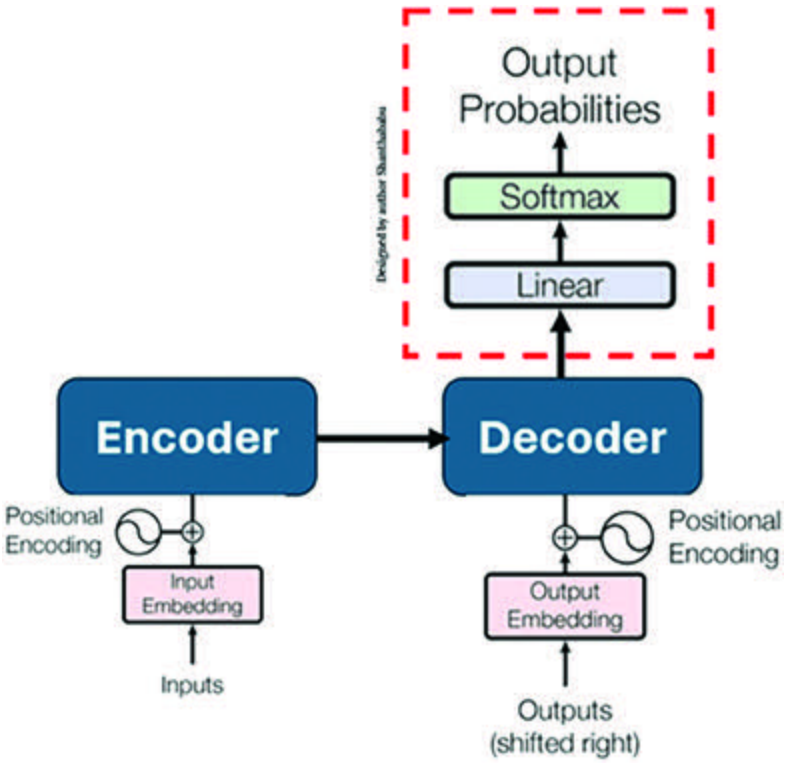

Figure 1.13: *Linear and Softmax Layer*

Transformers have a simple architecture consisting of an encoder and a decoder. In the encoder, the model first takes the input text and translates it as vectors, which use attention. The decoder works in the opposite direction and goes from the vectorized transformation into the sentence of the entire word. The output could be in any language.

The concept of self-attention is essential in this transformer's architecture. Self-attention captures the input of a sentence and understands the meaning of each word and the relationships

between them. This process is very similar to the human's understanding of verbs and nouns, how adjectives relate to nouns, and how words are associated with every other word in the input sentence.

Gen AI versus Classical AIML

AIML is a collection of technologies under one roof, such as Data Science, Machine Learning, Deep Learning, NLP, and Computer Vision. All these technologies utilize a specific dataset and format depending on the technology used. The output includes numeric text, images, and video. The output is mainly statistical-based, but in the case of Gen AI, it is completely independent of the input format, outputs a hundred percent new data, and is highly creative.

Machine Learning

As we know, **"Machine learning"** is a subset of artificial intelligence that uses various algorithms to analyze the data, insights of the data taken for analyses, and make predictions and informed decisions based on the nature of the data.

ML has the following algorithms and is designed to learn and improve from training datasets, using relevant statistical techniques to identify patterns, insights, and relationships between them.

Supervised Learning:

Regression

Linear Regression

Decision Tree Regression

Random Forest

KNN Model

Support Vector Machines (SVM)

Classification

Logistic Regression

Naive Bayes Classifier

Decision Tree

Random Forests

Support Vector Machines

K-Nearest Neighbor

K-Means Clustering

Unsupervised Learning:

Clustering

K-Means Clustering

K-Nearest Neighbor (KNN)

Hierarchal Clustering

Anomaly Detection

Neural Networks

Principle Component Analysis (PCA)

Apriori Algorithm

We are not going to explore each algorithm in the following list since it is out of the scope of this book, but we will undoubtedly discuss Gen AI and ML's futures, benefits, pros, and cons.

Certainly, ML can continue updating and improving its capability to make accurate predictions and classification problems. As the volume of data grows and large datasets

slow down, the difficulty of handling ML solutions increases. ML algorithms are difficult to train, time-consuming, and expensive.

Challenges with Interpretability:

ML Deep learning algorithms are incredibly complex in certain scenarios that are complex and challenging to interpret. This lack of clarity can prevent businesses from understanding how to make decisions and challenge them to build trust in ML systems. This is critical in the healthcare, banking, and finance industries, where decisions are high risk.

Costs and Resources Incurred:

It requires significant investment in technology, infrastructure, data acquisition, cleaning, preparation costs, and specialized and expert development and training.

Data Quality:

The accuracy and reliability of the ML model depends on data quality. The data should be complete and balanced for accurate predictions and decisions. Otherwise, this would require high

maintenance costs and effort for data cleansing, validation, and monitoring.

Data Security and Privacy:

Data collection raises data security and privacy concerns. Businesses are responsible for ensuring that this challenge is handled sensibly by customer data. Incidents such as data breaches and unauthorized access to customer-sensitive information are significant drawbacks of ML implementation.

Generative AI

Generative AI, or Gen AI in short, is a branch of AI that uses specifics to generate new content that mimics the data fed and trained on. The fundamental concept of Gen AI is to learn patterns and relationships in a given data (structured and unstructured) to create exclusively new text, summary, images, audio, and video as responses based on the underlying patterns in the input data.

As we understood from our earlier topics, the mechanics of Gen AI are powered by neural networks. These networks help analyze data to discover patterns and generate new, highly creative, and innovative content based on our requirements. This Gen AI technology utilizes several learning methods during training, including semi-supervised learning and unsupervised techniques.

Two major Gen AI models are available: **Generative Adversarial Networks (GANs)** and

GANs are best at creating visual and multimedia data. Transformer-based models, such as GPT and its family, specialize in generating text. They can understand the context of available public and private data. On the other hand, ML always relies on ML-specific algorithms. They allow the processing of large volumes of data collected from various sources based on the use cases, followed by learning the patterns and classification from the data and what is trained on.

Because of intelligence techniques such as Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Transformers, GenAI has become a possible solution provided space and enables models to develop rich thoughtful large datasets and produce extraordinary and genuine synthetic content based on the business needs. It opens up endless possibilities for several industries.

Multiple Gen AI tools are available and ready for business purposes.

Let us discuss the commonly available Gen AI tools. They have various powerful features and have emerged for different purposes in several industries.

ChatGPT is an extensive language model chatbot developed by OpenAI. It is predominantly used for natural language understanding and generating text, summaries, images, and more, mainly for tasks such as customized chatbot tools, content creation, and language translation. It largely depends on the use cases.

Gemini/Google Bard: Gemini, formerly known as Bard, is a Gen AI chatbot developed by Google. This tool can also understand human language and simulate a conversation with users. It analyzes a user request and then works to identify the input's intention. Initially, it was an experimental AI chatbot tool. It is designed for research, report generation, educational material creation, and writing coding tasks.

Copilot/Bing Chat: This is a Microsoft product formerly called Bing. You can ask any complex questions, and you can get comprehensive answers and summarized information. It is also a conversational AI language model focusing on information retrieval, task automation, and content creation.

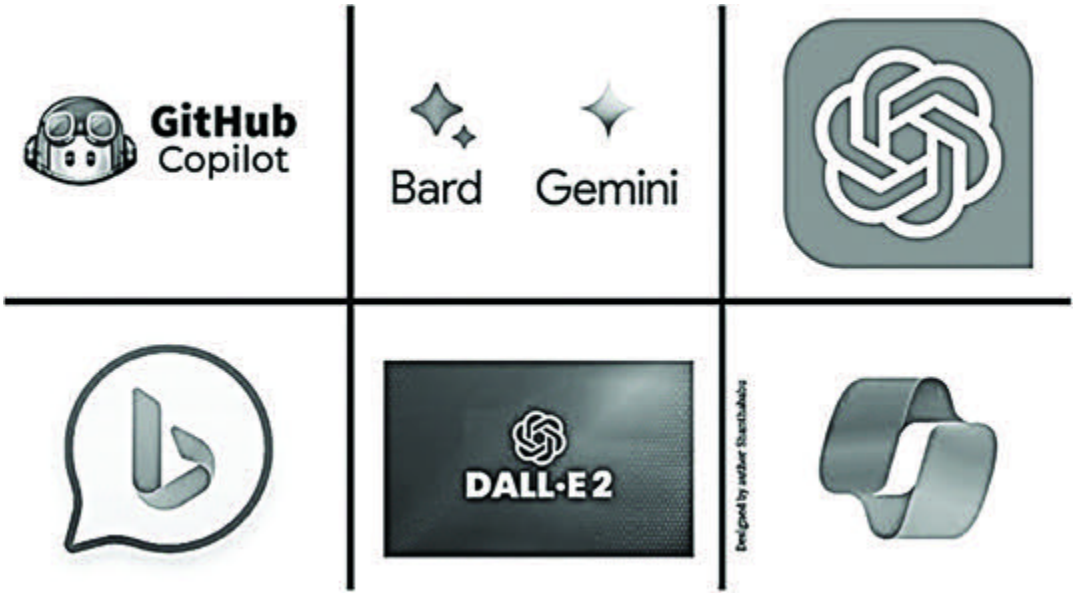

Figure 1.14: *Leading Gen AI Tools in the Market*

GitHub Copilot: GitHub Copilot differs from the earlier mentioned tools and is a specialized tool for code completion developed using **GitHub** and **OpenAI** Its primary objective is to assist users in autocompleting code for multiple IDEs, such as Visual Studio Code, Visual Studio, JetBrains, and more. Ultimately, it helps the developer with code assistance, enhancing code writing efficiency, reducing errors, and learning new programming languages.

DALL-E 2: DALL-E 2 is a recent advanced state-of-the-art AI tool for creative image generation on the OpenAI platform. It is more potent than DALL-E. Using neural network techniques, it can produce exceptionally realistic and comprehensive

pictures based on textual description as input. It can generate many more images for specific inputs and edit existing images with additional parameters.

The former-discussed AI tools help to reshape operations and consumer interactions across many industries, and many more advanced AI tools are evolving based on different use cases based on business demands irrespective of domains,

Let us deep dive into Gen AI and compare ML now:

Learning Experience (Gen AI and ML):

Gen AI models learn from data distribution and targeting to generate new data (text, summary, images, audio, and video) that recalls the input training data. The models are obtained and extracted from the input data, and new, innovative, creative instances are created from what they have learned.

In contrast, ML models are focused on learning patterns and classifications, followed by making predictions or classifications based on input data. Basically, this is focused on finding relationships and correlations within the data to make accurate predictions and patterns based on the requirements.

Data Usage/Utilization (Gen AI and ML):

Gen AI models typically require a large amount of data that accurately reflects the underlying data distribution. Without any questions, if we have more varied data, the model can capture the complexity and diversity of the actual data and produce the output. On the other hand, ML can also work with less data, which might lead to accuracy.

In both cases, insufficient or biased data can lead to poor generalization and generate unrealistic outputs. So, the dataset has to stratify the demand to produce the expected output.

Scope of Output (Gen AI and ML):

The critical distinction between ML and Gen AI lies in generating output capability. Gen AI models can create new data, such as text, summary, images, audio, and video, that are unconventional and resemble the outlines and styles they have learned from the input data. At the same time, ML algorithms make predictions or classifications based on available data in the given dataset.

In some situations, Gen AI can produce unrealistic or unreasonable outputs due to its confidence in learned patterns and the nature of the model; this is called “Hallucinations.” However, it can be controlled using techniques we will learn later in this book.

ML algorithms, on the other hand, are more focused on making precise predictions and classifications based on datasets.

Gen AI's Drawbacks

Of course, every technology and methodology has disadvantages. We will explore some of Gen AI's key drawbacks, including:

Amount of Data: One major disadvantage of Gen AI is the requirement for large amounts of data, which, too, compromises quality. Otherwise, the generated outputs may be deprived of stability and semantic meaning, making them inappropriate for targeted tasks.

Data Quality: Of course, data quality is a critical factor in AIML and Gen AI; there is no question about this. The impact will be reflected during the Gen AI training stage, even though extensive datasets will be needed to learn the pattern and build the solution.

Lack of Accuracy: Gen AI models usually struggle with maintaining consistency, accuracy, and semantic coherence in their outputs. This limitation makes Gen AI less suitable for tasks that expect precise output and adherence to rules in the

following domains: document generation in legal, financial, real estate, and research.

Because of the nature of Gen AI, it overcomes the drawbacks discussed in ML earlier.

Let us understand the key differences between ML and Gen AI:

AI: AI:
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Table 1.1: *ML versus Gen AI*

Foundation Models

With respect to Gen AI, many open-source "FOUNDATION MODELS" are available in the market, including large-scale, adaptable AI models, which can reshape your enterprise AI scope and bring huge business benefits. At the same time, there are risks and factors such as biases and security impacts.

These foundation models will form the basis of generative AI's future across the enterprise. The current buzzword around Gen AI is that **Large Language models (LLMs)** fall into this foundation model category.

The foundation models work with multiple data types, such as semi-structured, structured, and unstructured data. Because of this, businesses can make new associations across data types and increase the range of tasks that AI can use to provide solutions.

Large Language Model (LLM)

An LLM is an advanced generative artificial intelligence algorithm that applies typical neural network techniques with many parameters to process the desired output. It can understand any human language and any input text format.

LLMs are highly efficient in capturing complex entity relationships in the given input.

LLMs are capable of generating text using syntactics and semantics of the data from any language.

Ultimately, it helps to build conversational AI applications such as RAG, Text, Summary, Chat-bots, Image generation, Building coding, and so on.

Top LLMs are:

GPT-4

LLAMA 2

ChatGPT

FALCON

Mistral 7B

In the former list, the GPT (Generative Pre-trained Transformer) is the most potent and popular LLM model; it has its own history, as we discussed in the earlier topic of this chapter (Evolution of Gen AI 2018-2021 GPT Families).

2018 GPT-1 – 117 million parameters and 985 million words

2019 GPT-2 – 1.5 billion parameters

2020 GPT-3 – 15 billion parameters. Chat GPT is also based on this model

2023 GPT-4 – Trillions of parameters

Many use cases influence Gen AI. Let us discuss a few common use cases here:

Text Generation: Generally, this use case is used to generate stories and scripts in the form of text-based content that could be utilized in research, official, and social media perspectives.

Question Answering (Q&A Chatbot): This use case has a higher potential than others. It helps to generate the most relevant answer(s) based on input questions to help any size of organization, institution, and government clarify, explore, or understand the vast document collection for various initiatives.

Text In most scenarios, the organization, institution, or government entity needs a concise summary of a massive volume of documents to help explain a policy or process document based on the demand. Data would be available in public or private.

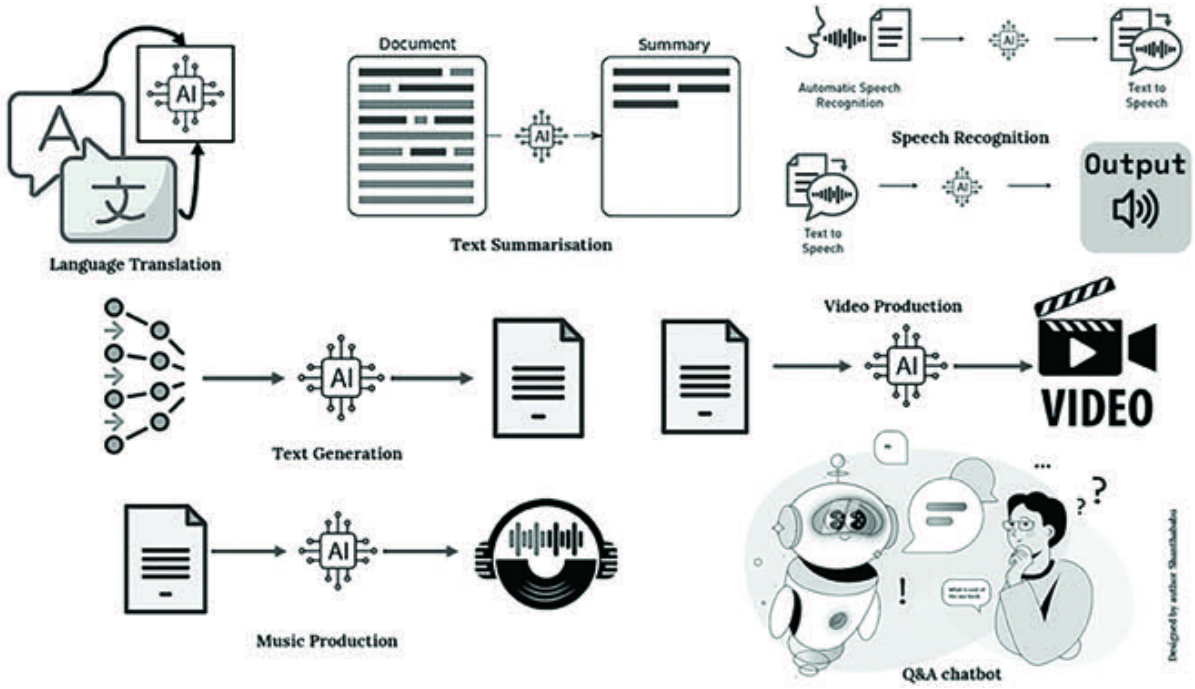

Figure: 1.15: *Gen AI – High-Level Use Cases*

Language This use case helps translate between languages. It is another high-potential use case next to text generation and Q&A, especially for helping a non-English speaker interact with a critical government service, law forum, legal document translation, and more.

Speech Recognition: This use case helps translate from one language to another using audio. It can process the audio. It would be any of the following conversions: audio-to-audio, text-to-audio, and audio-to-text. This helps many domains based on their needs.

Music—Video Production: This is a typically entertained specific use case for general purpose. It helps create new music or videos based on precise input content to produce the output in the required format. A musician or social media creator can utilize this feature and build content with practical and quality outputs in less time.

How Industries Get Benefits out of Gen AI

In general, all new technology benefits various industries to gain their market. Gen AI is no exception in its new journey. From an industry-specific point of view, there are huge lists. Let us take a look at the vital industries as follows:

Financial Functions: As we know, Gen AI helps produce content, generate ideas, and provide a Q&A perspective. Specific to financial functions, the contribution of Gen AI is enormous. Majorly written tasks include contracts, drafting responses for investor communications, reviewing general ledger entries, financial scenario generation, automated financial report generation, extracting business strategic insights, and supplementing a credit review.

Demand Forecasting: GenAI is an emerging transformative force in the current scenario and can provide real-time data insights and demand forecasting. This can help update forecasting models dynamically from previous market cycles quickly based on customer feedback, product reviews, market competition across different websites, access to market cannibalization, product loyalty on the customer side, and more. Using all this information, Gen

AI can make real-time amendments to an organization's marketing strategies and help it gain a competitive advantage.

GenAI uses unconventional forecasting models to derive demand sensing and forecasting from various data, such as consumer sentiment, price changes, demographics, seasons, and climate.

Healthcare Diagnosis and Prediction: Gen AI has rapidly become a significant factor in the healthcare industry. However, experts must understand how to use all technology to support and avoid the risks inherent in applying it to patient care. Multiple use cases across healthcare in different segments exist. At the same time, Gen AI brings about some clear uncertainties and risks. Alongside this, it promises to increase efficiency and quality and create new value for the healthcare domain, some of which are as follows:

Support real-time patient monitoring and building reports

Accelerating the drug discovery and development process

Improve clinical trial planning and execution

Building sophisticated medical imaging and analysis

Medical research and prediction of pandemic readiness

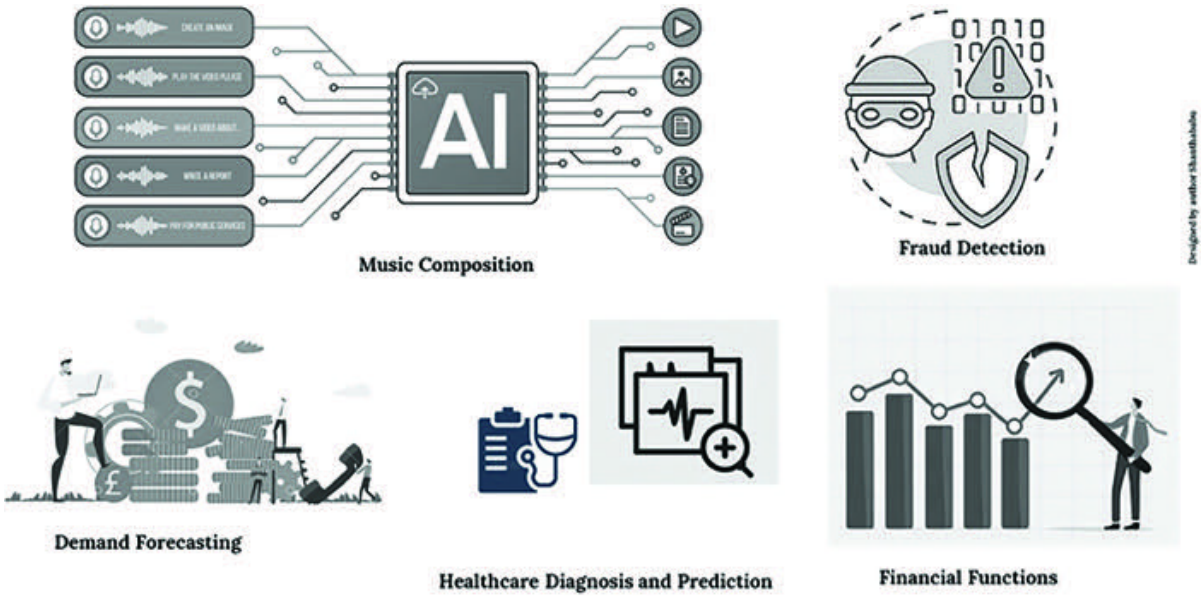

Figure 1.16: *Gen AI – Industries Benefits*

Fraud Detection: Gen AI for fraud detection is transforming risk decision-making. It offers dynamic, evolving solutions that address the challenges and obstacles in the existing approaches using classical AI solutions. Gen AI is thereby transforming the landscape of fraud detection with new energy.

It has the ability of near real-time analysis to capture fraudulent transactions and context-aware analysis capabilities, helping a lot in fraud detection.

Gen AI has excellent adaptive learning from historical and real-time data to detect fraud.

High predictive capabilities, customization of the output, and reducing huge manual oversight are more attractive features of Gen AI.

It is an excellent way to support synthetic data generation aspects, enhance privacy preservation, and ensure balancing and imbalanced datasets for further analysis.

Music Composition: Gen AI is highly innovative and creative in producing new data, summaries, and images, and it is also capable of producing excellent music. It efficiently generates new chimes, measures, trills, and whole songs.

Of course, this technology can potentially transform how music is created, using some samples to produce new and novel work. Two main approaches are used here.

Using an algorithm on a large music dataset as a sample so that the algorithm can learn the patterns and structures of music ultimately generates new music that closely resembles the training data provided.

They are inducing arbitrary sequences of notes and exploring the various musical combinations.

Across various industries, many Gen AI use cases have emerged in recent days for automating and simplifying workflows, which are effective time-intensive processes for knowledge workers, such as art and design, content creation, fashion design, legal firms, automating engineering and data processes, and supporting language translation services.

Gen AI Ethics and High-Level Security

Gen AI technology is becoming more advanced nowadays, and many ethical issues are associated with its implementation. On the other hand, it simulates human intelligence and decision-making. This brings many risks, including those to human safety.

Alongside the data requirements for Gen AI implementation, we need a massive volume of data to train the model, so there might be changes and threats in addressing customers' and patients' privacy and security concerns. If we collected the data with poor quality or synthesis, the results would be dangerous, and practically, we could not go for productization versions.

So, Gen AI ethics came into the picture to answer the preceding challenges. It consists of principles and guidelines for designing, developing, testing, and deploying products and for how end users can use Gen AI safely in their day-to-day lives and get the maximum benefits without any fear.

All these sets of principles and guidelines are bundled into Gen AI ethics policy packs; this is needed to ensure that the organization, development team, staff, and stakeholders follow ethical AI guidelines to minimize risks and, at the same time, focus on improving the lives of all human beings without any risk threats.

All industries must consider the issues listed down while handling Gen AI ethics appropriately:

Ethical Considerations Factors

Listed down are the key ethical considerations for building any Gen AI product:

Justice: Justification from the perspective of Gen AI, without any discrimination such as group, community, race, gender, or economic status.

Confidentiality: Gen AI products must ensure the protection of individual data and information.

Safety: It must protect against unauthorized access and manipulation of sensitive data.

Accountability: Showing who is accountable for the behavior and resolution provided by the Gen AI product.

Clarity: Explain how the Gen AI product operates, including data considerations, the choice of algorithms, how the model was trained, and the justification of the product's decision.

Figure 1.17: *Gen AI – Ethical Considerations Factors*

Let us focus on a few scenarios of ethically protecting Gen AI products, starting with:

Copyright and Legal Coverage:

Concerning Gen AI, the input for the model would be in any format; here, most of the feed is in the form of documents, text, and data from various internet sources. Building a model with internal data will not have issues concerning copyright and legal questions since all the documents and text are verified in advance of sharing for model training. If public data is used, then the Gen AI tool creates text, summary, and

images that can create a problem, specifically while handling banking and financial-customer data, pharmaceutical-drug formula, legal forum-clients and case details, and hospital and patient treatment details. So, questions about reputation and security might arise from the respective organizations. Hence, Gen AI product companies must look to validate the output of the models and get advice from SMEs and legal teams to provide approvals on copyright challenges, IP, and privacy concerns.

Safer and Unharmful Content:

Gen AI systems can create content based on the input text, summary, image, and more, and based on the prompt instructions; the outcome from the systems can generate enormous throughput and much-improved content, but they can also be used for harm, but we could not do this intentionally or unintentionally. Still, in some cases, Gen AI-generated emails sent to external organizations or government entities from the company's system in offensive language or unnecessarily harmful messages to the leadership team, customers, and employees will create significant issues. While handling this requirement, we must ensure that the content meets the company's ethical outlook and realize the reputation of individuals and organizations.

Disclosure of Sensitive Information:

Gen AI tools can make an organization more democratic, supporting a system and making it more accessible. The composition of democratization and accessibility, disclosing sensitive information about patient and consumer interests, is unintentionally exposed to the outside world. The outcome of unintentional occurrences could breach the patient, and customer trustworthiness could carry legal implications. So when we build Gen AI products, we get clear guidelines from the governance body and aligned communication across all pillars and approvals—emphasizing the responsibilities of safeguarding sensitive information and protecting personal data.

Employee New Roles and Opportunities:

As we know, Gen AI can perform many more daily tasks than knowledge workers, including writing, content creation, build summarization, and data analysis. Automation has always replaced workers in an ongoing situation, right from the many industry revolutions in history. So, Gen AI is not exceptional in this case.

The ethical responses included preparing the workforce for Gen AI-specific jobs and roles such as Gen AI engineers, Prompt engineers, and specialized test engineers. This impacts organizational design and hierarchy. This system indeed prepares the organization for a new growth path.

Along with the aforementioned ethical actions, Gen AI is responsible for the provenance of data and privacy destruction, the explainability and interpretability of the Gen AI product, and the strengthening of existing biases in the system.

Conclusion

In this chapter, we introduced Gen AI and Transformer Architecture. We studied each component and its role in this architecture, along with a clear pictorial representation. We explored how this architecture primarily adopts neural network techniques and natural language processing methodologies, such as text generation, sentiment analysis, question-and-answer, text summarization, and language translation

We explored very detailed walkthroughs of both technologies along with factors influencing them, such as challenges with interpretability, cost and resources, data quality, data security, and privacy. We covered the comparisons and drawbacks of Gen AI and touched upon the commonly available Gen AI tools, their features, and their tenacities for several industries' needs: ChatGPT, Gemini, Copilot, GitHub Copilot, and Dall-E 2. We explored how industries benefit from Gen AI in a very detailed fashion in the areas of financial functions, demand forecasting, healthcare diagnosis and prediction, fraud detection, and music composition.

In the next chapter, we will discuss LLMs and their capabilities in detail, provide an overview of the essential components of LLMs and architecture, and explain how models work.

CHAPTER 2

Exploring LLMs and Its Capabilities

Introduction

In the previous chapter, we delved into Gen AI, its unique traits, and the advantages it brings. We then took a closer look at the evolution of Gen AI and examined its Transformer Architecture and its key elements: the ENCODER, DECODER, SELF-ATTENTION, INPUT, and OUTPUT. We also compared Gen AI to traditional AI and ML, highlighting the former's limitations. To illustrate the real-world impact of Gen AI, we provided classic industry examples and discussed the ethical implications of its use in high-level security.

Now, let us dive into the fascinating world of LLMs. We will explore what LLMs are, their networks, and the key components that make them tick. We will take a detailed look at the primary components of LLMs, such as embedding layers, transformer blocks, multi-head self-attention mechanisms, feedforward neural networks, positional encoding, and decoders. We will also unravel the mysteries of output layers, loss functions, and fine-tuning layers, as well as explain the concept of attention masks. Finally, we will discuss the importance of fine-tuning and its role in optimizing performance.

Structure

In this chapter, we will discuss the following topics:

Overview of Large Language Models (LLMs)

Exploring Available LLMs in the Market

Types of Large Language Models (LLMs)

Applications of Large Language Model (LLM)

Working Principles of Large Language Models

LLMs: Architecture and its Essential Components

Transformer Blocks

Embedding Layers

Positional Encoding

Encoder

Multi-Head-Self-Attention Mechanisms

Feedforward Neural Networks

Decoder

Fine-Tuning in LLM and its Importance

Fine-Tuning Process of LLMs and their Types

Fine-Tuning Benefits

Output Layers, Loss Functions, and Fine-Tuning Layers

Domain Adaptation Methodology

When to Opt for Fine-Tuning

Advantages of Large Language Models

Challenges in Large Language Models (LLMs)

Overview of Large Language Models (LLM)

Large language models are typical AI systems designed to process and analyze substantial amounts of data and then process required information as a desired output in response to user strategic prompting techniques.

Without any skepticism, these types of AI systems are trained on gigantic volumes of data sets using advanced ML algorithms to learn the patterns among the dataset irrespective of the data structure, which means any data format.

LLM is a neural network family with trillions of parameters trained on enormous amounts of data using supervised learning and multiple layers of an algorithm. These models can perform a wide variety of tasks, from simple text generation to drug design and legal advisory documents.

In recent years, LLMs have become increasingly important in various applications and products across different use cases.

Large Language Models (LLMs)

An LLM is an AI algorithm that applies typical neural network techniques with numerous parameters to process and understand human languages or text as input. It uses self-supervised learning techniques to generate text, translation, summary writing, creative image generation from texts, developing codes, and creating chat-bot applications (RAG and Q&A). Classic examples of such LLMs are GPT, Chat, BERT, LaMDA, and so on.

LLM is purely based on deep learning methodologies, and these models are highly efficient in capturing complex entity relationships in the given input. The input could be text, summary, structured, or unstructured, and the output would be text, summary, image, audio, and video. The text uses the syntactic and semantics of any language.

Exploring Available LLMs in the Market

Let us discuss common LLMs in the market, such as GPT-4o, T5, and BERT. These models showcase outstanding capabilities, including language understanding and generation, and open up new possibilities in various domains. Google, OpenAI, Meta, and others introduced all these LLMs for their specific tasks.

Bidirectional Encoder Representations from Transformers (BERT): Google introduced it based on the transformer architecture and follows the deep learning model technique. This means every output element is connected to another input element, and the weightings between the two elements are dynamically calculated based on their connections. Moreover, the framework was trained using text from Wikipedia data.

Robustly Optimized BERT Pretraining Approach (RoBERTa): This model is an improvised performance of the BERT transformer architecture, which Facebook AI Research develops.

Generative Pre-trained Transformer 4- (GPT-4o): This is the latest product from OpenAI that came into the market in 2023.

This model is much different from the earlier version; it can consider input in not only text but also images and also shows impressive performance. The maximum input length was increased to 32,768 tokens, which is an additional benefit of the model. This model is not open source and is accessible through API only. This means you have to pay to access the model for your needs.

ChatGPT 3.5 and 4: ChatGPT has two versions. 3.5 (Free Tier) is free to access, and 4.0 (20 USD per month) is paid. The earlier version supported a text-only model owned by OpenAI. It can take input from around 20 billion parameters. ChatGPT is a sibling model for InstructGPT. This means that it was designed specifically to receive prompts and provide very detailed responses in the form of paragraphs, bullet points, and classified formats for user benefit.

LLaMA 2: LLaMa is Meta's open-source product. It has various sizes, from 7B to 70B parameters, and offers two model variants, 1 and 2, respectively.

LLaMa Chat (or LLaMA 1) is a general-purpose model similar to the GPT family, capable of tasks such as Q&A, generating text, and language translation.

Code LLaMa (or LLaMa 2) is the best fit for challenging tasks, such as reasoning, coding, and proficiency tests. It utilizes the reinforcement learning methodology from human feedback and can handle 2.2 trillion tokens.

The models are open source, so they are free for research purposes and commercial purposes. Both models have performance capabilities for fine-tuning scripts.

FALCON: This was developed by Technology Innovation Institute. It has various ranges of parameter capabilities, such as 180B, 40B, 7.5B, and 1.3B. Based on its xxB, it has powerful performance, which varies in aspects of linguistic accuracy, versatility, and efficiency. On the other hand, it has the following issues: complete data dependency and concerns about ethical facets.

Language Model for Dialogue Applications This Google model can handle 173 billion parameters and is trained explicitly on dialogue. LaMDA is specifically designed to excel in dialogue-based interactions, allowing it to engage in more natural and contextually rich conversations with users.

The model's potential use cases vary, ranging from customer service-based chatbots to supporting personal assistants. Google's earlier chatbot version was called Meena, and the conversational service is much more powerful and renamed into BARD.

Pathways Language Model (PaLM): This model is also developed by Google. It can handle 540 billion parameters. Basically, this was designed to build and handle many different tasks and learn new ones more quickly than other models. It is capable of performing hundreds of language-related tasks and has achieved state-of-the-art performance. Its extraordinary feature is that it generates explanations for scenarios requiring multiple complex logical steps, such as explaining how to solve problems.

As discussed, the various LLMs are text generation models, such as GPT, that can generate consistent and grammatically precise text in different languages and formats.

LLMs aim to predict the next character, word, or sentence based on the previous ones in a running sequence. In this sense, language modeling encodes the rules and structures of a language in a way that a machine can understand.

LLMs can capture the structure of human language in terms of grammar, syntax, and semantics. These features help the model create content, translate languages, summarize, and perform text-editing tasks, including spelling correction.

Types of Large Language Models (LLMs)

We have discussed commonly available LLMs in the market, which can be broadly classified into three types:

Pre-training models

Fine-tuning models

Multimodal models

Pre-Training Models: LLMs are initially pre-trained on a massive amount of data from the internet, which is a vast amount of public data. During pre-training, the model can analyze the context and learn to predict the next word in a sentence with the help of surrounding words. This capability helps the model learn grammar, reasoning, and facts.

Commonly available models, such as the GPT family, are trained on vast amounts of data. During the training, they learn a wide range of languages, their patterns, and structures.

They are used as a starting point, and this will involve further training and fine-tuning for specific tasks along with a small amount of data.

Fine-Tuning After pre-training, the model undergoes fine-tuning on more specific tasks and a considerable size of the dataset. Fine-tuning is an additional process to enhance the model's performance. Basically, the set of parameters is updated on a new data set. It can be translation, summarization, sentiment analyses, and Q&A.

These models are often used in leading industrial-specific applications, where task-specific language models are essential to use cases.

Models such as BERT and RoBERTa are pre-trained on a large dataset and then fine-tuned on a small set of data for a specific task perspective. Fine-tuned models are highly effective and qualified for tasks such as text classification, sentiment analysis, and Q&A chat.

Multimodal The models are focused on a single type of data. However, this multimodal model is much more capable than

the other models. It can understand and process the details from multiple modalities and handle text, images, audio, and videos.

Moreover, this model utilizes the multimodal neural network, which effectively enhances overall performance. Combining visual and textual indicators has enhanced the understanding of data. It can handle uncertain situations better than unimodal models.

These models can understand the relationships between images and text, allowing them to generate text descriptions of images or even generate images from textual descriptions.

Models such as DALL-E 1, DALL-E 2, and CLIP combine text with other data types, such as images or video, to create more innovative and robust language models.

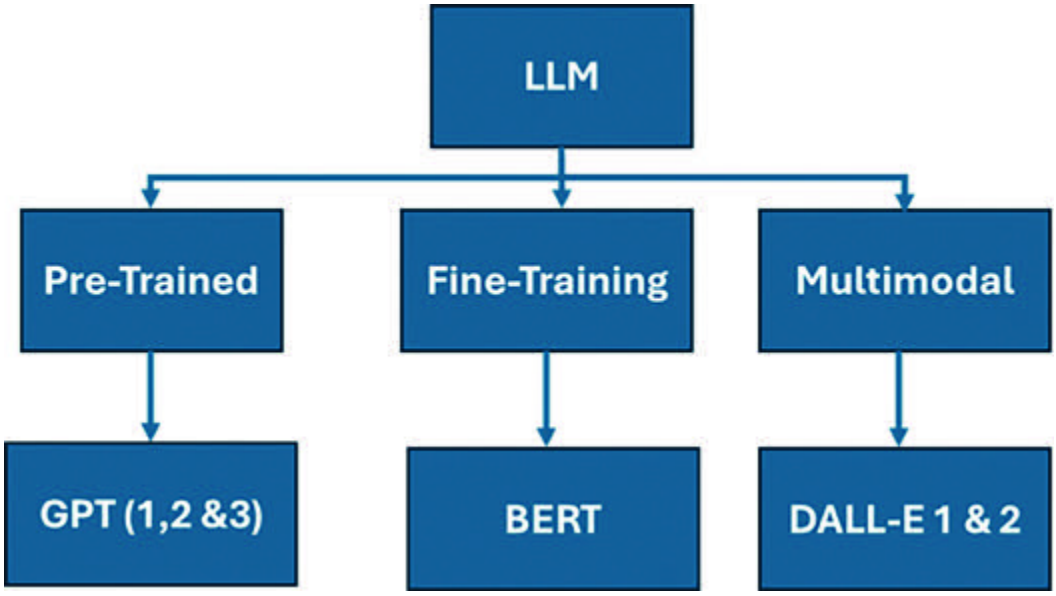

Figure 2.1: *Types of Large Language Models (LLMs)*

Applications of Large Language Model (LLM)

LLMs have a wide range of applications in specific domains; there are no limitations. Let us explore some notable applications.

Content LLMs can enable automated content generation for research papers, advertisements, and articles related to product marketing, technical and non-technical blog posts, new product descriptions and benefits, workflows and procedures, and social media marketing and captions.

Language LLMs can provide accurate and more confident translations across different languages based on the demand for Language X into Y. This ability allows them to support global-level languages, ease communication, and advance multilingual collaboration views. Specifically, government document translations for board support and legal aspects to build a smooth relationship between countries and ease border security and developments.

Chatbots and Virtual LLMs are potent conversational agents. This allows businesses to provide customer support, handle issues and inquiries about products, and assist end-users, ultimately reducing the need for human assistance and improving customer experience in a timely manner and on a highly flexible timeline. They can also close queries and issues quickly.

Text One of the great features of LLMs is that they can extract essential information from massive textbooks and generate concise summaries. This capability results in quick and fast news generation, aggregation, research analysis, and document analysis and processing.

Q&A: The best use case for LLMs is Q&A. They are really capable of understanding the context of the questions and getting precise answers, which is valuable for building Q&A systems and retrieval applications across many industries.

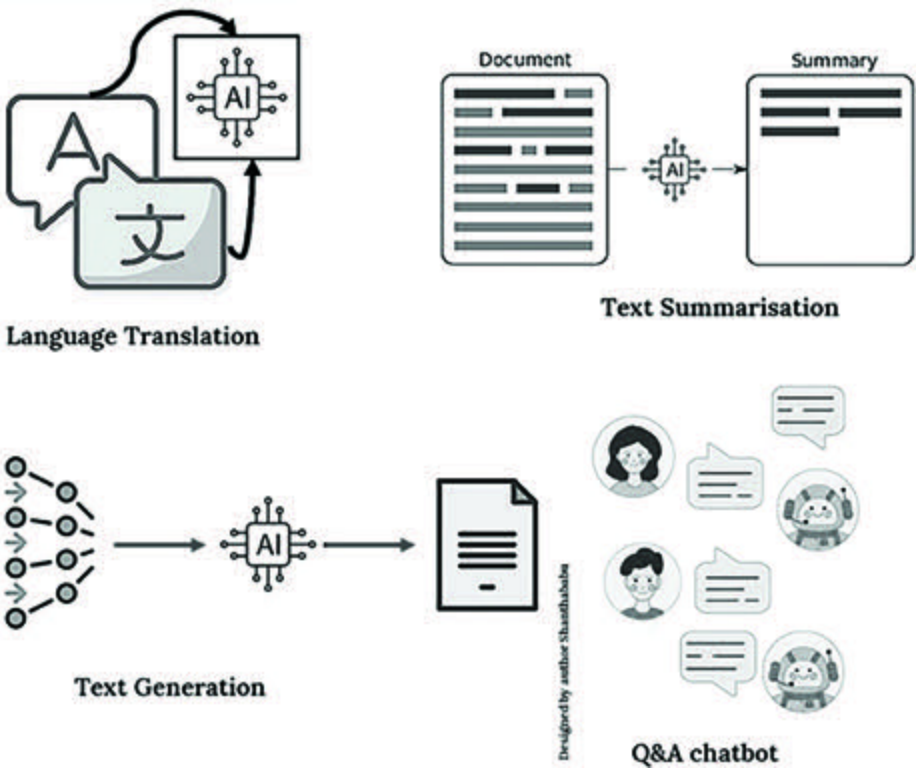

Figure 2.2: *Large Language Model (LLM): Applications*

Working Principles of Large Language Models

In simple terms, "*The LLMs work on neural network architectures and deep learning principles to process and understand tasks.*"

All these models are trained on a huge volume of data using self-supervised learning techniques, understanding the patterns and relationships among the training data during the training.

From an architectural point of view, Transformer-based LLM Model architecture consists of multiple layers, including **feed-forward layers, embedding layers, and attention**

Of course, they employ attention mechanisms, such as self and multi-attention.

The architecture of LLM is the same as the transformer architecture, which we have discussed in *Chapter 1, Introduction to Generative*

Here, let us discuss the architecture of LLM from a high-level perspective without replicating.

The following factors determine LLM's architecture:

Objectives of the model design

Computational resources

Types of Language processing tasks

Transformer-based LLM models typically follow a general architecture that includes the following components:

Input Layers

Input Embeddings

Positional Encoding

Encoder Layers

Self/Multi Head-Attention Mechanism

Feed-Forward Neural Network

Normalization Layer

Decoder Layers

Multi-Head Attention

Feed-Forward Neural Network

Normalization Layer

Linear and Softmax

Output Probabilities

Output Layers

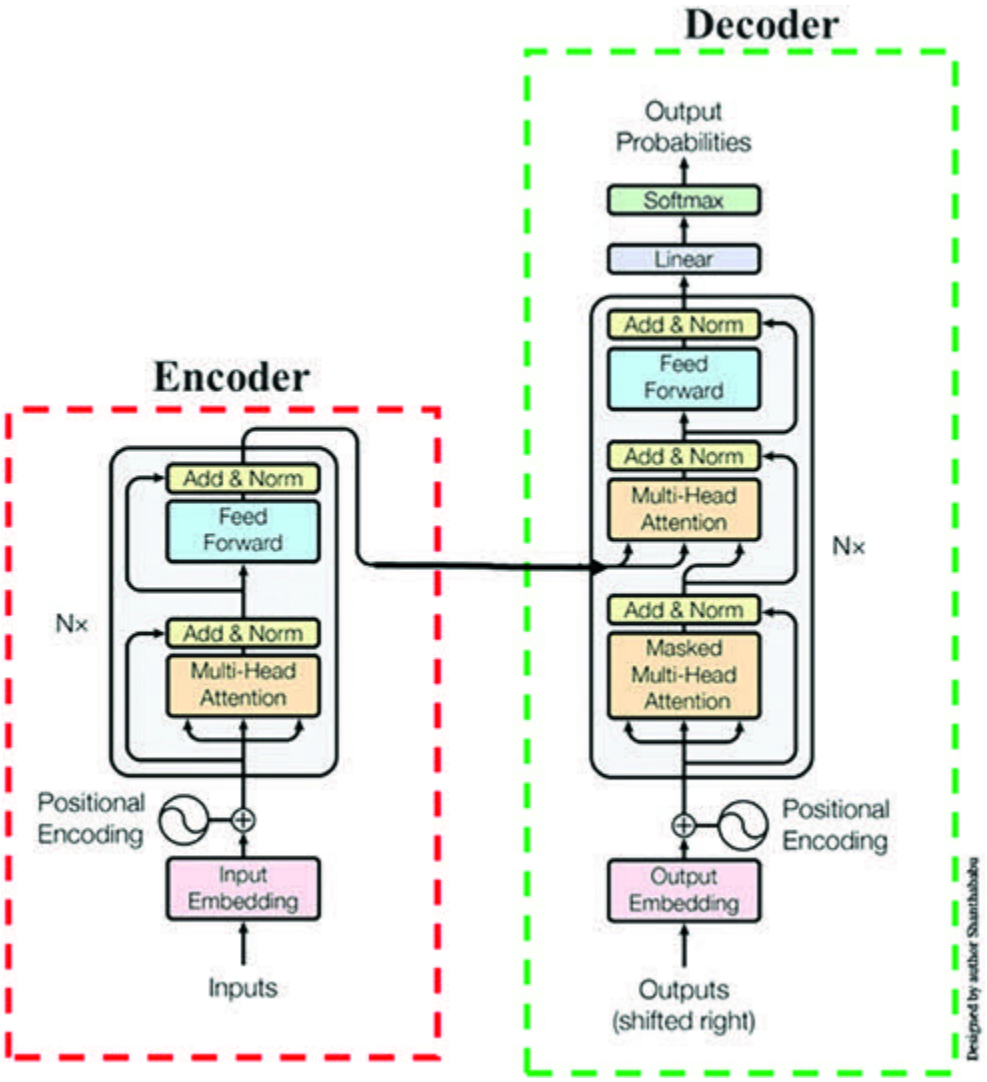

Figure 2.3: *Transformer-based LLM Architecture (Universal Picture)*

Input Layer: This is just an input for the system.

Output The output is the sequence generated by the transformer.

Input Embeddings: In this architecture, the input is not considered as word(s) directly; each word will be converted into numbers before it is fed into the system. The mechanism of this conversion process is to build an input sequence into the system, called

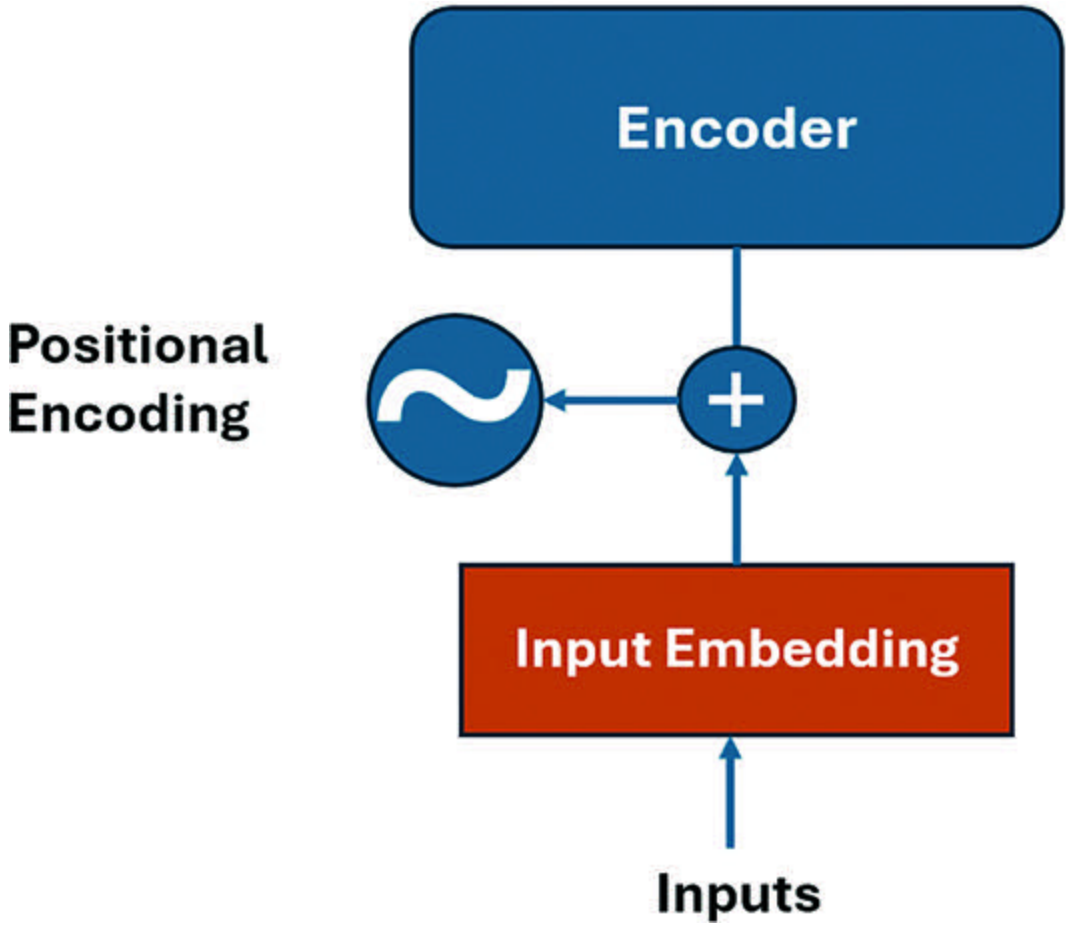

Figure 2.4: *Transformer-Input Embedding and Positional Encoding*

Positional Encoding: Positional encoding is the process of adding additional information to the input embeddings. The positions of the tokens are additional details because the transformers do not encode their order, unlike how LSTM functions. This enables the model to process the tokens while considering their sequential input order.

Based on a neural network technique, the encoder analyzes the input text and creates hidden states that protect the context and meaning of text data.

Encoder: The encoder layers comprise the transformer architecture's core. It consists of multiple layers and fundamental sub-components of each layer, such as:

Self-Attention/Multi-Head mechanism

Feed-forward neural network

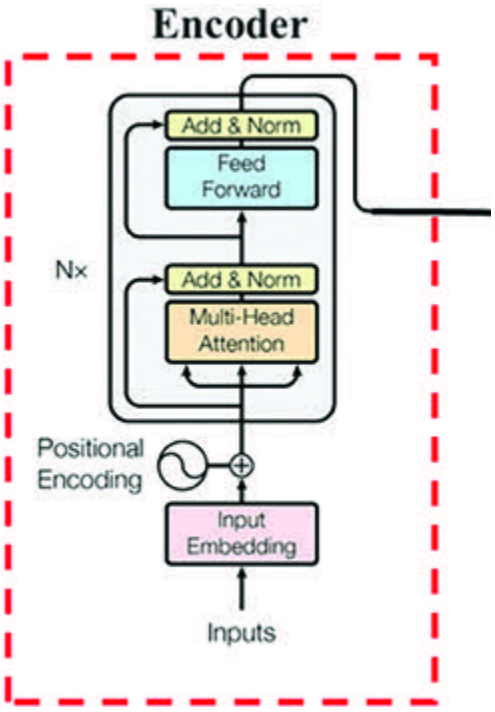

Figure 2.5: *Transformer-Encoder Layer*

Self-Attention Mechanism: This is a key mechanism used in transformer architecture. It allows the model to pay attention to different parts of input sequences when processing each element, count the significance of other tokens in the input sequence, and consider the dependencies and relationships between different tokens in a context-aware methodology, enabling each token to learn to understand all the surrounding tokens.

The masked MHA layer masks some tokens randomly to make this system learn situations.

For example, consider the following input sentence:

Hello, how are you doing?

If we have just reached the word the input of the encoder could be as follows:

Hello, how sequence>

Self-Attention will run dot-products between word vectors and determine the most substantial relationships in the given input sequence of the word and all the other words, as represented in *Figure* This attention mechanism will work out well and provide a more significant relationship between the given input sequence of words, resulting in better outcomes. If x=8, then there are N8 blocks and layers inside the encoder; the transformer model runs eight attention mechanisms in parallel to speed up the overall calculations. This process is named **"Multi-Head Attention (MHA)."**

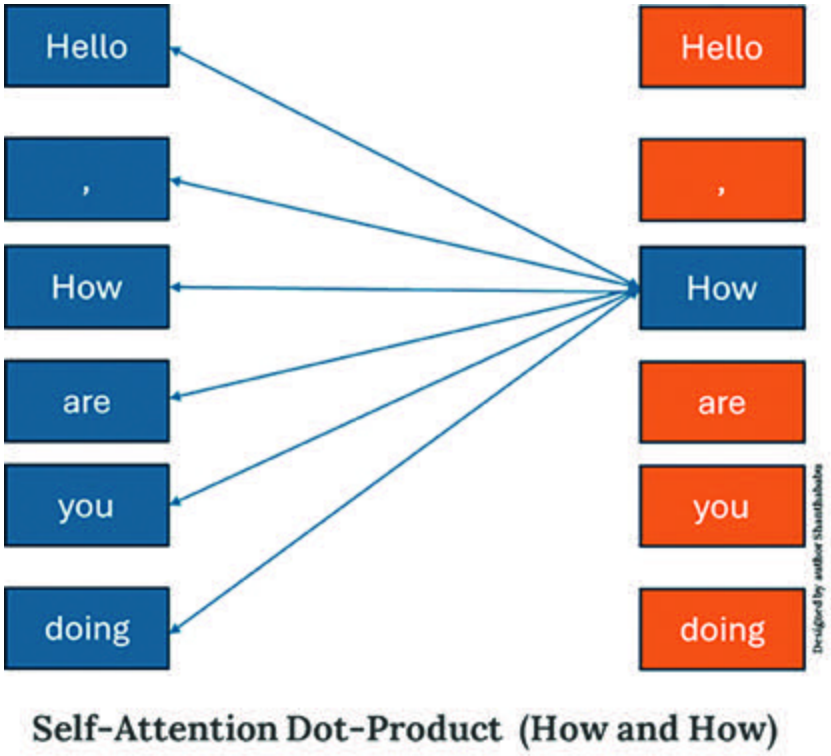

Figure 2.6: *Self-Attention - Including Itself ("How" and "How")*

Feed-Forward Neural Network: FNN fully connects layers with non-linear activation functions, enabling the model to capture complex relationships between tokens. This layer is applied to each token individually after the previous step, such as the self-attention step.

FNN are nodes and do not form loops. They are also known as multi-layer neural networks. As the name implies, all information is only passed forward. During data flow operations, the input nodes receive data, which travels through

hidden layers, and exit through the output nodes, similar to deep learning concepts.

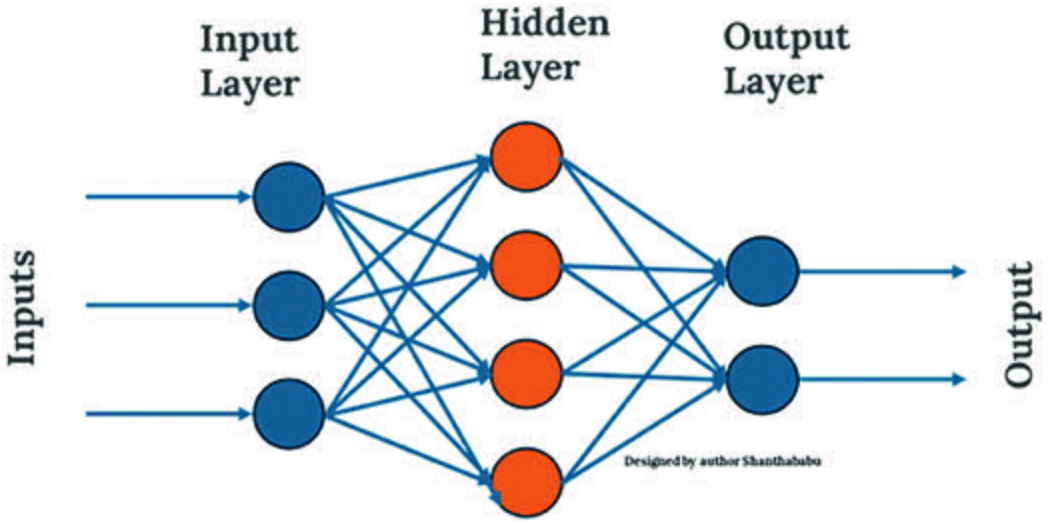

Figure 2.7: *Feed-Forward Neural Network*

Decoder This layer consists of the following components and the encoder's input:

Multi-Head Attention

Encoder-Decoder Attention

Linear Normalization and Softmax Layer

Output Layers

It enables autoregressive generation, where the model can generate sequential outputs by attending to the previously generated tokens.

Multi-Head Attention: In this component, self-attention is performed concurrently with uniquely learned attention weights. This enables the model to capture different types of relationships and concurrently attend to various parts of the input sequence.

Encoder-Decoder Attention: This encoder-decoder attention is the second sublayer that receives the multi-head mechanism input/queries from the decoder side from the previous decoder's output (the first sublayer's output) along with the keys (K) and values (V) from the encoder's output (the left side of the architecture diagram). This layer is also MHA. In this layer, the decoder attends to all the words in the input sequence concerning the MHSA mechanism, similar to layer one in the encoder.

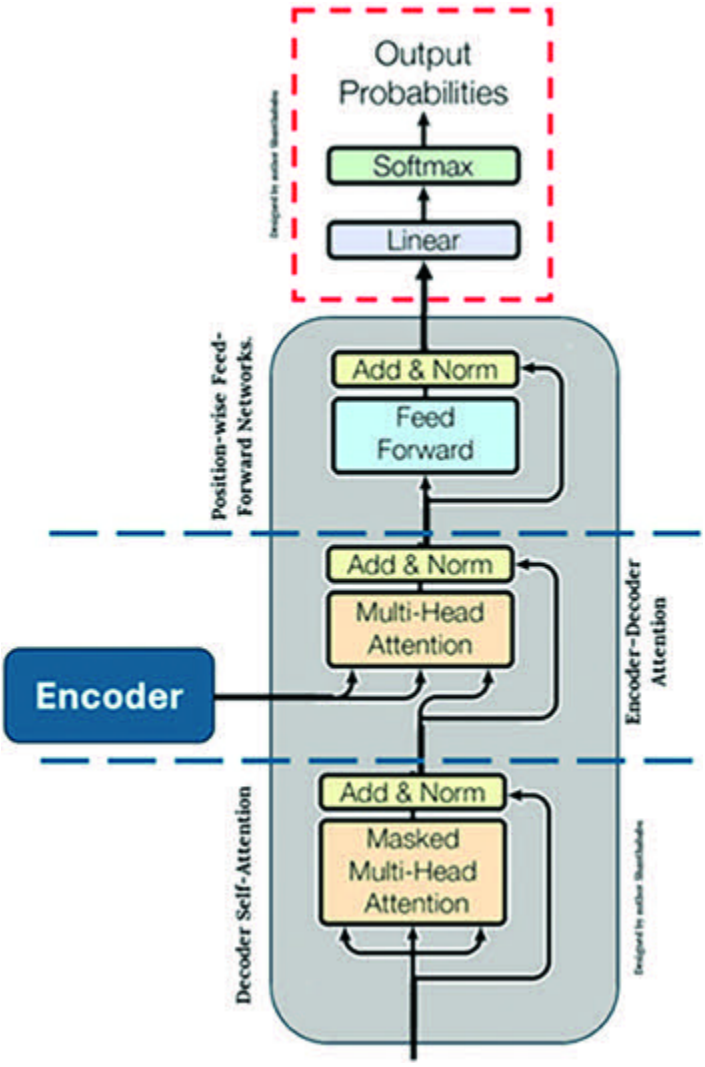

Figure 2.8: *Decoder Layer*

Layer Normalization: This layer came into action after each sub-component in the transformer architecture at the decoders block. It helps the learning process, steadily improves the model's capacity, and generalizes across different inputs.

SoftMax: SoftMax activation is generally used to generate a probability distribution over the next token feed.

Output Layers: This can vary depending on the specific task.

Factors Influencing LLM Architecture Functions: We have discussed the layers and elements of LLM architecture. Even though the components function with their own roles and responsibilities, the following factors would influence the architecture's overall functions.

Input interpretations

Model size

Parameter count

Self-Attention mechanisms

Objectives of the training

Efficiency of the computation

Encoding, decoding, and output generation

Types of Large Language Models (LLMs) include pre-training, fine-tuning, and multimodal models.

Large Language Model Content Generation, Language Translation, Chatbots and Virtual Assistants, Text Summarization, and Q&A.

The following factors determine the LLM's architecture, objective of the model design, computational resources, and type of language processing tasks.

Transformer-based LLM models typically follow a general architecture that includes input layers, encoder layers, decoder layers, and output layers.

Controlled Output of LLMs and their Significance

The quality of the LLM output is always discussed in the context of Uncontrolled output can lead to incorrect content, potentially misleading the end user and, in some situations, to threats, unsecured privacy concerns, and harmful results.

We can observe the LLM hallucination index, choose the appropriate temperature to achieve quality and reasonable output from the LLM, and correct it depending on many factors. Of course, considerable costs are incurred during the controlled output, and if a weaker LLM is used for costing purposes, it leads to serious consequences while dealing with critical decision-making. Therefore, we must control the occurrence of hallucinations and confirm that information generated using LLM is truthful, relevant, and consistent.

We can adopt the following techniques to control and enhance the LLM's output. In the following chapters, we will discuss all these techniques in detail.

Embeddings and text chunks

Vector and Index optimization

Building memorization and caching

Adapting parallel processing

Monitoring, logging, and profiling

Fine-Tuning Process of LLM and their Types

Fine-tuning is a process for the pre-trained model, which has already been trained and learned patterns and features on a large dataset and needs to be tuned further on a smaller dataset and domain-specific task.

Any model requires fine-tuning before entering the production environment. The fine-tuning process typically involves feeding the task-specific data to the model, where we have to adjust its parameters throughout the forward-back propagation. The major objective is to minimize the loss function, which measures the difference between the actual facts of the dataset and the model's predictions; ultimately, this process updates the model's parameters based on the requirements and makes it more aligned with the specialized task we are targeting.

The pre-trained model is a transfer learning method in which a pre-trained model is trained on a large dataset and adapted to work for a precise task. However, for fine-tuning, the dataset required for the entire process is much smaller compared to the pre-training model.

The fine-tuning process allows developers to customize pre-trained language models for dedicated tasks.

Types of Fine-Tuning

Various approaches to Fine-Tuning LLMs are available; let us explore them. We should remember that not all methods for fine-tuning LLMs are created equally. Each technique serves its unique purpose based on different applications and scopes. In some scenarios, you aim to adapt a model for a new task.

At a high level, there are two methods for fine-tuning:

Primary fine-tuning

Feature extraction, Feature-based, or Repurposing

Full fine-tuning I and II

Potential fine-tuning

Supervised fine-tuning

Instruction fine-tuning

Feature Extraction, Feature-based, or Repurposing: This is one of the primary approaches, as the name implies, that uses a pre-trained LLM, which is treated as a feature extractor component, and transforms the input text into a fixed-sized array list. Additionally, a separate classifier network has to be applied to predict the text's class in terms of probability. The classifier's weight continues changing during the training, making it resource-friendly. In this method, the model, having been trained on a sizable amount of data, has already learned the substantial features that can be repurposed for the specific task, so it is also known as **or** It offers a cost-effective way to fine-tune LLMs, but potentially less effective. Please note that only the final layers will be adjusted for fine-tuning purposes.

Full Fine-Tuning: This is another primary approach to fine-tuning LLMs. Here, the entire model has to be trained on task-specific data; this means all the model layers are adjusted during the training process. As per the process, we allow the entire model to learn from the task-specific dataset; this can lead to a more insightful adaptation to the specific task and potentially perform better than the feature extraction method. The downside is that it requires more computational resources and time consumption when compared to the feature extraction tuning method.

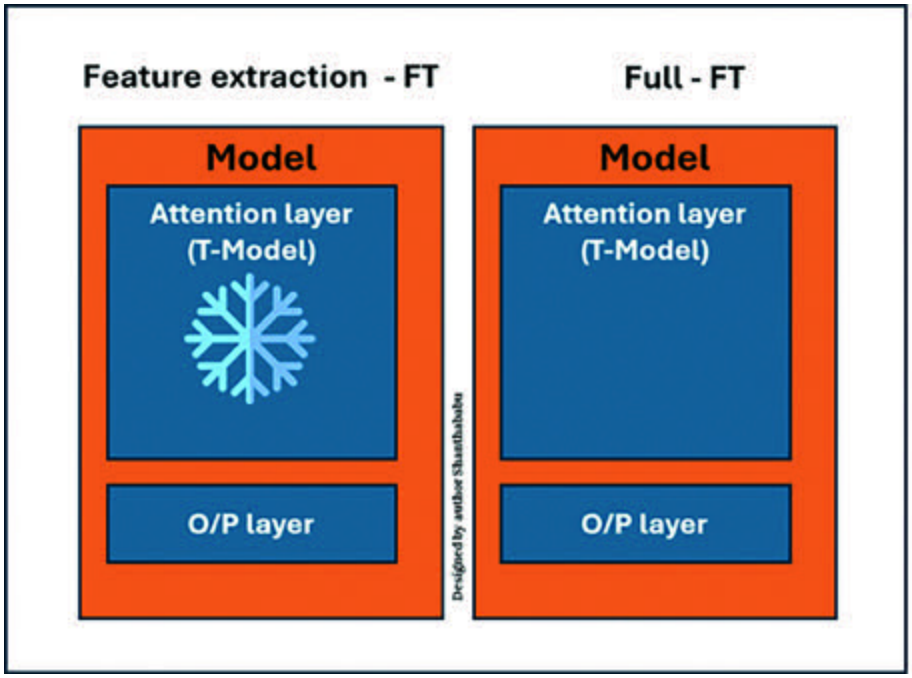

Figure 2.9: *Fine Tuning (Feature Extraction and Full FT)*

The full fine-tuning method is further classified into two types: Full fine-tuning I and Full fine-tuning II.

Full Fine-Tuning I: In this method, a dense layer is added during training to enhance the performance of pre-trained LLM. The weights of the newly added layers are adjusted, and the weights of the pre-trained LLM layer are frozen during the training. Performance-wise, it shows marginally better performance than the feature-based method.

Full Fine-Tuning II: In this method, the entire model is unfrozen during training, and all the model weights are kept updated, where new features overwrite old knowledge and lead to catastrophic forgetting in the model. This model delivers superior results and is more resource-intensive.

Supervised Fine-Tuning: As we know, the supervised model is well-trained with labeled data and will be trained further on a task-specific basis. Since the model is naturally supervised and the dataset has input-output pairs, the model can learn to map inputs to matching outputs using this tuning method.

Functionally, in this method, the pre-trained LLMs are trained for specific downstream tasks using labeled and validated input and output data combinations. During the training, the model's weights are adjusted based on the gradient factors derived from the task-specific loss, which represents the difference between the LLM's predictions and the base truth of labeled data.

The supervised fine-tuning method also learns task-specific patterns and nuances present in the labeled input and output data.

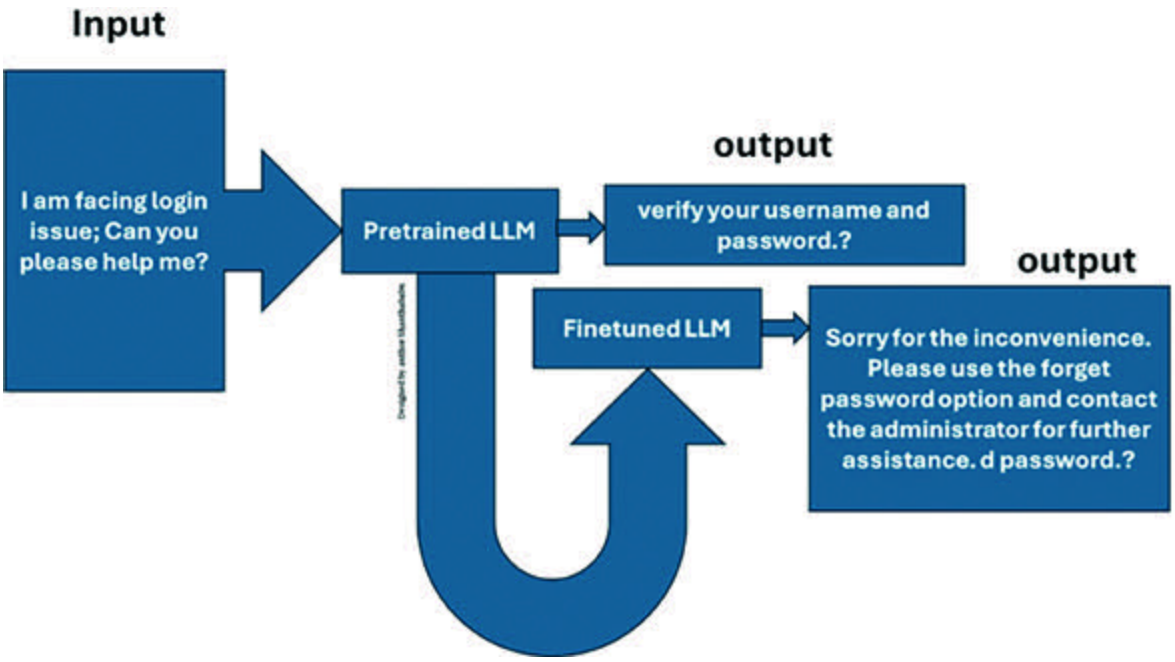

Figure 2.10: *Fine Tuning (Supervised Fine-Tuning)*

The supervised fine-tuning method is further classified into the following types based on the tuning approach.

The most common are:

Basic Hyperparameter Tuning: This is a straightforward approach that manually adjusts the model's key hyperparameters specific to an algorithm until reaching the desired performance, such as the batch size, epochs length, and learning rate.

Multi-Task Learning: This method fine-tunes the model for multiple related tasks concurrently. The goal is to influence cohesion and variation across the list of tasks considered during the training and is expected to improve the model's performance.

Few-Shot Learning: In this method, the model is exposed to a few classical tasks in terms of "shots" in the idea of models to understand the new task. This method helps the models to learn, enable, and adapt to a new task with considerable data. Basically, we will try to utilize the vast knowledge from the earlier training and use it for new tasks to reduce expenses.

Instruction Fine-Tuning: This method is something special; even though the huge amount of data trains the models, it cannot respond to specific modes of prompts. In such situations, the instruction fine-tuning method is applicable to change the behavior and response of the model, in which the input and output are augmented with given instructions in the prompt template with multiple options. This exercise will enable the models to generalize more quickly to new tasks using an instruction-tuned-based method.

Terminology Closer to LLM During Training

Multiple terminologies are emerging regarding LLMs, and the following terms are recommended. It is essential to know the theory behind them and understand the concepts.

Loss Function: This is the ratio of the predicted output and the actual output as inputs, computed as a numerical value representing the error. The objective is to minimize this error factor by adjusting the model's parameters through an optimization process. LLMs utilize cross-entropy as a loss function during training to measure the difference between the predicted and the actual probability distribution factors observed in the training.

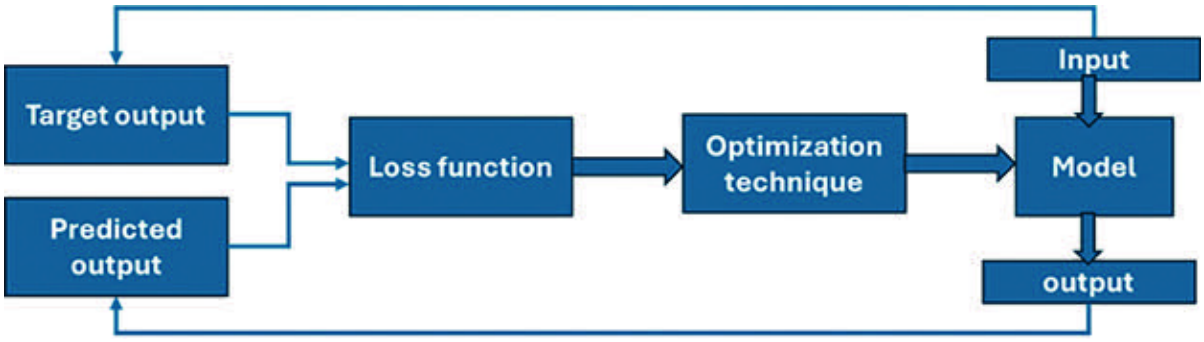

Figure 2.11: *Fine-Tuning (Loss Function)*

Domain Adaptation: In the AIML domain, many techniques exist to adapt and improve the performance of a model in which annotation-related details play a major role. This exercise is called domain adaptation, and it enhances this capability by using the knowledge learned from another related domain with adequate labeled

The domain adaptation technique helps by exposing the common hidden factors across the source and target domain data and reducing the marginal and conditional mismatches in the observation of the feature space between domains.

Source Domain: This data was analyzed and distributed in the frame on which the model is trained using labeled data.

Target Domain: The model has been pre-trained on a different domain to perform a similar task.

Let us say samples are collected from two different but similar domains.

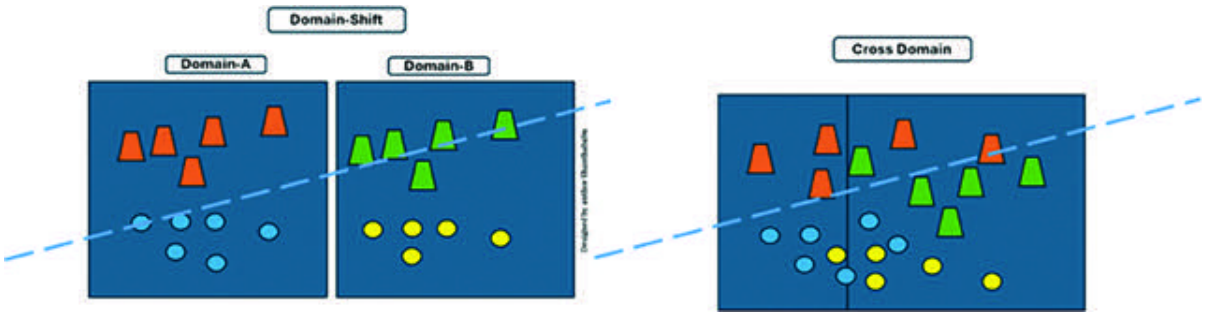

Figure 2.12: *Domain Adaption (Shift and Cross)*

LLMs need to adopt a specific domain to satisfy the needs of a particular task and improve the model's accuracy.

There are different techniques that help adapt an LLM for a domain-specific task:

Prompt Tuning: Feeding the appropriate instructions to the LLM and generating desired output text. We will discuss this in detail in the upcoming chapters.

Retrieval-Augmented Generation (RAG): In some scenarios, the feed content to the prompt might be too large, and the prompt could not accommodate it. In this case, the retriever step can be added as an additional step. In this step, the relevant content is retrieved to augment the context of the prompt to an LLM.

Supervised Fine-Tuning (SFT): Please refer to the earlier details in this chapter.

Note:

Loss Function: This is the ratio of the predicted output and the actual output as inputs, computed as a numerical value representing the error.

This exercise is called domain adaptation, and it enhances this capability by using knowledge learned from another related domain with adequate labeled data.

When to Opt for Fine-Tuning

We always have to justify our actions, so let us review some crucial and industry-specific scenarios when considering fine-tuning.

Data Availability and Limitations: We need to understand one thing: fine-tuning is particularly favorable when you have limited labeled data for your domain-specific task. We cannot expect to train the model from scratch in the pharmaceutical, healthcare, and legal domains. We cannot go for a larger dataset for various reasons, such as data privacy, security, and so on. So, instead of training a model from scratch, we could leverage the knowledge of pre-trained models and adapt it to our domain-specific tasks using a smaller dataset to train effectively.

Resources and Time: As we know, deep learning models require very high computational resources and time. In addition to a pre-trained model, additional fine-tuning provides a more efficient model. If we avoid the initial training and follow the fine-tuning process, we will find a solution faster. These

approaches will help the pharma industry, specifically during drug design.

Transfer Learning: The fine-tuning process is a critical component of transfer learning, especially whenever a pre-trained model's knowledge is transferred to a new and specific task for demand. To avoid training a model from scratch, we can pre-train and fine-tune it. These kinds of processes are generally adopted for a larger volume of image analysis in the healthcare and medical device industries to diagnose disease.

Data Security and Data is always a sensitive entity, irrespective of any domain. So, while working with sensitive data, we cannot leave or leak information and follow up on security and compliance concerns. Under these conditions, we might need to fine-tune our model to secure data with highly secure infrastructure and policies in place. This activity would ensure that the model never leaves the data and controlled environment and would follow up on governance. The following domains are highly recommended: banking, finance, legal, government policies, and insurance.

Fine-Tuning Benefits

Fine-tuning is an additional process to enhance the model's performance. After pre-training, the model must undergo fine-tuning on tasks that are more specific to the dataset, which is considerable in size. During this process, there are multiple benefits; let us focus on a few of them here.

Improved Performance: Since the model learns the new data for a specific task, it automatically increases the overall performance and output stability.

Domain Knowledge This process allows the model to adapt to the specifics of a domain and its related domain-specific vocabulary. It helps the model to generate text, summaries, and other outputs in domains such as law, finance, real estate, and medical.

Model Efficiency: By adapting a pre-trained process for a specific task along with a set of data, the model avoids training needs, which saves the computational cost very effectively.

Generalization: Due to this process, the model would generalize and exhibit better performance on relevant training data and bring

out patterns and insights present in the data.

Once the models are fine-tuned, they gain the following capabilities and build the following tools or products for domain-specific use. Indeed, this would enhance many more competencies than the earlier version.

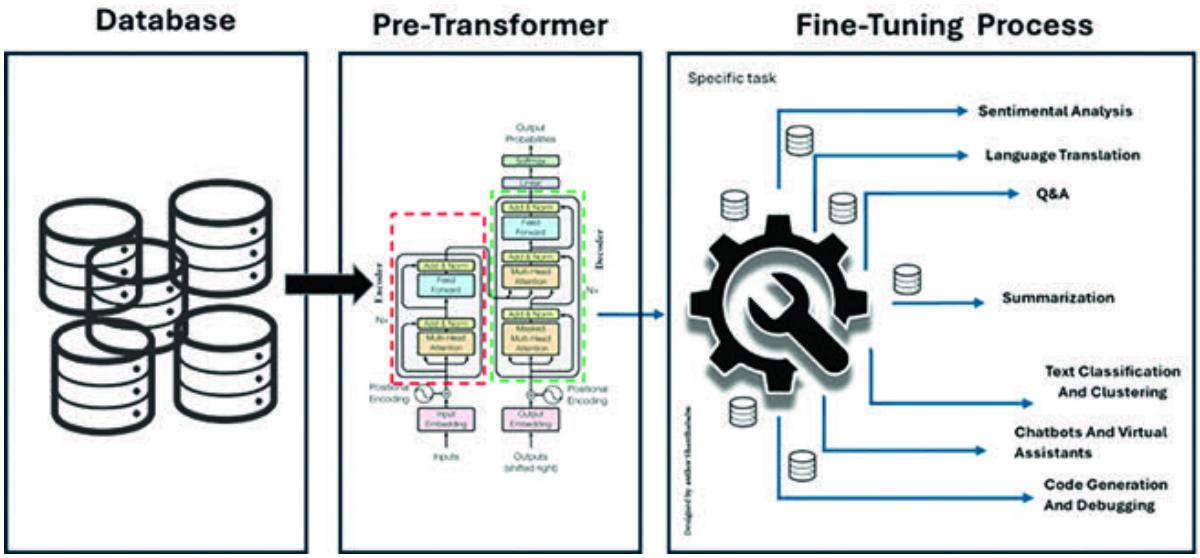

Figure 2.13: *Fine-Tuning the Models*

Advantages of Large Language Models

Overall, LLMs represent a significant advancement in NLP and Gen AI technology. They offer unparalleled performance, flexibility, and scalability across a wide range of applications and domains, along with their challenges. However, computational costs and ethical considerations are major drawbacks. Even though industries are building systems with high computational speeds and excellent ethical considerations, the advantages must be tapped into for industry-specific use cases to carefully specify how we can interact with and understand natural language.

Improved Performance: LLMs have demonstrated state-of-the-art performance on a wide range of tasks, including text generation, text classification, language translation, text summarization, Q&A, and more. Depending on their size and scale, they capture complex patterns and semantic relationships among the input text, leading to more accurate, contextually relevant, and required outputs.

Versatility: LLMs are versatile and can be fine-tuned for specific tasks with relatively minimal additional training data. This flexibility makes them suitable for a wide range of applications across different industries, including healthcare, finance, education, customer service, and more.

Zero-Shot and LLMs are capable of zero-shot and few-shot learning, which means that they can perform tasks without explicitly being trained on examples of that specific task. This capability enables them to generalize across different tasks and domains, which makes them adaptable to new and unseen scenarios with minimal guidance.

Language LLMs have a strong understanding of natural language semantics, syntax, and context, allowing them to generate coherent, contextually relevant responses back to the system. This enables natural and engaging user interactions in conversational systems, chatbots, and virtual assistants.

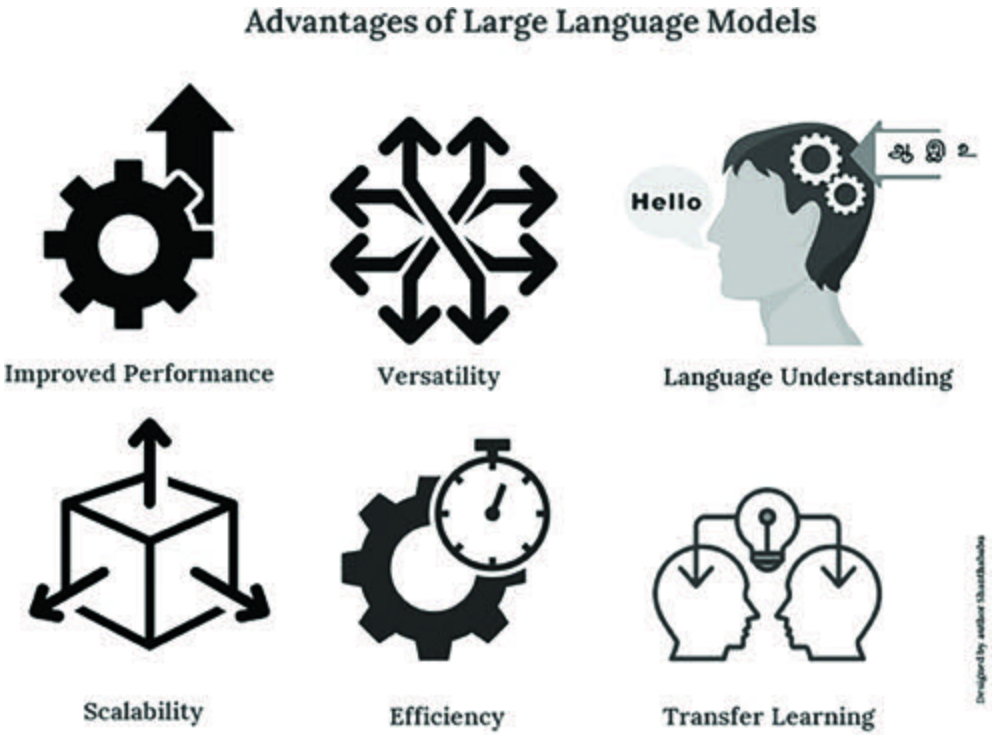

Figure 2.14: *Advantages of Large Language Models*

Scalability: LLMs can scale to accommodate vast amounts of data and parameters, which is why they need vast amounts of data. This enables them to capture various phonological nuances and handle diverse languages and domains. It also ensures they can continue improving performance as more data becomes available for training and tests.

Efficiency: Despite their size, LLMs can be deployed efficiently using modern hardware accelerators, such as Graphics Processing Units (GPUs) and Tensor Processing Units (TPUs),

in a cloud environment. This allows them to process large volumes of text data and generate responses in real-time or near real-time, making them practical for various real-world applications.

Transfer Learning: LLMs are pre-trained on massive datasets, which enables them to capture a vast amount of general language, patterns, and knowledge from various sources of text data. This pre-trained knowledge can be transferred to downstream tasks with fine-tuning, reducing the need for extensive task-specific training and helping the development of innovative Gen AI applications.

Challenges in Large Language Models (LLMs)

So far, we have discussed the exponential abilities of the LLMs in terms of current trends and future technology as part of AI-powered applications across industries. On the other hand, LLMs also consist of challenges and drawbacks.

What are those challenges? Yes, they are end-to-end, right from Gen AI tools' design, development, and deployment, which also come with significant challenges—and not stopping there, beyond that, such as privacy, security, and ethics issues.

Here are some of the key challenges associated with LLMs:

Resource As we know, LLMs require huge amounts of data for training and fine-tuning; significantly, they require massive computational resources with high power, such as GPUs or TPUs and, of course, a large-scale dataset. This sets a barrier to the entry of smaller organizations that want to get into this journey, and even researchers, who used to have limited access to such resources for their research purposes due to cost and sponsors.

Prejudice and LLMs can inadvertently amplify the biases in the given training data, leading to unexpectedly biased outputs with hallucinations. Addressing these issues is a complex challenge, and the volume of the data is not analyzed well, such as in machine learning or data science programs. Because these technologies require careful curation of analysis and training data and the development of bias detection and mitigation techniques earlier in the process, we can also follow those steps in Gen AI.

Ethical Use: This is a significant concern across all industries because LLMs have the potential to generate harmful or misleading content due to their nature, which includes high fabrication, hate speech, irrelevant content, deepfakes, and so on. Ensuring the ethical use of LLMs involves implementing security and privacy aspects, such as content restraint, fact-checking, impact assessments, and responsible deployment in practices.

Privacy Concerns: This is closer to ethical implications. LLMs may be trained on sensitive or private data that accidentally reveals confidential information through generated outputs publicly. Protecting user privacy and data security is essential

to the code of ethics, so appropriate governance must be in place to control all these elements.

Strength of LLMs are robust, but their output might be sensitive and threaten data privacy, even if the impact is small due to input or adversarial attacks. It is our response that by enhancing the robustness of LLMs and improving their interpretability, much ongoing research is in place, focusing on increasing trust and reliability in AI systems.

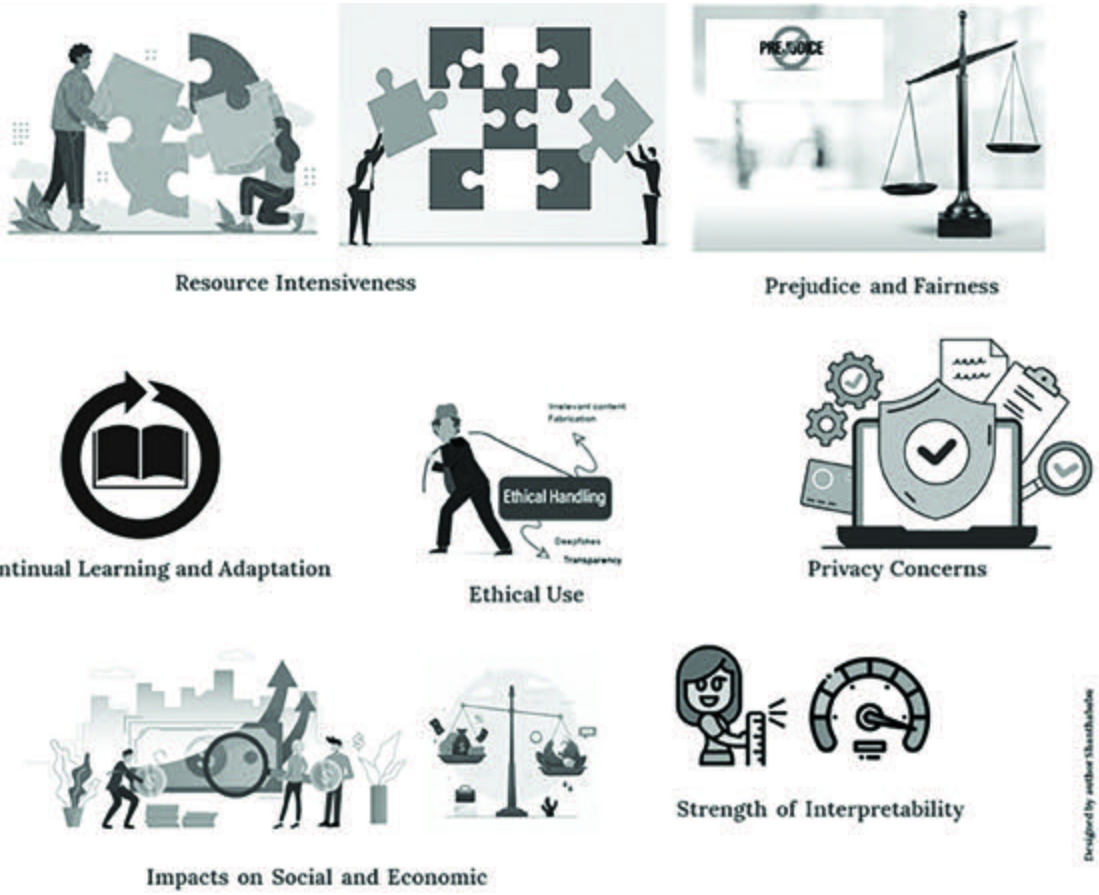

Figure 2.15: *Challenges in Large Language Models (LLMs)*

Continual Learning and Adaptation LLMs typically undergo daily pre-training on large datasets, the volume of which is increasing exponentially, either publicly available or private data (in-house), followed by fine-tuning on domain-specific data. Enabling LLMs to learn and adapt to new information without devastating and, at the same time, continual learning is also a challenging area with implications for real-time applications.

Impacts on Social and The adoption of LLMs could impact social and economic factors, including job shifts, transformations in the labor pool, and movements in power dynamics. Of course, serious market policies and stronger stakeholder engagement should be implemented to address all these impacts.

Collaboration will be required across all disciplines without questions or arguments, including data collection, AI research and development, aligning with ethics, building policy packages, and grooming domain-specific expertise to guide the entire portfolio.

By responding to and aggressively easing all these challenges, we can tackle LLMs' significant transformative potential, minimize potential risks, and maximize societal benefits.

Conclusion

In this chapter, we understood the Large Language Models (LLMs), detailing their definition and functionality. LLMs leverage neural network techniques with numerous parameters to process and understand human languages, enabling tasks such as text generation, translation, summarization, creative image generation from text, and coding development for various applications. We also explored various types of LLMs along with their specific applications in content generation, virtual assistants, text summarization, and language translation.

We delved into the architecture and components of LLMs, providing a comprehensive understanding of their working principles. Additionally, we examined the significance of controlled outputs and detailed the fine-tuning processes. Benefits such as enhanced performance, efficiency, and generalization were contrasted with challenges, including resource intensiveness, bias, ethical considerations, privacy, interpretability, continual learning, and socio-economic impacts.

In the next chapter, we will shift our focus to vector databases, exploring their mechanisms, distinguishing characteristics, and theoretical foundations. Topics will include types of similarity measures, operational mechanisms, and a review of prominent vector databases such as Pinecone, Faiss, Chroma, and Azure Search, providing an analogy-driven understanding of their role in modern data systems.

CHAPTER 3

Vector Database and Embedding Techniques

Introduction

In the previous chapter, we explored LLMs, their key components, and the available LLMs in the market—GPT, Chat, BERT, LaMDA, and others. We understood the types of Large Language Models (LLMs), their applications, working principles, and the general architecture. We explored the advantages and challenges of Large Language Models (LLMs) and related them with industry-specific examples. We also went through the varied aspects of fine-tuning.

This chapter will explore vector database mechanisms, including indexing, querying, and other features. It will also discuss the differences between vector databases and traditional database systems, as well as the characteristics of vector databases. It will present an analogy to understand vector databases, delve into its theories, similarity measures, and its working mechanism. The chapter will explore the different categories within the vector database, understand what serverless vector databases are and their scope, and learn about the multiple vector databases available in the market, including Pinecone,

Faiss, and Chroma. Detailed steps with code will be provided for working with the Pinecone and FAISS vector databases.

Structure

In this chapter, we will discuss the following topics:

Defining a Vector Database

Need for Vector Database

Characteristics of Vector Databases

Vector Database versus Traditional Database

Embeddings and Techniques

Working Mechanism of a Vector Database

Serverless Vector Databases

Vector Embedding and DB-Specific Use Cases

Semantic and Similarity Search

Types of Similarity Measures

Analogy to Understand Vector Databases

Classification of Vector Databases

Popular Vector Databases on the Market

Working with Pinecone and FAISS Vector Database

Defining a Vector Database

A vector database is one of the new generations of database management systems. It can store data as fixed-length lists of numbers or high-dimensional vectors and implement approximate nearest-neighbor algorithms. Because of this feature, it can search the database with a query and retrieve the closest matching data from the database records stored in the vector database.

It is in the form of high-dimensional mathematical representations, such as features or attributes. In the current scenario, this is explicitly designed to accelerate artificial intelligence application development and simplify the operationalization of AI-powered applications and their workload requirements. It helps remembering inputs using semantic and similarity searches, perform high-computational and powered searches, and make recommendations for various use cases concerning Gen AI in trends across all industries. It can be used predominantly for similarity and multi-modal searches, large language models (LLMs), and so on.

The vectors are generated by applying a selected form of transformation or embedding function to raw data, such as structured and unstructured data types, including text, images, audio, and video. The LLM framework provides various methods to build the embedding based on the requirements. All these embedding functions can be based on multiple methods, in which word embeddings play a vital role in Gen AI-based use cases.

Figure 3.1: *Vector Database (Dimensional Vector Space)*

Need for a Vector Database

You might ask why a new database form suddenly appeared in the database world. This section will help you understand the importance of vector databases and why they are unavoidable when dealing with a massive volume of data in multidimensional space.

Traditional databases are typically designed for well-structured data with high integrity between columns and rows in terms of relationships. Of course, we need this setup to build concrete database models to handle enterprise-level applications.

The need for vector databases emerged when we started storing and dealing with semi-structured and unstructured data types, such as images, audio, and video for specific use cases at industry-level problem statements and high utilization of social media platforms.

Handling data in terms of vector data representation is extremely resource-intensive. However, vector data representation can be highly scalable, searchable, extendable, and dimensional,

making it well-suited for Gen AI-specific products and applications.

Characteristics of Vector Databases

It is capable and compact enough to store data in all formats (text, images, video, and so on).

It adapts embedding techniques and has highly indexed features.

It offers a complete solution for complex and vast volumes of datasets.

It organizes the dataset through high-dimensional vectors with high retrieval speed.

Each dimension in the vector space corresponds to a specific feature or attribute of the data it represents in the storage.

It enables advanced analytics and insights by leveraging vector-based representations much faster than traditional databases.

It offers flexible schema designs to accommodate varying vector dimensions and different types, such as sparse and dense vectors.

It empowers applications such as content recommendation, anomaly detection, semantic search, and personalized user experiences.

It is optimized for real-time or near-real-time query processing. It can quickly retrieve similar vectors and perform similarity searches on large datasets.

It is suitable for applications requiring low-latency responses due to its features.

Vector Database versus Traditional Database

As we know how the traditional databases work, let us explore how a vector database differs from traditional database systems. It is primarily related to its design, capabilities, assessment methodology, storage format, and so on.

Foundational Data Model:

Traditional databases are table-based structures with defined rows and columns and well-organized database systems. A traditional database model enforces a predefined schema and ensures data integrity between tables by primary and foreign key relationships and concrete constraints.

Conversely, vector databases depart from the relational model and incorporate multidimensional vectors to store and manage the data. This feature system allows for a more flexible description of complex relationships between data, encompassing all types of data such as text, summaries, images, audio, and video. It supports unstructured or semi-structured vector data, representing each entity as a numerical vector (for example, embeddings and feature vectors).

Storage and Retrieval Efficiency:

As we are familiar with traditional database systems and their tabular structure, we may need help handling data efficiently while building longer query and retrieval times, increased storage, and computational requirements.

The storage and retrieval efficiency in the vector database is improved by efficiently storing multidimensional arrays, while features such as similarity search enhance the storage and retrieval efficiency.

Traditional databases are typically designed to scale vertically by adding more resources to a single server. However, vector databases are often built with horizontal scalability, allowing them to distribute data across multiple nodes in a cluster for parallel processing and improved performance, even for complex queries.

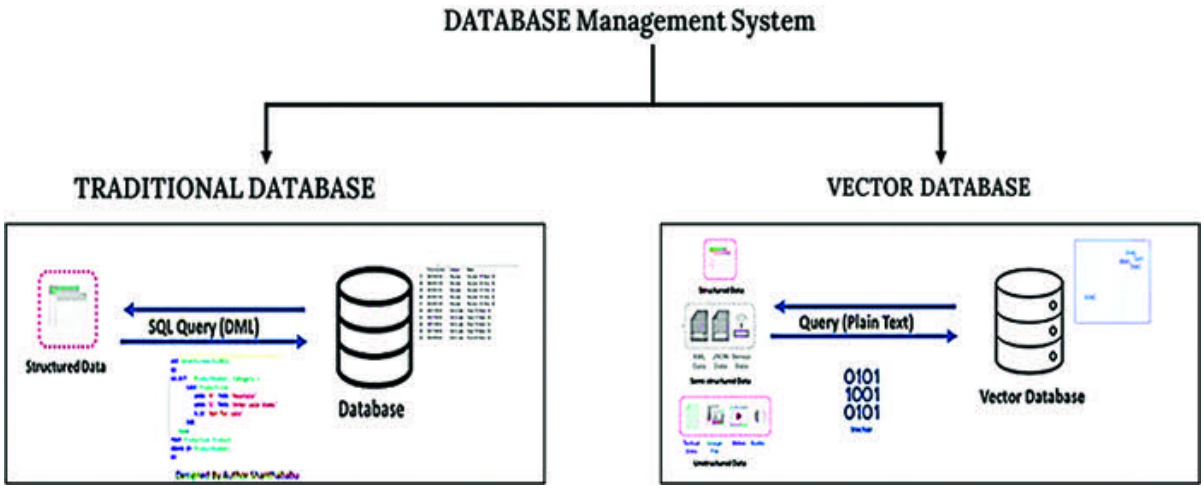

Figure 3.2: *Traditional Database versus Vector Database*

Query Processing:

In both systems, query processing and its outcome play a significant role. As we have experienced, traditional databases are highly constructed using relational database management concepts.

Traditional databases are notable for their ability to ensure data consistency and integrity. However, for large-scale, high-dimensional data, there may be better choices, as complex queries are required to find the most similar records in a database with millions of records and high computational speed.

Ultimately, relational queries involve very composite joins, a combination of multiple aggregation functions, and critical

business-specific filters. Monitoring query performance and optimization is always challenging for developers.

Vector databases store data in a high-dimensional format. They focus on vector-oriented operations such as calculating distances, finding nearest neighbors, and performing vector-based computations such as dot product and cosine similarity methodology.

Here, the query involves simple text in English; there is no need to worry about building queries using joins, aggregation functions, and data filtering. All of these are taken care of by the vector database itself.

Moreover, vector databases can also process unstructured or semi-structured data such as images, audio, video, and text (JSON, XML), which relational databases cannot.

Comparison of Vector and Traditional Databases

We have discussed how vector databases differ from traditional databases in terms of operational aspects. Let us discuss them here in terms of functional elements.

Database/Collection: In traditional databases, the database is a collection of tables, but in vector databases, it is a collection.

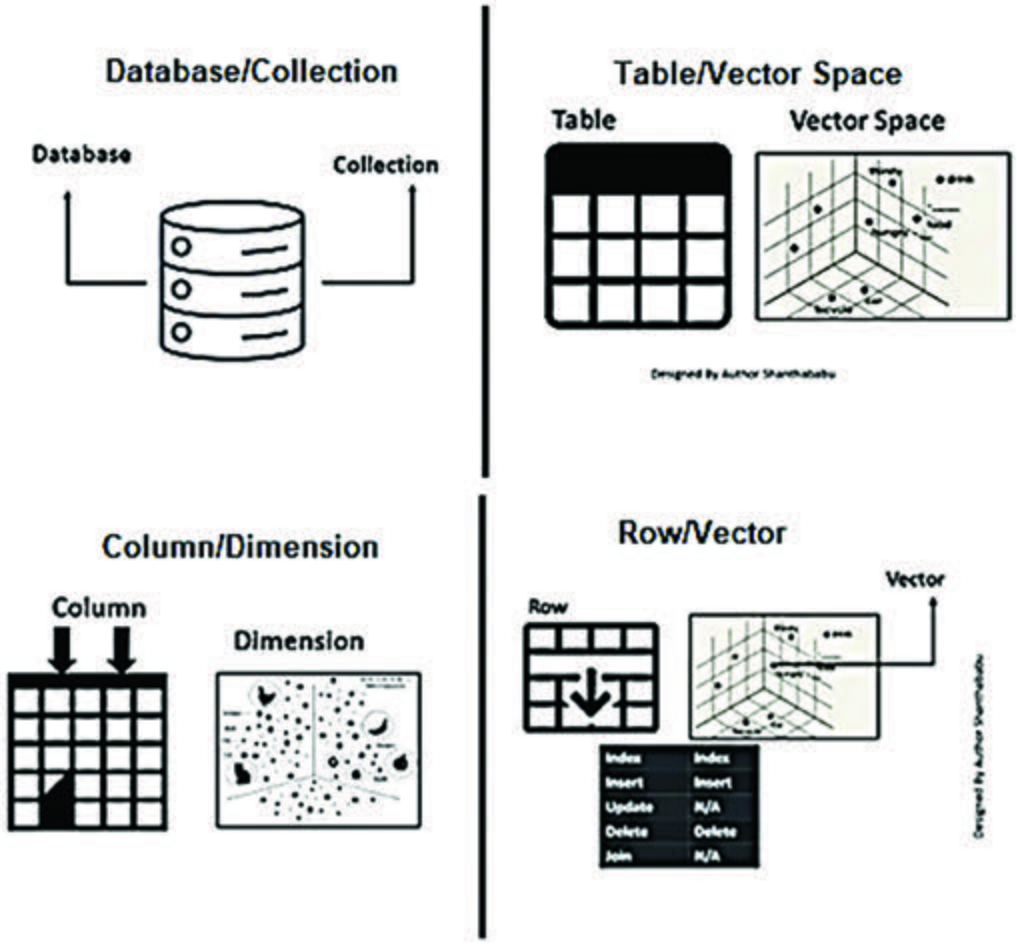

Figure 3.3: *Traditional Database versus Vector Database (Functional)*

Table/Vector Space: In traditional databases, tables are collections of entities with rigid constraints, such as primary and foreign keys and other limitations. However, in vector databases, the space is a multidimensional vector space.

Column/Dimension: In traditional databases, columns, attributes, features, and well-defined type and length are related to other tables as RDBMS concepts. But in the vector database, it is called a dimension (n-dimensions).

Row/Vectors: In traditional databases, the row is the information stored in the table, but in the vector database, information is stored in the form of an embedding vector, which is mathematically represented in the vector space as numerical values, irrespective of the data stored in the vector database.

Embeddings and Techniques

Let us understand how vector databases store data and the logic behind embedding techniques. In vector databases, data is stored using vector embeddings, demonstrating objects in a multidimensional space.

Each object is assigned to a vector point that captures various characteristics of that object and converts them into numerical vector values. It is done in a compressed format that captures their characteristics during the embedding process. When searching (querying) the data, it uses these embeddings to efficiently retrieve very similar and appropriate objects based on their positions in the multidimensional vector space, according to the requirements of the query.

The simple mechanism behind this is that similar objects are assigned to some vector space and have vectors closer to each other in the vector space. In contrast, dissimilar objects have vectors that are apart. This results in clusters in the vector space, which can be represented visually or mathematically to identify relationships or patterns among the data. This is represented in *Figure*

This capability enables vector databases to perform fast retrieval and high-similarity search operations, with excellent features such as CRUD operations, metadata management, horizontal scaling, and serverless operations.

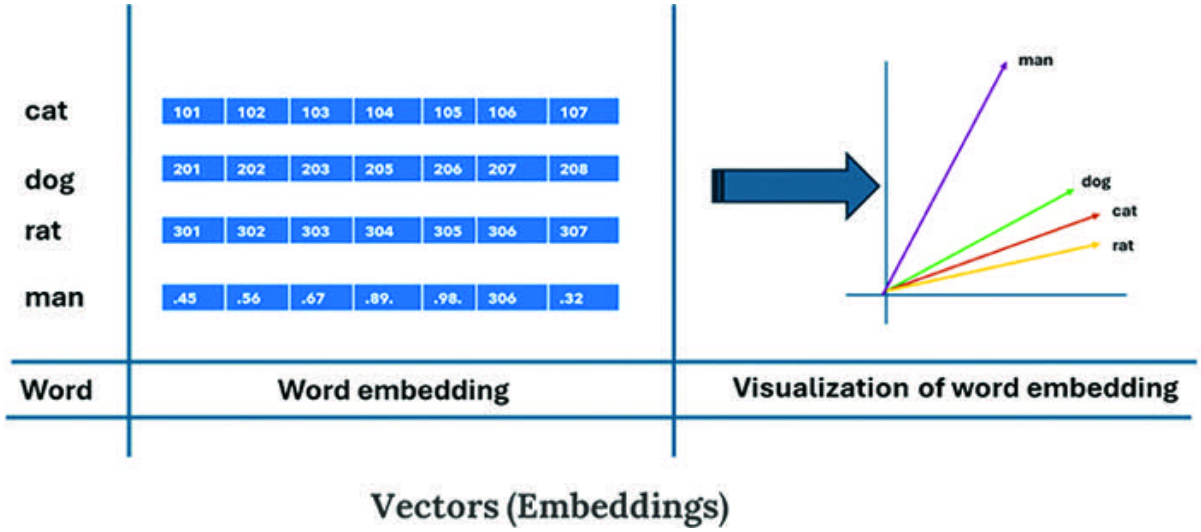

Figure 3.4: *Vector Embedding and Visualization*

Gen AI-based applications are demanding and rely on vector embedding techniques because they carry semantic information in the features. This allows LLMs and frameworks to gain an understanding of the data and maintain long-term memory that they can adopt when implementing complex tasks on massive volumes of data stored in the vector database.

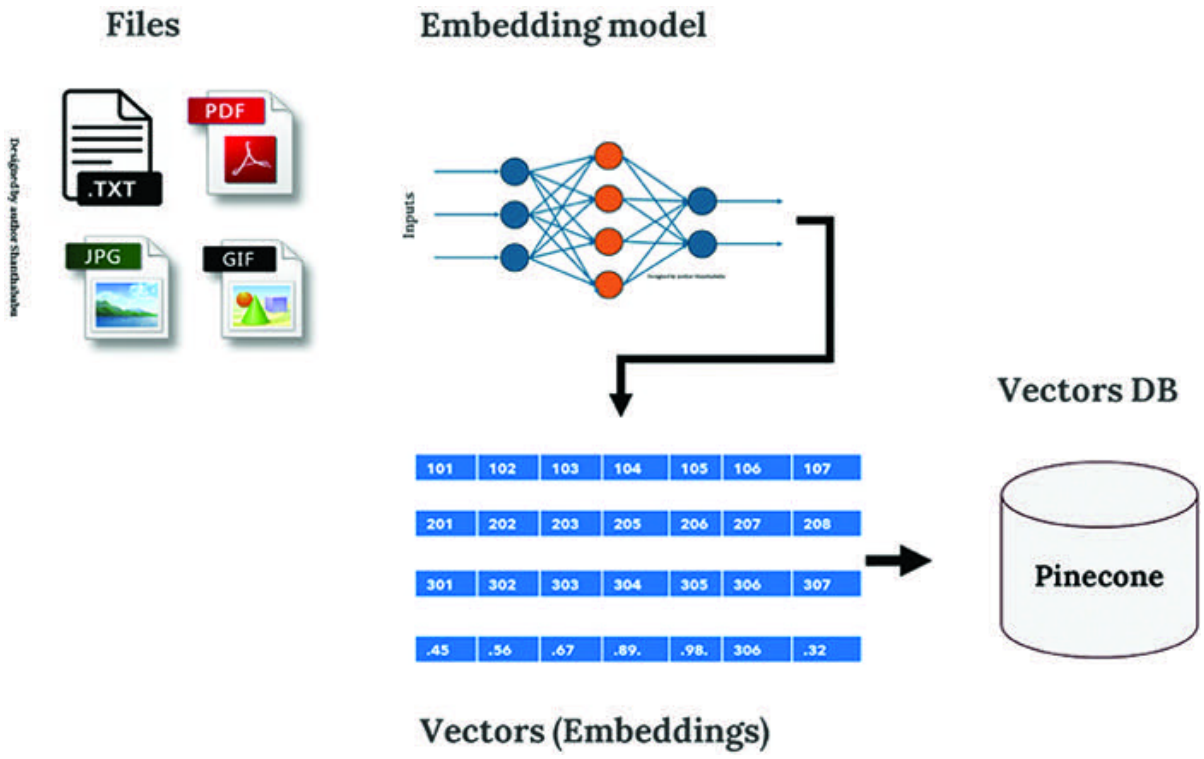

Figure 3.5: *Vector Embedding and Storage in a Database*

Working Mechanism of a Vector Database

The vector database helps the entire Gen AI application enhance performance at different levels. The working mechanism is simple and consists of the following components:

Input Query

Embedded Modeling Building

Similarity Search Operations

Output Generation and Response

Chain of Queries

Input Query: This occurs when the user inputs a question or query into the Gen AI application, which could be a chatbot or agent.

Embedded Modeling Building: As mentioned earlier, the inputs are converted into a compact numerical representation and stored in

a database with high-dimensional space. This process is called **vector**

Similarity Search Operations: The input vector embedding is compared with other embeddings stored in the database, and similarity search measures are executed to identify closely related embeddings based on the content available in the database.

Output Generation and Response: After the appropriate search, closely matching data is retrieved from the database, and a response in the form of embeddings is generated back to our query/questions as output for the application. This response contains relevant information linked to the recognized embeddings in the given vector space.

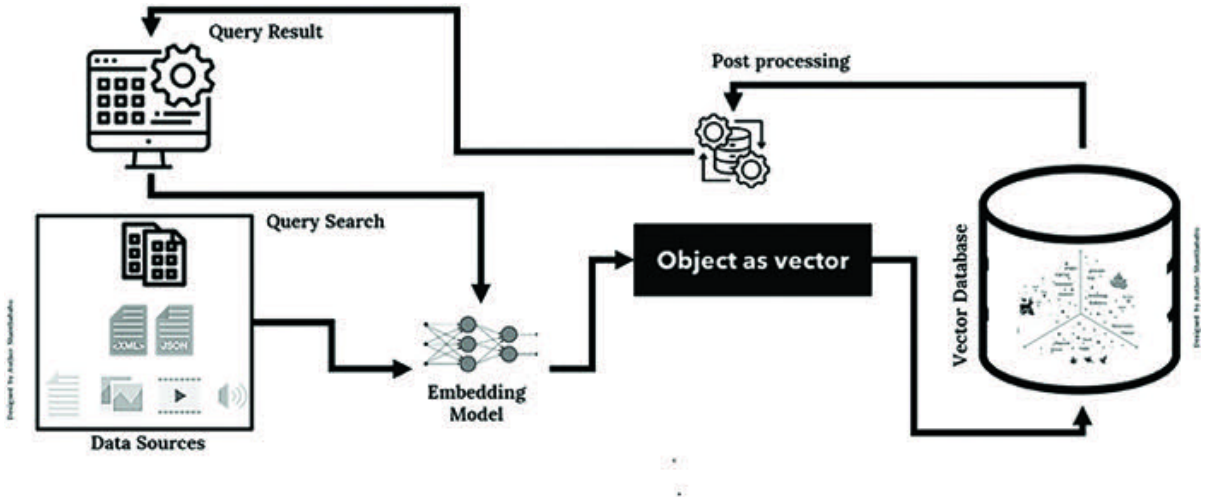

Figure 3.6: *Vector Database (Working Mechanism)*

Chain of Queries: The output generates and responds to the application, keeping the answer in cache memory for the following questions and forming the sequential chain operations.

In simple terms, the working mechanism consists of the following steps:

Vector databases first convert data into embedding vectors.

Embedding vectors store them in vector databases.

Configuration of indexing based on the required dimensions.

Input text by interacting with the embedding vector.

Internally, it searches for the nearest neighbor in the vector database.

Use the index to retrieve the relevant results.

The retrieved data would be fine-tuned, formatted, and shared for action in the next feed.

Serverless Vector Databases

We could call the vector database the "Next Generation" of databases, with more advanced architectures to handle fast retrieval, semantic and similarity search, CRUD operations, filtering metadata for specific scenarios, and horizontal scaling to enhance performance. Typically, they fall into serverless categories. Moreover, this comparatively cost-effective storage and high computing power support highly complex AI-based applications dealing with complex data.

Traditional database architecture, where we manage and monitor servers and infrastructure, is highly challenging and expensive. However, vector databases are becoming serverless, away from the underlying infrastructure and maintenance. This allows us to focus on data and AI applications without the need for database server management.

Serverless vector databases combine significant benefits, such as the capabilities of serverless computing and vector database features. They differ from traditional databases in that they eliminate the need for server management, allow resources to

scale up and down, and handle infrastructure on a regular basis.

In simple terms, Serverless Vector Databases have the following benefits:

High Scalability: It can automatically scale the resources up and down based on the demand.

It supports and allows AI applications to handle dynamic workloads without manual interference.

No worries about managing servers or infrastructure; the team can focus on other critical tasks.

Cost-Effectiveness: It eliminates infrastructure investment and maintenance costs.

Vector Embedding and DB-Specific Use Cases

The vector database is a high-potential source and instrumental for Gen AI implementation; this vector embedding feature helps us perform multiple tasks irrespective of domains and industries.

Let us explore a few of them.

Information Retrieval Generation (RAG): Vector embedding and database contributions to information retrieval are significant tasks in RAG-based applications. These powerful techniques influence search engine operations phenomenally, exponentially extending demand and fulfilling requirements.

Similarity Search Operations: Since the vector database is a high-dimensional indexed system, the indexing and indexer are well-organized within the vector database architecture itself. This feature helps us find the similarities between different occurrences in the vector data more quickly and efficiently than in a traditional system.

Classification and Clustering: We know that classification and clustering use cases play a role. These are significant problem statements for Machine Learning and Data Science-specific tasks, and we also have dedicated algorithms. Additionally, vector embedding techniques help classify and cluster use cases precisely and with well-enhanced approaches due to index, similarity, and retrieval features. This allows us to train and build machine learning algorithm models to group and classify datasets and provide solutions.

Recommendation As we know, recommendation systems are a classic use case in many NLP and Advanced NLP solutions across various industries, specifically in retail, sales, and marketing. To accomplish all these requirements, vector databases and their architectures already have many features, such as index, indexer, and similarity, which lead to exponentially accurate recommendation systems that relate products, items, descriptions, and more.

Sentiment Analysis: This type of analysis is prevalent and in high demand in retail and marketing industries to assess a product or item among critical customers. Based on the response, the manufacturer would enhance the product quality and features in the next release, or the product might be recalled to satisfy the customer. This embedding model helps us categorize and derive sentiment solutions efficiently.

Many applications require vector databases and embedding techniques to enhance Gen AI product performance; we will discuss those in the coming chapters with detailed steps.

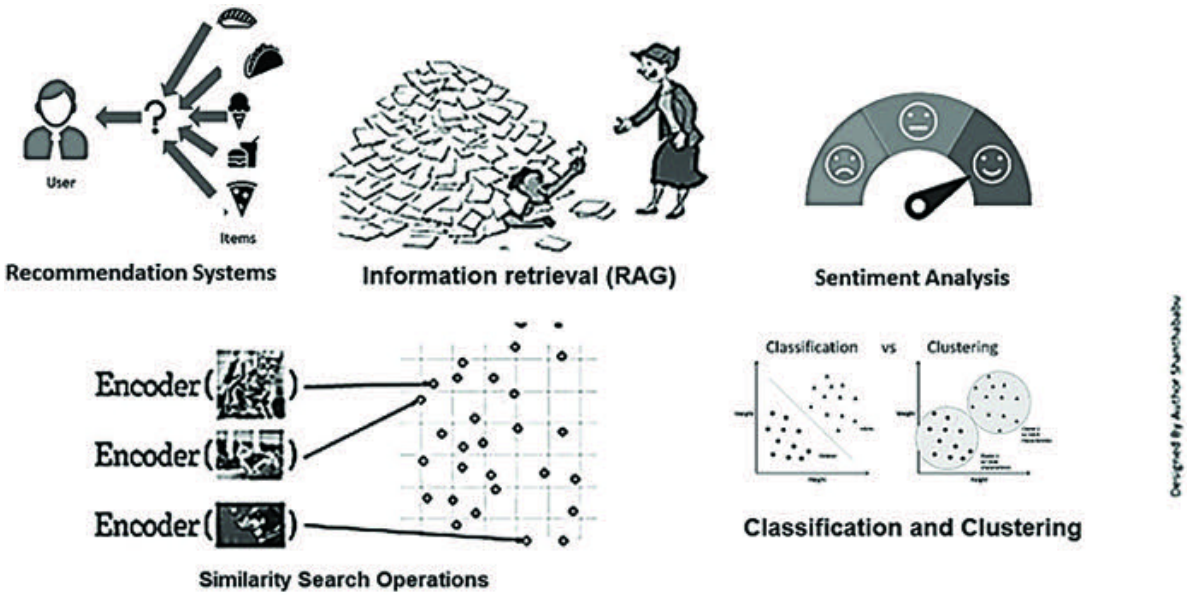

Figure 3.7: *Vector Embedding and DB-Specific Use Cases*

Semantic and Similarity Search

Vector databases' unique features include semantic and similarity search. These two features simplify life and reduce the burden of complex query building in production development, ultimately replacing complex SQL queries.

Semantic Search: This is basically semantic search-based technology. During the operation, this technology helps search for keywords using eloquent conversation practices. This means it interprets the meaning of words and phrases, and the results will return the content that matches the sense of an input query. It also carefully avoids not matching content with the input query. It can understand words from the input query's intent and the end user's search context. It improves the quality of search results by understanding natural languages accurately and focusing on the context of purpose. This search predominantly helps with text generation and summarizing application purposes.

The following figure shows the Semantic Search for Input query: Vitamins A and C-based fruits.

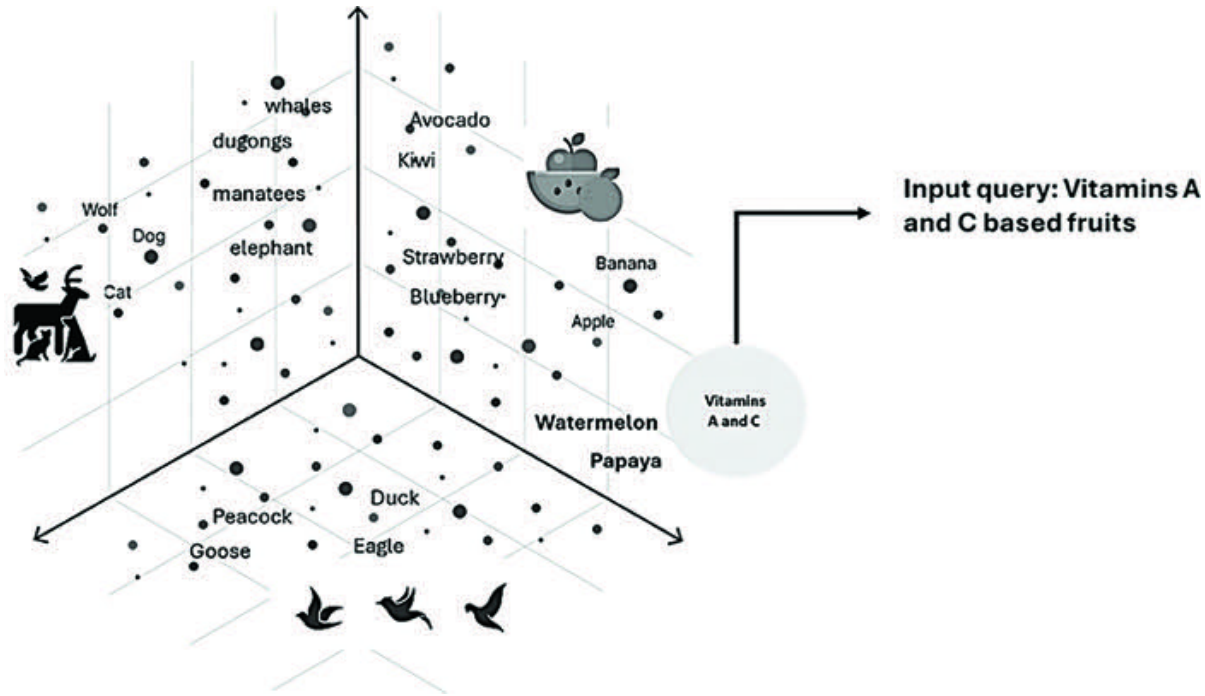

Figure 3.8: *Semantic Search: Input Query: Find Vitamins A and C-Based Fruits*

Similarity Search: This is a critical feature of vector databases that helps us discover the closeness of two vectors and assess how they resemble each other. This specific feature enables us to find similarities and generate the expected context as text or a summary. It allows us to identify similarities in images, audio, and video files.

Understanding the similarities between vectors enables us to extract the similarity between the data objects stored in the vector database. This process helps us understand patterns and insights

and leads to decisions for specific use cases and product perspectives.

The mechanism of similarity search is not keyword-based. Instead, it goes with the meaning in the multidimensional vector space.

It uses the **Navigable Small World** graph indexes for vector similarity search operations. This is a highly acceptable technology with fast search speeds and quick recall, ultimately exhibiting excellent performance.

When user queries are received through the application interface as input, they are cascaded into the database's vector space. Prior to this, the input query is also converted into a query vector to align with the vector database. This enhances the search and retrieval speed. Additionally, it enables the database to understand what is expected from the user's query and response.

Generally, the vector search starts at the top layer of the HNSW index, and the search operations gradually move on, narrowing down to the closely related vector space. Internally, it builds the graph representation to ease the search operations. This space contains the most relevant data points for our results or the required response back to the system.

The algorithm helps us to identify or locate the area of the right graph position, which contains the vectors closest to the input query.

Internally, it compares the query vectors to all the other data points in the vector space, using "distance" and a concept called **SIMILARITY** for the outside world to estimate how close they are. After finding the best match between the user's input query and the stored vector embedding data in the vector database, the LLM helps us generate the appropriate response as text, summary, or new documents specific to requirements back to the application. This process further trains the LLM over time and improves its capabilities.

We will briefly discuss the various metrics similarity search operations used to determine the distance between the vector spaces, such as Euclidean distance, Dot Product, and Cosine similarity.

Figure 3.9 illustrates the Similarity Search for Input query: Find fruits with higher water content. Input query: Animal with four legs and one tail, able to brake, and domestic animal in the multidimensional vector space.

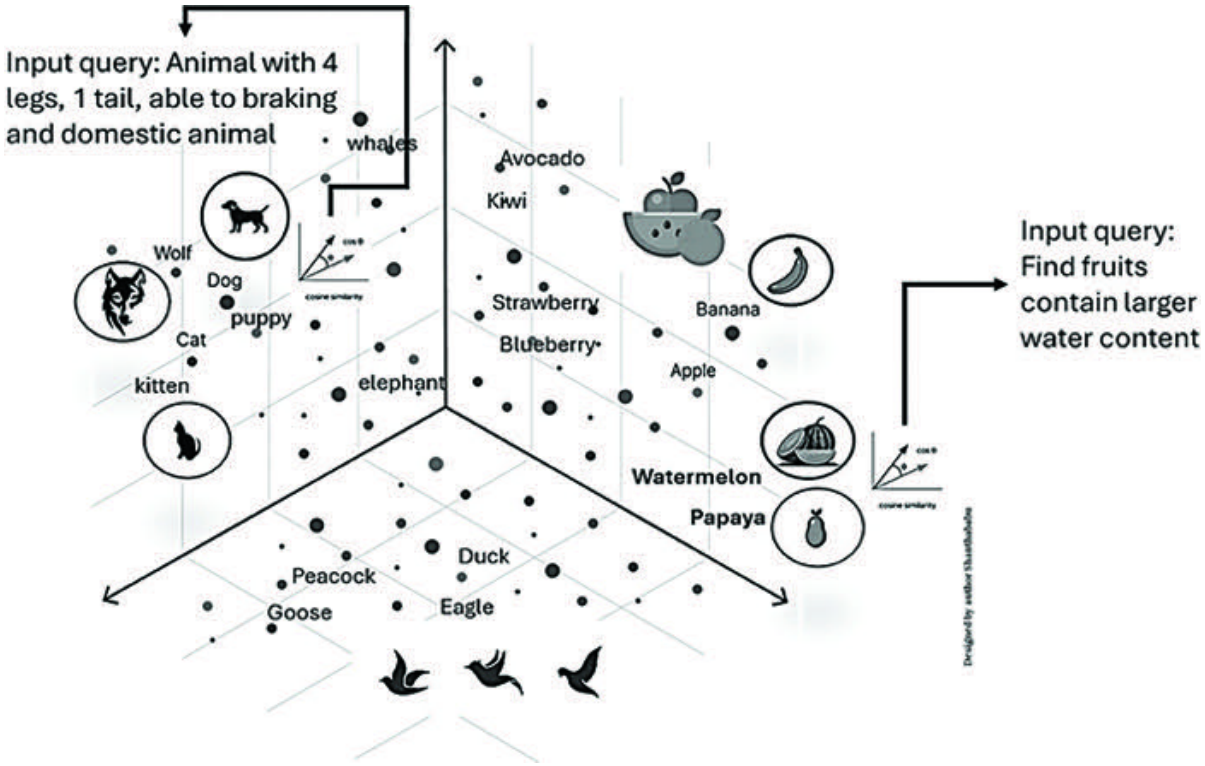

Figure 3.9: *Similarity Search*

Types of Similarity Measures

Fundamentally, these measures are mathematical in nature. These measuring methods form the basis of how a vector database compares the closeness of vector space and identifies the most relevant data points as results for the given input query. Different methods are used, and each method depends on the nature of the data, such as Euclidean Distance, Dot Product, and Cosine Similarity.

Euclidean Distance: In the vector space, this measures the straight-line distance between two vectors, such as A and B. The range spans from o to infinity.

If it is then the two vectors are identical;

If it is both vectors are dissimilar for larger values.

Dot Product: This method discusses the alignment of two vectors' positions and offers different possibilities of alignment, such as in the same direction, opposite directions, or perpendicular to each other.

It ranges from -∞ to ∞.

Positive value represents vectors that point in the same direction.

0 value represents orthogonal vectors.

Negative value represents vectors that point in opposite directions.

Cosine Similarity: It measures the similarity of two vectors using the angle between them. In this measure, the angle is only considered for the similarity calculation.

It ranges from -1 to 1:

1 represents identical vectors.

0 represents orthogonal vectors.

-1 represents vectors that are entirely opposed.

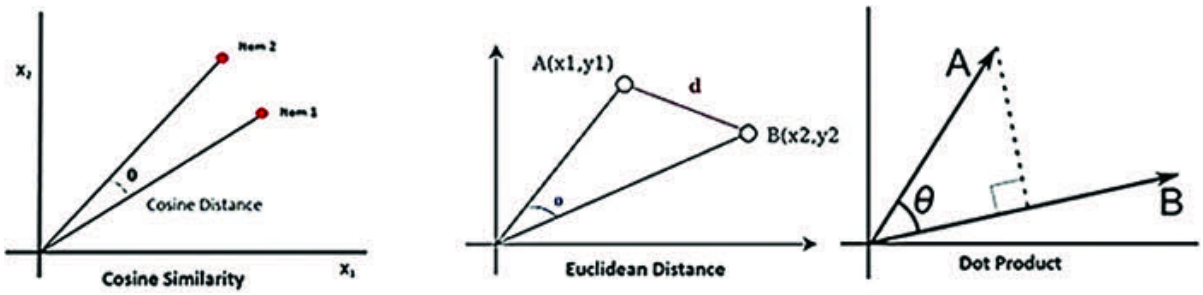

Figure 3.10: *Types of Similarity Measures*

Analogy to Understand Vector Database

To understand better, we can consider examples such as libraries, department stores, vegetable shops, and so on.

In the library, the books are arranged based on age group, literature, drama, poetry, math, science, history, and languages.

In the department store, the products are arranged based on their categories, physical and chemical nature, purposes, manufacturer's specifics, customer usages, and quantity bases—such as rice, wheat, oil, soap, and snacks. They are further divided into multiple sub-categories.

In the vegetable shop, the vegetables, such as potatoes, tomatoes, brinjals, onions, and beans, are arranged.

You can imagine all these analogies and the purpose of arranging them appropriately for easy access and to get things done faster and more accurately from specific places.

In the same fashion, the data is automatically arranged congruently based on content that has the same similarity. Let us

consider the preceding analogies for vector databases, similar to organizing books/products/vegetables on shelves based on their characteristics. In similar conduct, data is automatically arranged in the vector database space within the given dimension, while searching internally in the vector database by applying the search mechanism, relevant data is brought back to the model. The model can articulate the content and pass it on as application output. Otherwise, it will display a message such as 'Not Available' or 'Unavailable,' and the LLM will communicate appropriate statements back to the system.

Figure 3.11: *Analogy to Understand Vector Database*

Classification of Vector Databases

As the demand for vector databases is increasing exponentially, various organizations are building their own vector database products to support and enhance their market stability. Some are new to the market, while others are adding new vector database capabilities to their existing products.

Pure Vector Databases

Text Search Databases

Vector-Capable NoSQL Databases

Vector-Capable SQL Databases

Vector Libraries

Pure Vector Databases: This category is designed to effectively store and supervise vector embeddings and metadata.

Architecturally, it is intended to separate the derived embeddings from the data source. **Pinecone and Chroma.**

Text Search Databases: This category is famous for its full-text search and rich documents. The search index is a dedicated data structure that supports efficient and fast textual data searching. **Example: Elasticsearch and Solr.**

Vector-Capable NoSQL Databases: This category is very new to the market and small. It has not been utilized and tested for real-time purposes, so its performance is lower than that of other vector databases. **Examples: MongoDB and Azure Cosmos DB.**

Vector-Capable SQL Databases: This category is comparatively small and typically based on the relational database model. It supports vector search with similarity searches such as dot-product, cosine, and Euclidean distance. **Examples: SingleStoreDB and**

Vector Libraries: This category is a dynamic vector representation model that uses multiple templates. It enables us to perform a vector similarity search using the "ANN algorithm," and the implementation is more accessible than in

another category. **Examples: Facebook Faiss, Google ScaNN, and Spotify Annoy.**

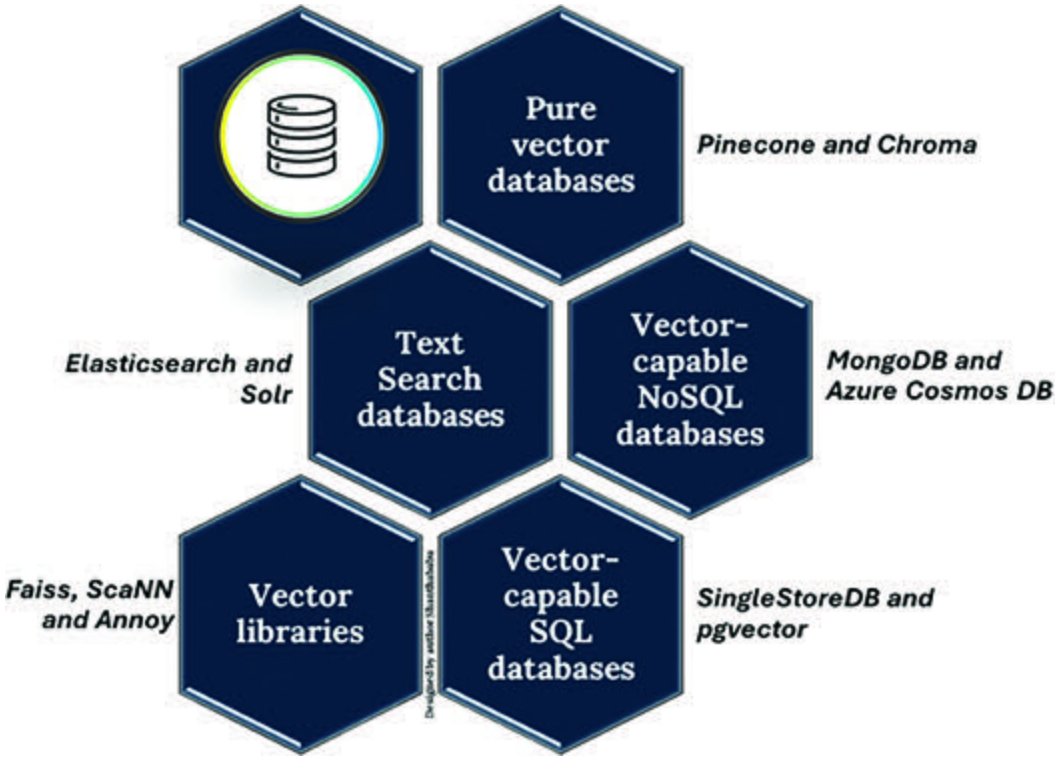

Figure 3.12: *Classification of Vector Databases*

Popular Vector Databases in the Market

As demand increases, the industry introduces several popular vector databases into the market. Based on market demand, let us discuss the best leading vector databases available and choices from a Gen AI application development perspective.

All these databases and their libraries provide very specialized features, performance, and optimizations for handling vector data efficiently by enabling various tasks, such as search engine operations, namely semantic, similarity, classification, and clustering systems at different values of scales.

Besides the following, other vector databases are available in the market:

Pinecone

Faiss

Chroma

MongoDB Atlas

Elasticsearch

Pgvector

Apache Cassandra

OpenSearch

Let us discuss the top five here:

Pinecone: This is a managed, cloud-native vector database, not open-source. We can connect to the database through the API from the code itself; no infrastructure is required. It can process the data promptly, with high indexing and filtering expertise. It supports high-quality consequence responses, securing fast and accurate results across a wide range of search operation requests from the applications.

Gen AI and AI developers can build AI solutions without any infrastructure, maintenance, monitoring, and more. This is an easily configurable pay-as-you-go model, ideal for individual users' purposes or smaller projects.

Additionally, it has features such as classification, rank tracking, and duplicate detection.

Faiss: This was developed by the Facebook AI Research team. It is an open-source and dense-vector similarity search pattern using a set of vectors, and it offers a "Dot Product" vector comparison mechanism. This includes high-performance vector searches, exceptionally fast search speeds, and clustering library models. It is also highly efficient at handling CPU and GPU computations for large-scale data. However, the major drawback is that it only allows offline index add/deletion updates.

Chroma: This open-source vector database facilitates data storage and retrieval and employs semantic search operations on text data. It has rich features such as queries, filtering, density estimates, and building visualization.

It helps the LLM application perform enhanced NLP operations and significantly reduces hallucinations while generating text, summaries, facts, and more.

It also specializes in generating visualization and discovery from high-dimensional vector space. Using native interfaces, raw data points are transformed into visuals and patterns.

MongoDB Atlas: This fully multi-cloud-managed data platform has an extensive array and vector search facilities, such as text or vocabulary. It can handle various workloads, such as transactional and search operations. It has high availability, data durability, high archival, and backup features.

It has a specialized vector index and independent scaling ability that is automatically synchronized with the database and easily configurable without additional infrastructure to perform the operations.

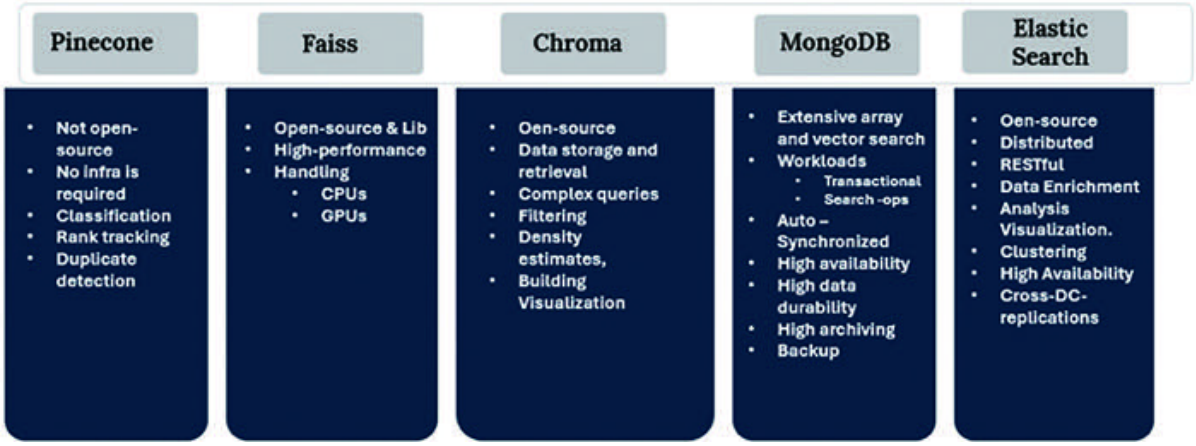

Figure 3.13: *Popular Vector Databases Comparison*

Elasticsearch: This open-source vector database facilitates distributed and RESTful analytics capabilities. It is part of the "Elastic Stack," which includes open-source tools for various

activities such as data collection and storage, data enrichment, analysis, and visualization.

It has excellent features, including clustering, high availability, cross-datacenter-specific replications, fast searching, fine-tuning, and complex analytics assessments. It can also expand horizontally. This will help identify errors to keep clusters highly secure and accessible for LLM applications.

Working with Pinecone Vector Database

Let us discuss how to work with the Pinecone vector database using simple steps to create a vector database through a browser-based interface. By referring to the API key, we can connect it using Python code.

Step 1:

https://www.pinecone.io/solutions/semantic/

New user sign-up. Click on **Up**

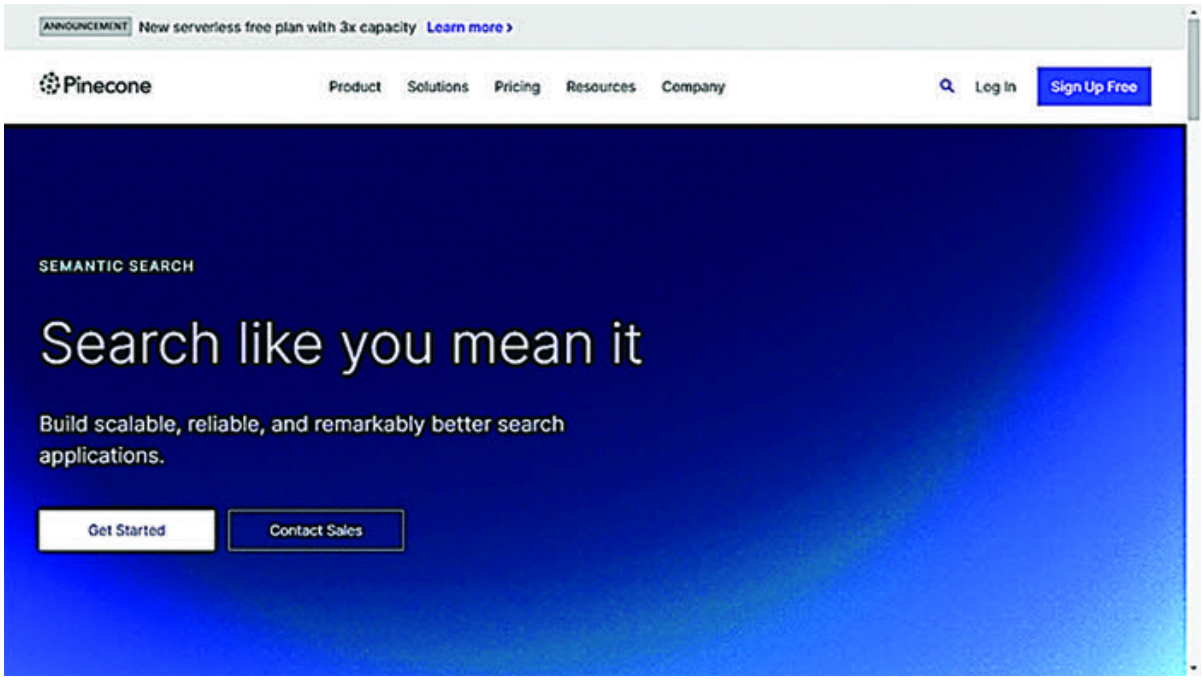

Figure 3.14: *Pinecone Main Page*

Step Select Google/GitHub/Microsoft login or use Pincone-based login.

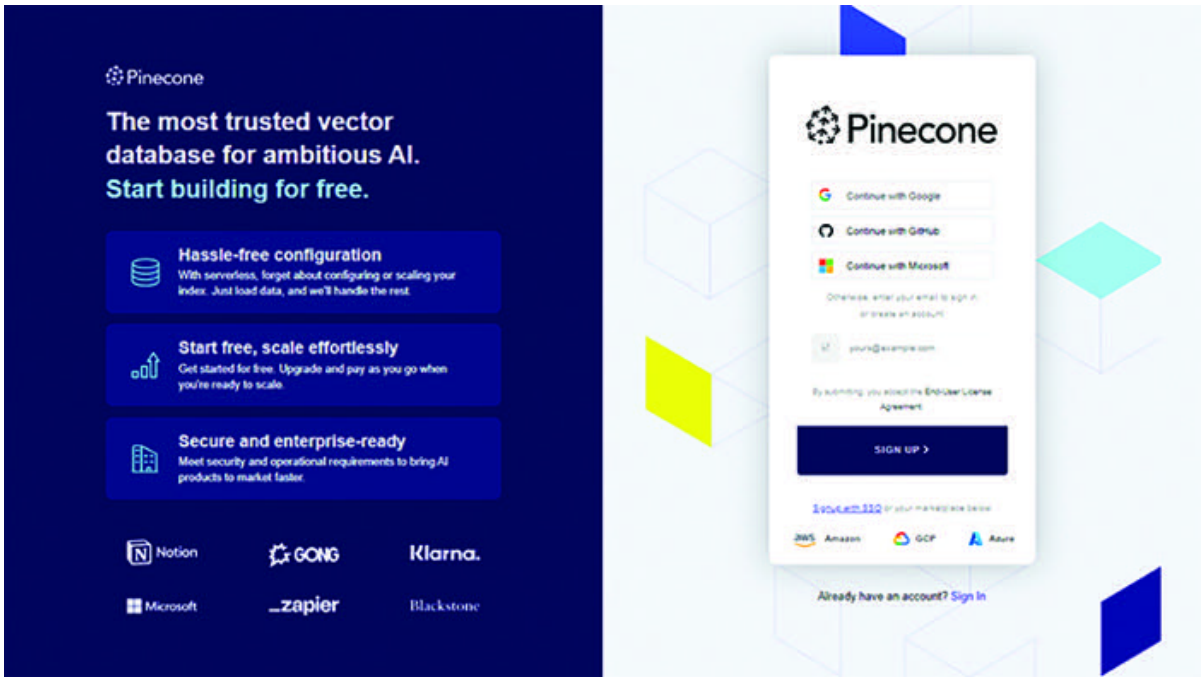

Figure 3.15: *Pinecone Login Page*

Once you successfully log in, you will be placed on the starter package by default. If you want to enhance your utilization, you can go with Standard or Enterprise; it will be charged based on the features, capacity, and support.

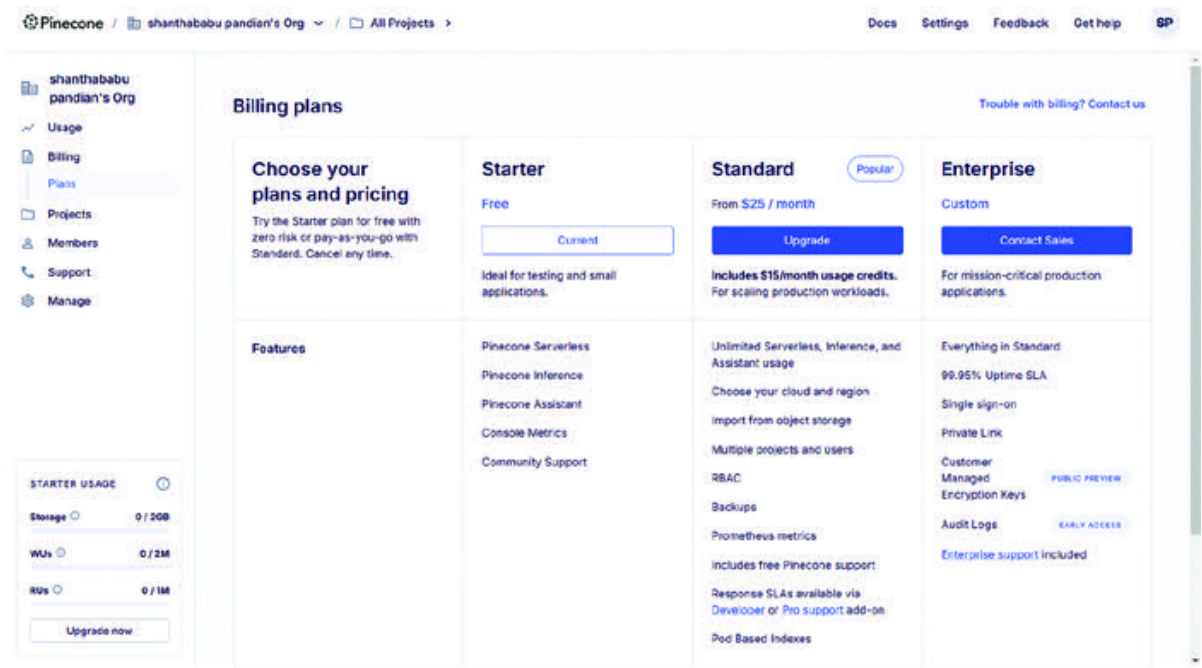

Figure 3.16: *Pinecone Billing Page*

After a successful login, the application will take you to the index page.

Step Click the button to create a new index.

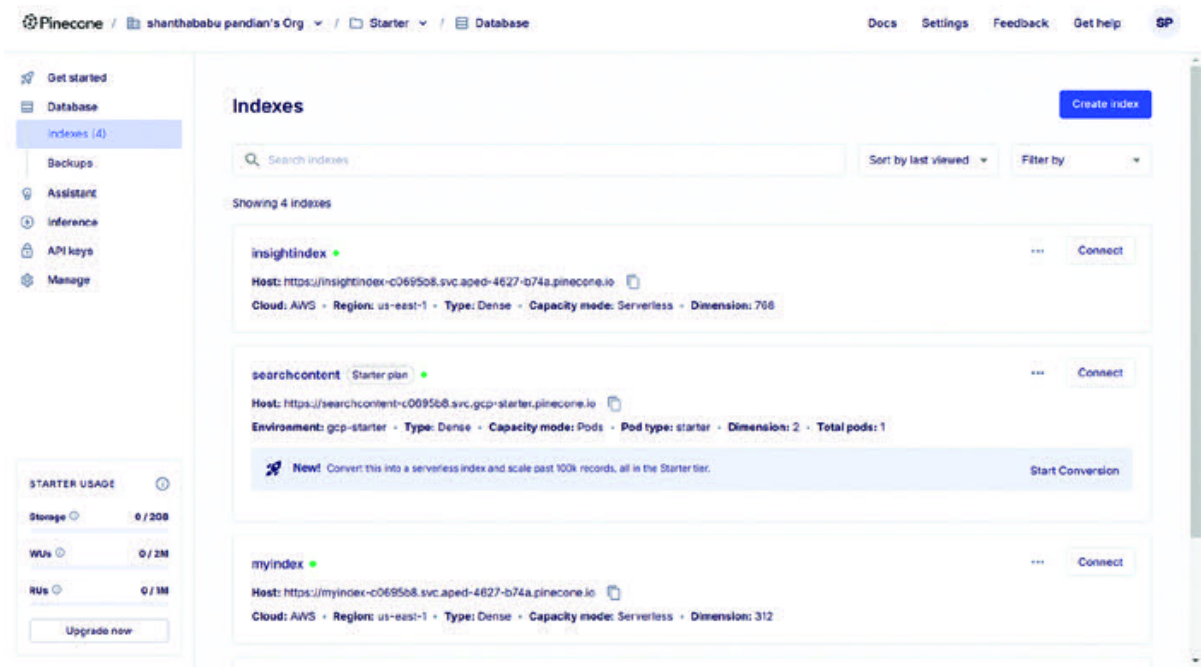

Figure 3.17: *Pinecone Index Creation Page*

Step Enter the name of the index.

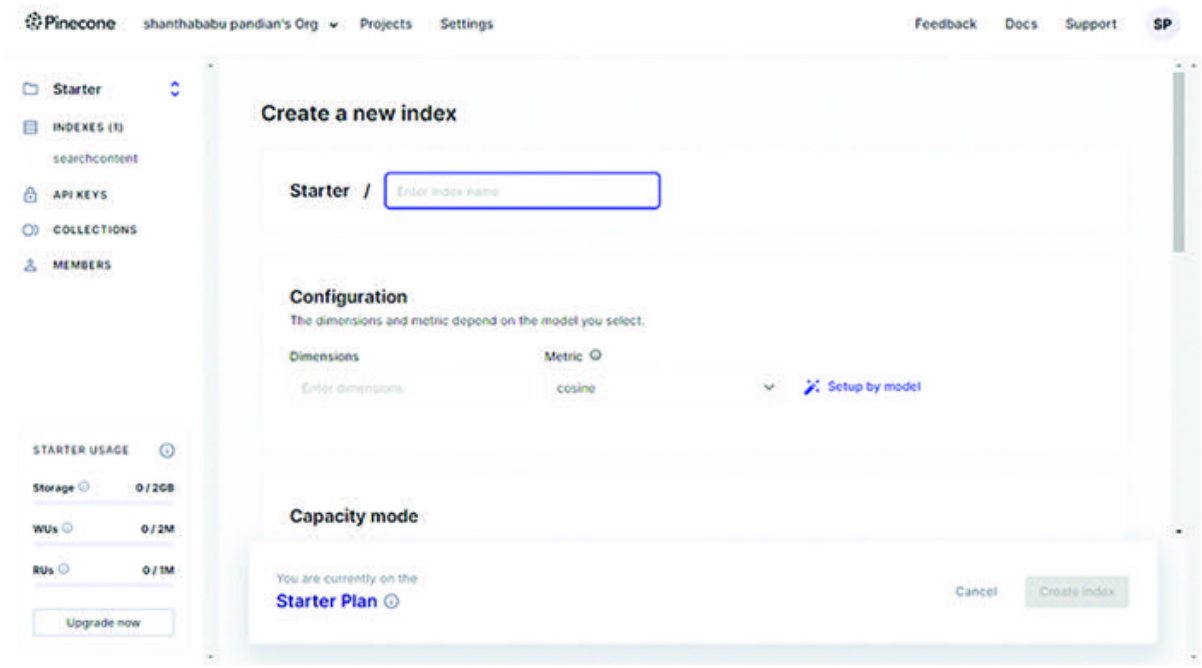

Figure 3.18: *Pinecone Index Configuration Page-I*

Step Enter the dimensions and **Metric** (from the drop-down list box); your index is ready. You can select the **REGION** (Example: us-east-1).

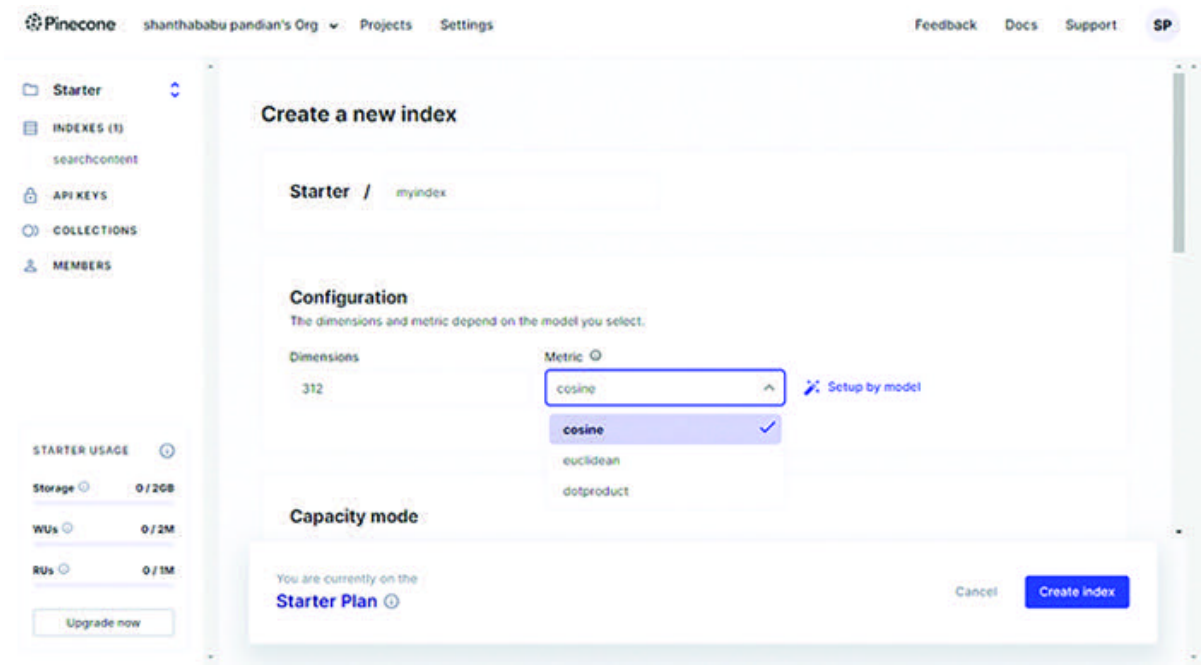

Figure 3.19: *Pinecone Index Configuration Page-II*

Step Create API Key; click the **API** button.

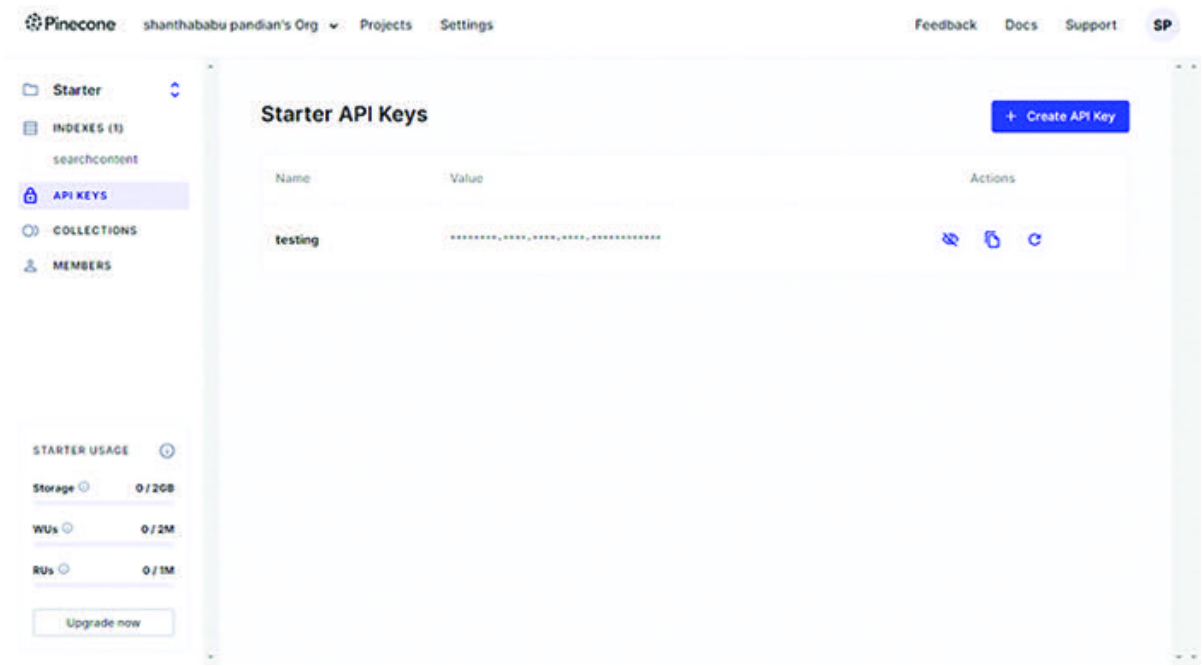

Figure 3.20: *Pinecone API Key Configuration Page-I*

Step You will see a small pop-up screen asking you to enter the API key name and fill in the information.

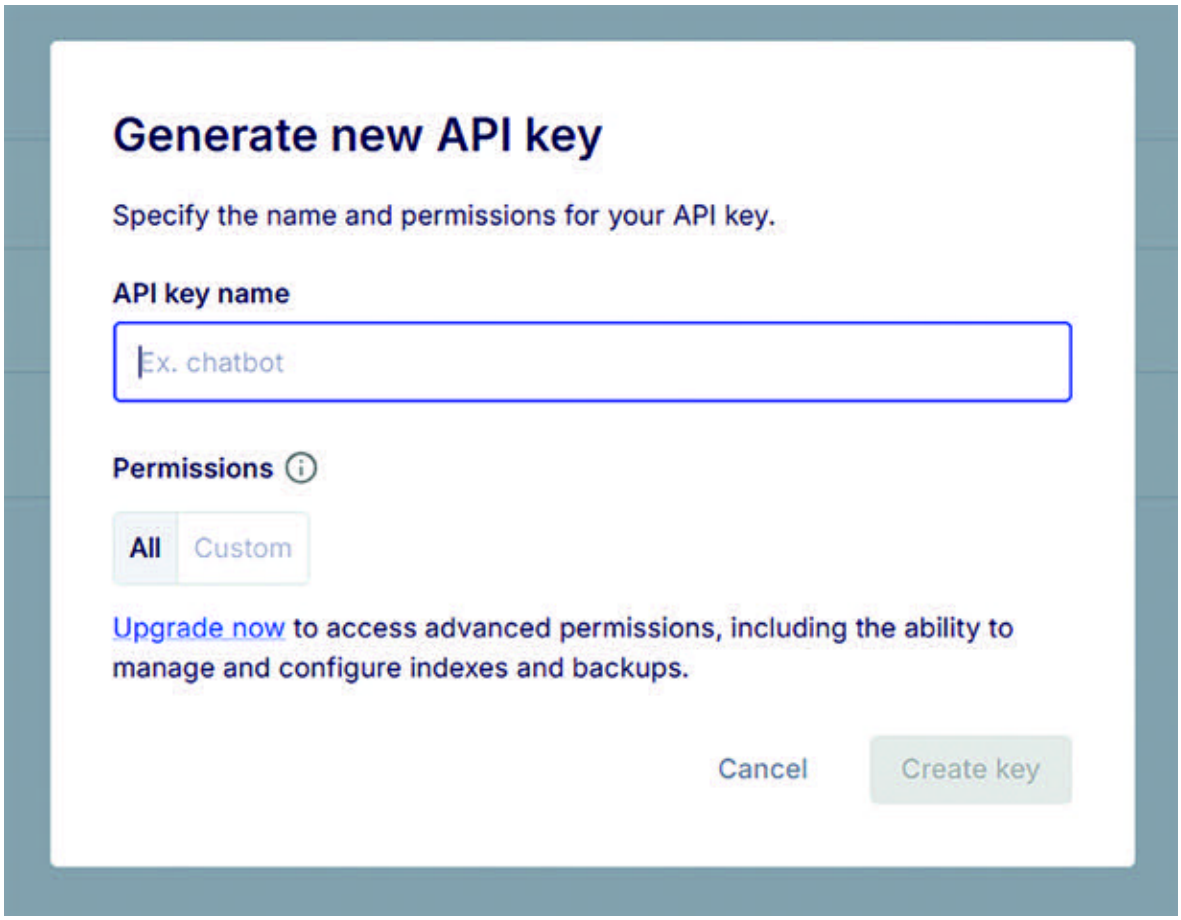

Figure 3.21: *Pinecone API Key Configuration Page-II*

It should be in lowercase, and you can include numerals as well.

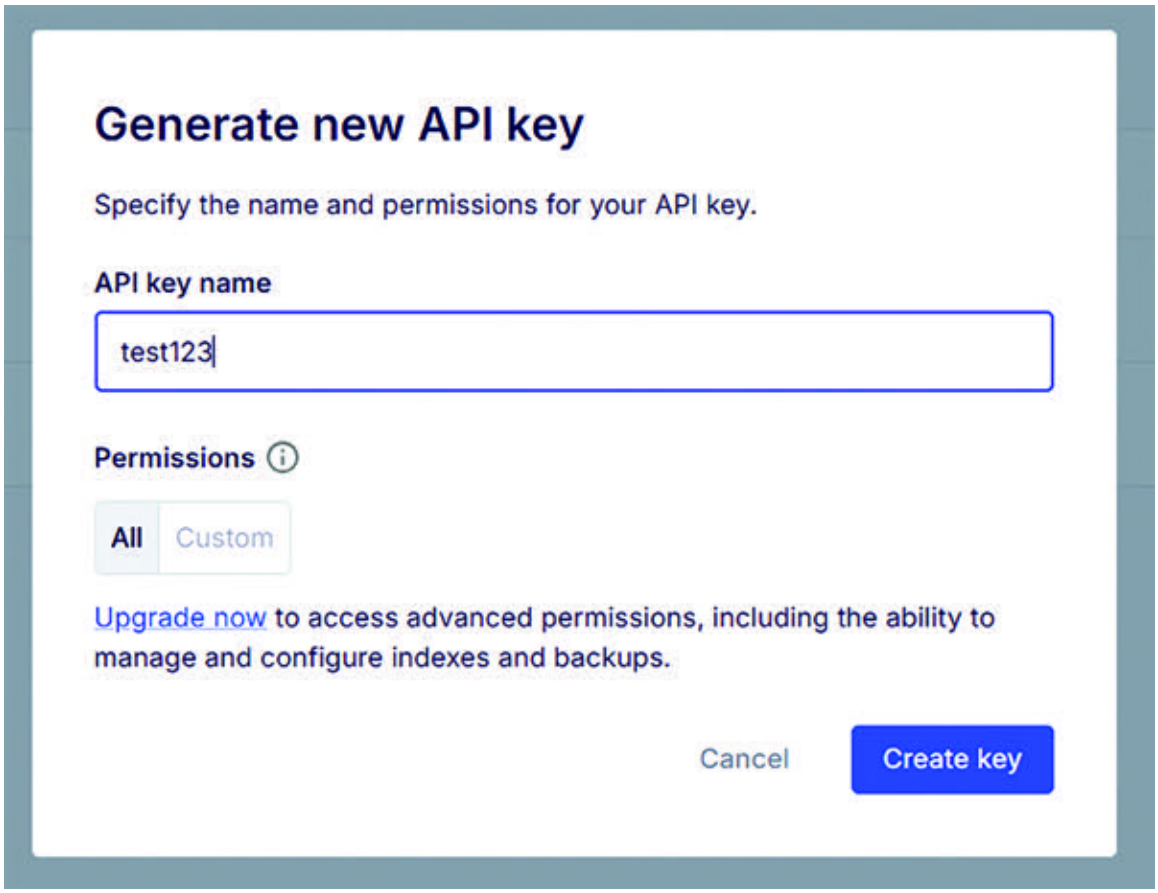

Figure 3.22: *Pinecone API Key Configuration Page-III*

Once it is done, your new API key will appear on the list. Near the API key, you will see four icons, each representing an individual purpose.

Figure 3.23: *Pinecone API Key List Page*

Step If you want to delete the API key, click on the **Delete** icon, which is the leftmost icon. A popup screen will appear to confirm the deletion by entering the same key.

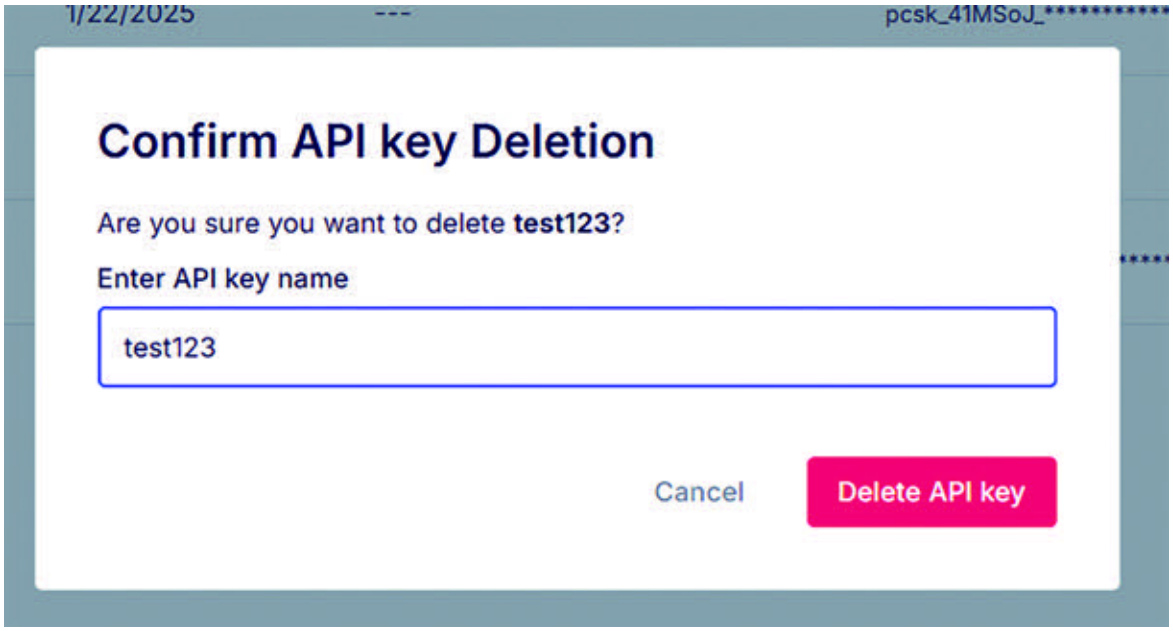

Figure 3.24: *Pinecone API Key Deleting Process*

If you want to explore more, you can visit the document page.

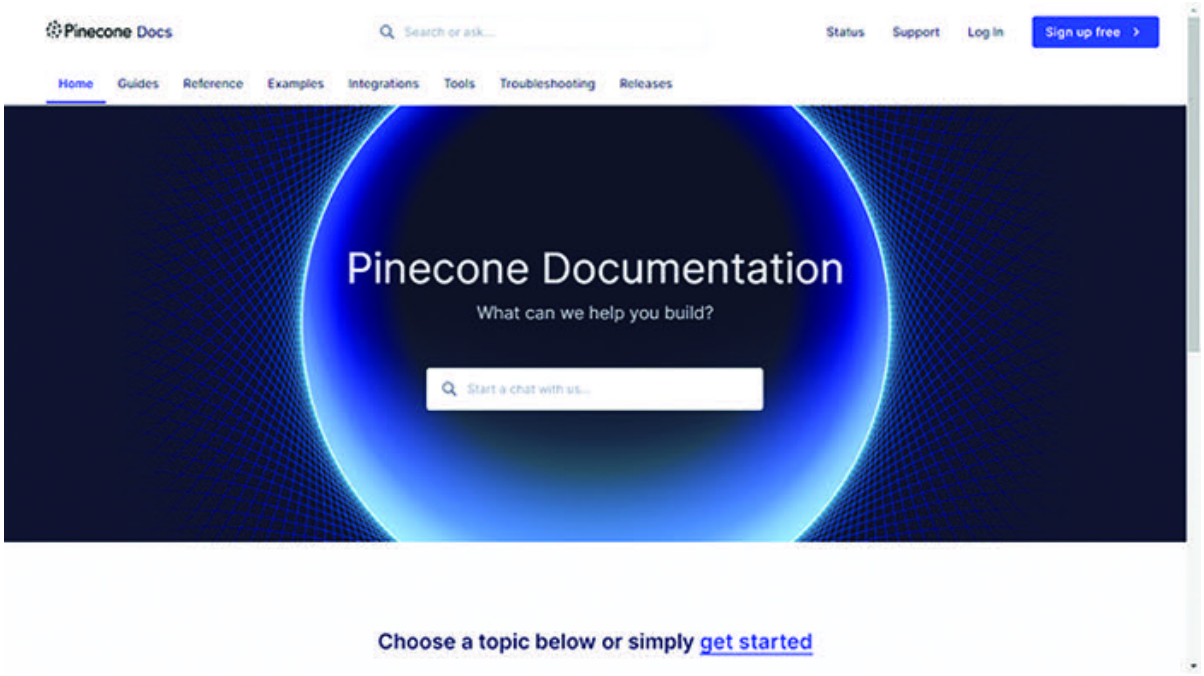

Figure 3.25: *Pinecone Documentation Page*

So far, we have discussed creating the vector database index and providing dimensions, API Keys, and settings for the Pinecone web application. Let us use a simple code snippet to access the document through the vector database by passing a query (Prompt) to get a response. Remember, this is not open-source.

Technical Requirements:

OpenAI: https://platform.openai.com/docs/overview

Requirements: You must create an account to get the Open API Key if you haven't already created one.

Colab : https://colab.research.google.com/

Pinecone: https://www.pinecone.io/solutions/semantic/

Create API Keys, indexes, and environments.

Langchain framework

Google Drive: Create a folder and upload the document required for our exercises to ask Q&A (RAG).

Install the required Python libraries for this implementation.

Install the following libraries:

```
print("**********************************************")
print("Libraries are installed successfully. ")
print("**********************************************")
!pip install langchain
```

```
!pip install openai
!pip install PyPDF2
!pip install pinecone-client
```

Output

```
**************************************
Libraries have been installed successfully.
**************************************
```

Ready to code for simple vector database implementation using Pinecone, OpenAI, and Langchain.

```
# Mount your drive to access the files from Google Drive.
from google.colab import drive
drive.mount('/content/drive/')
```

Output

```
Mounted at /content/drive/
```

```
print("*****************************************")
print("Libraries are imported successfully. ")
print("*****************************************")
import os
from PyPDF2 import PdfReader
```

```
from langchain.embeddings.openai import OpenAIEmbeddings
from langchain.text_splitter import CharacterTextSplitter
from langchain.text_splitter import RecursiveCharacterTextSplitter

from langchain.llms import OpenAI
from langchain.vectorstores import Pinecone
from langchain.document_loaders import TextLoader
from langchain.chains.question_answering import load_qa_chain
from langchain.llms import OpenAI
```

Output

```
**************************************
Libraries are imported successfully.
**************************************

print("*****************************************************")
print("Environment variables are created successfully. ")
print("*****************************************************")

# Creating environment variables.
os.environ["OPENAI_API_KEY"] =
"XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX"
os.environ["PINECONE_API_KEY"] =
"xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx"
```

```
PINECONE_API_KEY = os.environ.get('PINECONE_API_KEY',
xxxxxxxxxxxxxxxxxx')
PINECONE_API_ENV = os.environ.get('PINECONE_API_ENV', 'gcp-
starter')
```

Output

```
Environment variables are created successfully
```

```
# llm is ready with minimal temperature.
llm = OpenAI(OpenAI=api_keys, temperature=0.1)
```

```
# Initialize pinecone
import pinecone
pinecone.init(

api_key=PINECONE_API_KEY, # find at app.pinecone.io
environment=PINECONE_API_ENV # next to the api key in the
console
)

index_name = "abcdefg" # put the name of your pinecone index
here

# Provide the path of pdf files.
pdfreader = PdfReader(filepath)
```

Loop through the file pages and get the content.

```
from typing_extensions import Concatenate
# Read text from pdf.
raw_text = ''
for i, page in enumerate(pdfreader.pages):
content = page.extract_text()
if content:
raw_text += content
```

Print the content

```
raw_text
```

Output

```
'Machine Learning Engineering  \non AWS\nBuild, scale, and secure machine learning systems and MLOps \npipelines in production\nJoshua Arvin La
ngineering on AWS\nCopyright © 2022 Packt Publishing\nAll rights reserved . No part of this book may be reproduced, stored in a retrieval syste
y means, without the prior written permission of the publisher, except in the case \nof brief quotations embedded in critical articles or revie
reparation of this book to ensure the accuracy of the information \npresented. However, the information contained in this book is sold without
either the author, nor Packt Publishing or its dealers and distributors, will be held liable \nfor any damages caused or alleged to have been c
k.\nPackt Publishing has endeavored to provide trademark information about all of the ...'
```

Figure 3.26: *Sample Document Output from the Object*

```
# We need to split the text using Character Text Split.

text_splitter = CharacterTextSplitter(
separator = "\n",
chunk_size = 800,
```

```
chunk_overlap = 200,
length_function = len,
)

texts = text_splitter.split_text(raw_text)

# Find the length of the overall text.
len(texts)
```

Output

```
1286

# Creating embeddings of objects from OpenAI.
embeddings = OpenAIEmbeddings()

# Connecting embedding object and text object along with
Pinecone index.
print("****************************")
print("Pinecone object is ready.")
print("****************************")
vectordb =
Pinecone.from_texts(texts,embeddings,index_name="demoindex")
```

Output

Pinecone objects are ready.

```
# Create a retriever object to query the vector database.
retriever = vectordb.as_retriever(score_threshold = 0.3)

chain = load_qa_chain(OpenAI(), chain_type="stuff")

query = "Who is the author?"
docs = vectordb.similarity_search(query, k=3)
chain.run(input_documents=docs, question=query)
```

Output

John Alan

```
docs = vectordb.similarity_search("What is age of the author")
chain.run(input_documents=docs, question=query)
```

Output

The name of the book is not given in the context provided. (The age of the author is not mentioned in the book)

Combining Multiple Chains Using simple Sequential Chain

```
from langchain.chains import LLMChain
master_template=PromptTemplate(input_variables=['Technical
requirements'],template="Please tell me about {Technical
requirements}")
level_1_chain=LLMChain(llm=llm,prompt=master_template)

References_template=PromptTemplate(input_variables=['Further
reading'],template="Suggest me some References and further
reading {Further reading}")
level_2_chain=LLMChain(llm=llm,prompt=References_template)
from langchain.chains import SimpleSequentialChain
comb_chain=SimpleSequentialChain(chains=
[level_1_chain,level_2_chain])
comb_chain.run("Deep Learning Containers")
```

Output

```
'\n\nReferences:\n\n1. Grogan, P. (2020). Deep Learning Containers: What They Are and How to Use Them. Retrieved from https://t
-are-and-how-to-use-them-3c8d1d7bac84\n\n2. KubeFlow. (2020). Deep Learning Containers. Retrieved from https://www.kubeflow.org
S., & Liu, Y. (2020). A Comprehensive Guide to Deep Learning Containers. Retrieved from https://towardsdatascience.com/a-compre
\n4. Zaremba, M., & Wawrzyniak, M. (2020). Deep Learning Containers. Retrieved from https://www.michalzaremba.pl/deep-learning-
earning Container? Retrieved'
```

Figure 3.27: *Sample Document Output from Pinecone Vector Database Object*

Working with the FAISS Vector Database

The FAISS vector database is much simpler than Pinecone. It is an open-source library model.

Install, import, and create the relevant objects so we can start working with the data.

Technical Requirements:

OpenAI: https://platform.openai.com/docs/overview

Requirements: You must create an account to get the Open API Key, if you haven't already created one.

Colab : https://colab.research.google.com/

Langchain framework

Google Drive: Create a folder and upload the document required for our exercises to ask Q&A (RAG).

Install the required Python libraries for this implementation.

Install the following libraries:

```
print("*****************************************")
print("Libraries are installed successfully. ")
print("*****************************************")
!pip install langchain
!pip install openai
!pip install PyPDF2
!pip install faiss-cpu
!pip install tiktoken
!pip install load_dotenv
```

Output

```
*****************************************
Libraries have been installed successfully.

*****************************************
```

Ready to code for simple vector database implementation using Faiss, OpenAI, and Langchain.

```
# Mount your drive to access the files from Google Drive.
from google.colab import drive
```

```
drive.mount('/content/drive/')
```

Output

```
Mounted at /content/drive/
print("****************************************")
print("Libraries are imported successfully. ")
print("****************************************")
from dotenv import load_dotenv
import os
from PyPDF2 import PdfReader
from langchain.text_splitter import CharacterTextSplitter
from langchain.embeddings.openai import OpenAIEmbeddings
from langchain import FAISS
from langchain.chains.question_answering import load_qa_chain
from langchain.llms import OpenAI
from langchain.callbacks import get_openai_callback
load_dotenv()
```

Output

```
****************************************
Libraries are imported successfully.
****************************************

print("************************************************")
```

```
print("Environment variables are created successfully. ")

print("**************************************************")
```

Output

```
Environment variables are created successfully.

# llm is ready with minimal temperature.
llm = OpenAI(OpenAI=api_keys, temperature=0.1)

# Provide the path of pdf files.
pdfreader = PdfReader("file path")
# Load the data
data = pdfreader

Loop through the file pages and get the content.

from typing_extensions import Concatenate
# Read text from pdf.
raw_text = ''
for i, page in enumerate(pdfreader.pages):
content = page.extract_text()
if content:
raw_text += content
```

Print the content

raw_text

Output

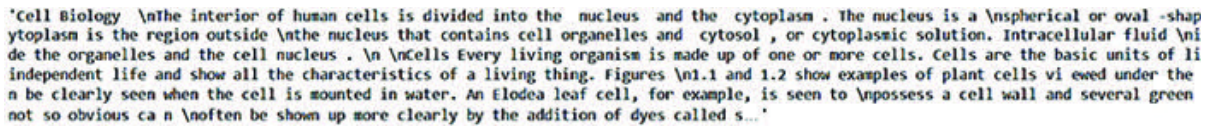

Figure 3.28: *Sample Document Output from Object*

We need to split the text using Character Text Split.

```
text_splitter = CharacterTextSplitter(
separator = "\n",
chunk_size = 800,
chunk_overlap = 200,
length_function = len,

)

chunks = text_splitter.split_text(raw_text)

# Creating embeddings of objects from OpenAI.
embeddings = OpenAIEmbeddings()
```

```
# Connecting embedding object and text object along with FAISS index.
print("**************************")
print("FAISS object is ready.")
print("**************************")
vectordb = FAISS.from_texts(chunks, embeddings)
```

Output

```
The FAISS objective is ready.
```

```
# Create a retriever object to query the vector database.
retriever = vectordb.as_retriever(score_threshold = 0.3)

chain = load_qa_chain(OpenAI(), chain_type="stuff")

query = "What is Cell Biology?"
docs = knowledgeBase.similarity_search(query, k=3)
chain.run(input_documents=docs, question=query)
```

Output

```
Cell Biology is the study of cells and their structures, functions, and processes.
It focuses on the organization and properties of cells, as well as how they interact with their surroundings.
```

Figure 3.29: *Output from FAISS Vector Database Object-Sample Query-I*

```
query = "What is Diffusion?"
docs = knowledgeBase.similarity_search(query)
chain.run(input_documents=docs, question=query)
```

Output

```
Diffusion is the movement of molecules from an area of high concentration to an area of low
concentration until the concentration becomes equal.
It is a passive process that does not require energy and is important
for the exchange of substances in cells,
such as oxygen and carbon dioxide, and in multicellular organisms for
the movement of respiratory gases and dissolved food.
The cell membrane plays a role in controlling the passage of
substances during diffusion.
```

Figure 3.30: *Output from FAISS Vector Database Object-Sample Query-II*

```
query = "Explain Sterile Petri dish"
docs = knowledgeBase.similarity_search(query)
chain.run(input_documents=docs, question=query)
```

Output

```
A sterile Petri dish is a container used in microbiology to culture microorganisms.
It is commonly referred to as a plate and can be made of either plastic or glass.
Glass Petri dishes can be sterilized in an autoclave,
while plastic Petri dishes are already sterilized during their manufacture.
This ensures that no unwanted microbes are present in the dish,
creating a sterile environment for the growth of specific microorganisms.
```

Figure 3.31: *Output from FAISS Vector Database Object-Sample Query-III*

Conclusion

In this chapter, we discussed the definition of a vector database and its role as part of the new generation of database management systems. We explored how it stores data with fixed-length lists of numbers and high-dimensional vectors, its need for dealing with multiple file formats, how all these are helping with specific use cases at industry-level problem statements, and how social media platforms are utilizing them. We discussed the characteristics of vector databases precisely with classical evidence followed by exploring the differences between vector databases and traditional databases in terms of design, capabilities, assessing methodology, storing format, and more.

We understood the functional differences between the vector and traditional databases by comparing them across different aspects. We gained insight into the working mechanism of vector databases with neat diagrams illustrating their components that help to build Gen AI applications, such as input queries, embedded modeling building, similarity search operations, output generation and response, and chain of

queries. We also understood how vector databases act as serverless vector databases and their advantages.

We have reviewed significant vector embeddings and DB-specific use cases, such as information retrieval (RAG), similarity search operations, classification and clustering, recommendation systems, and sentiment analysis.

Using a classic example, we exclusively discussed and experimented with working with the Pinecone vector database and the FAISS vector database.

In the next chapter, we will explore prompt engineering and its importance in Generative AI, components prompts, and their categories. We will discuss various prompt engineering techniques and how to generate knowledge prompting.

CHAPTER 4

Prompt Engineering and Its Significance

Introduction

In the previous chapter, we explored the definition and significance of vector databases as part of the new generation of database management systems. We discussed how, particularly for Gen AI products and applications, vector data representation is resource-intensive yet incredibly scalable, searchable, expandable, and dimensional. With an emphasis on the function of embeddings in vector databases, we examined the design, capabilities, evaluation techniques, and storage formats of vector databases in comparison to standard databases. Additionally, we looked at well-known vector databases like Pinecone, Faiss, and Chroma and included comprehensive instructions along with code for using Pinecone and FAISS.

This chapter will explore prompt engineering and its importance in Generative AI, component prompts, and categories. We will discuss various prompt engineering techniques such as N-shot prompting, Chain-of-Thought (CoT) prompting, Directional Stimulus Prompting (DSP), React prompting, Multimodal CoT prompting, Graph prompting, and how to generate knowledge prompting. Understanding the

problem, how to craft the initial prompt, evaluating the model's response, iterating and refining the prompt, testing the prompt on different models, and scaling the prompt performance. We will explore the prompt engineering aspects through examples such as Q&A, text summarization, chatbot responses, document classification, and named entity recognition, along with the best practices and guidelines for writing prompts while building strategic models.

Structure

In this chapter, we will discuss the following topics:

Prompt Engineering and Significance in Gen AI

Components and Elements of Prompt

Instructions, Context, Input and Output Data, and More

Role Prompting

Professional, Expert, Fictional Roles, and More

Effectiveness of Role Prompting Techniques

Role Prompt's Flip Side

Prompting: Voice Specific

Prompt Engineering Techniques

Single and Multiple Prompt

Best Practices for Building Effective Prompts

Prompt Engineering and Significance in Gen AI

Let us understand prompt engineering first, then discuss its significance in Gen AI.

We can describe prompt engineering in various ways.

Prompt engineering is the process of designing and improving the input prompt(s) for LLMs and instructing the vector database to get the desired outputs for Gen AI, which could be chatbots, agents, and so on.

Although Gen AI's objective is to replicate human thoughts and feelings, it needs accurate guidance and directions to produce high-quality and relevant output for the given task. So, in prompt engineering aspects, we must select the correct role and formats of output, including words, sentences, phrases, and indications to the LLMs to generate the appropriate and desired output for the selected fields or domains. This helps Gen AI products to interact more meaningfully with users. Of course, every prompt engineer must apply their creative and

innovative thoughts through trial and error by creating rational input text pools to operate an application's Gen AI effectively.

Prompt engineering involves a single and/or set of prompts, which are simple instructions and examples using any language to steer LLMs in their behavior in the environment. This allows us to align the LLMs' output with the user's objectives without the model being retrained expensively.

Effective design-engineered prompts can make LLMs suitable for a comprehensive variety of tasks for which they were initially trained.

In simple terms, prompts act as instructions to demonstrate LLM, using vector database details and providing the desired input-output mapping. This is an alternative to fine-tuning.

Before getting into Prompt Engineering, let us recap LLMs, which have a neural network learning ability-based architecture pattern that helps analyze the prompt. Internally, multiple components help the application build the right content. (Please refer to *Chapter 1, Introduction to Generative*

The first component focuses on the most appropriate words for the context based on its abilities, finding the most relevant word for the given context or prompt.

The next component helps the system remember the order of the words and ensures that the prompt makes sense of the right context. Following this, it generates an appropriate and logical list of words for the set of lines by probabilistically combining the words to meet the expectation of context for the given prompt.

In summary, an LLM's unique brain architecture helps it pay attention to the following factors: the right words, the order of remembrance, and probabilities of assignments to the next words in the context.

LLMs have proven to be a remarkable advancement in understanding and generating tasks specific to Natural Language Processing (NLP).

Significance of Prompt Engineering

So far, we have discussed and understood facts about prompt engineering. Prompt engineers bridge the gap between Gen AI product users and LLMs. They identify scripts and templates your users can customize and complete to receive the best results from the language models. These engineers conduct experiments using diverse inputs to construct a prompt library that application developers can utilize in different situations.

Prompt engineering helps make AI applications more efficient and effective. Application developers typically include open-ended user input within a prompt before sending it to the AI model.

For instance, think about AI chatbots. Users might type an incomplete issue such as, "Where to buy a shirt?" Internally, the application's code incorporates an engineered prompt saying, "You function as a sales assistant for a clothing company." A user from Alabama is asking you where they can purchase a shirt. Provide the three closest store locations with

a shirt in stock." The chatbot then generates more relevant and accurate information.

Components and Elements of Prompt

Regarding prompt engineering, prompts consist of the following major components or elements, each with its own importance, such as instructions, context, input data, and output data.

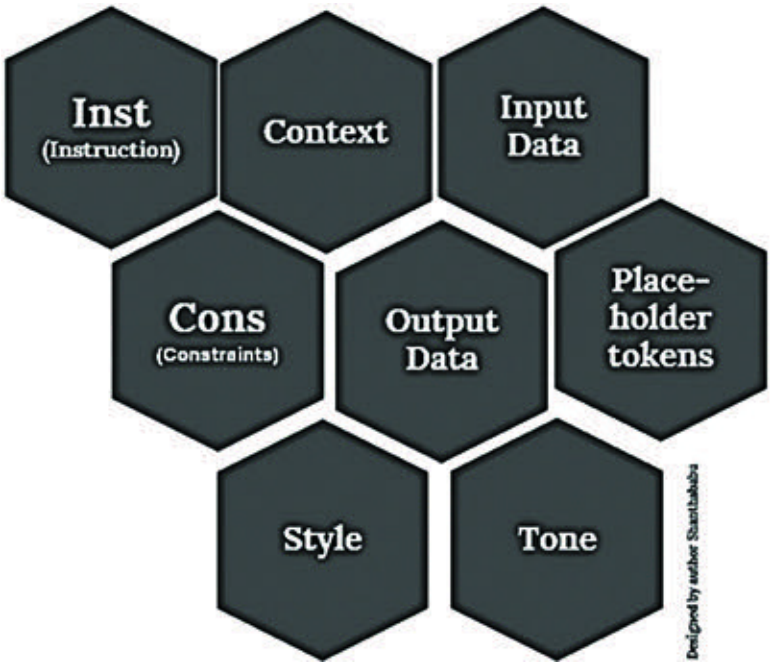

Figure 4.1: *Elements of Prompt*

Instructions: This is a major element that helps with specific tasks or sets of instructions that the model wants to perform precisely

during the operations. If there is any unclarity in this instruction, the overall system will react differently and send unexpected or unwanted output back to the user. This instruction describes the requirements, objectives, and format of input and output for the specific task. Focus on the point where it explicitly explains the task requirements for the model.

Here is an example:

Summarize Sachin Tendulkar's achievements from the following passage clearly in five-point sentences.

Context is a supporting element that can help steer the LLM to produce better responses by focusing on certain logical guidance. The desired context is input as an example, and the desirable output will be obtained. It provides distinct illustrations of how different contexts are inputs and how they should map to outputs.

Here is an example:

In the context of the major achievements of Sachin Tendulkar, provide a concise
summary of the main findings from the given input.

Input The input (query, prompt, or question) that we are interested in finding a response for; it can be a set of questions or statements. This is the input data and the actual content feed for the model, and we have to make sure that the input data is relevant to the given instructions. The objective is for the input to be intended to generate the required response based on instructions and context.

Here is an example:

Sachin Tendulkar is a cricket icon known as the "God of Cricket." He holds the record for the highest run-scorer in Test and ODI cricket, with 100 international centuries. Tendulkar's 24-year career was marked by exceptional batting skills, versatility, and numerous contributions to Indian cricket, including a pivotal role in winning the 2011 ICC Cricket World Cup. Off the field, his humility and dedication have made him a respected figure, and his legacy continues to inspire cricketers worldwide.

Output Data: Based on the instructions, context, and input data, when this was executed in the OpenAI playground within the system panel, the following answer was obtained:

1. Sachin Tendulkar is known as the "God of Cricket" and holds the record for the highest run-scorer in Test and ODI cricket with

100 international centuries.

2. Over his 24-year career, he showcased exceptional batting skills and versatility, impacting Indian cricket.

3. Tendulkar played a significant role in India's 2011 ICC Cricket World Cup victory.

4. His humility and dedication off the field have earned him respect worldwide, making him a revered figure in the cricketing community.

5. Sachin Tendulkar's legacy continues to inspire cricketers globally, solidifying his status as one of the sport's all-time greats.

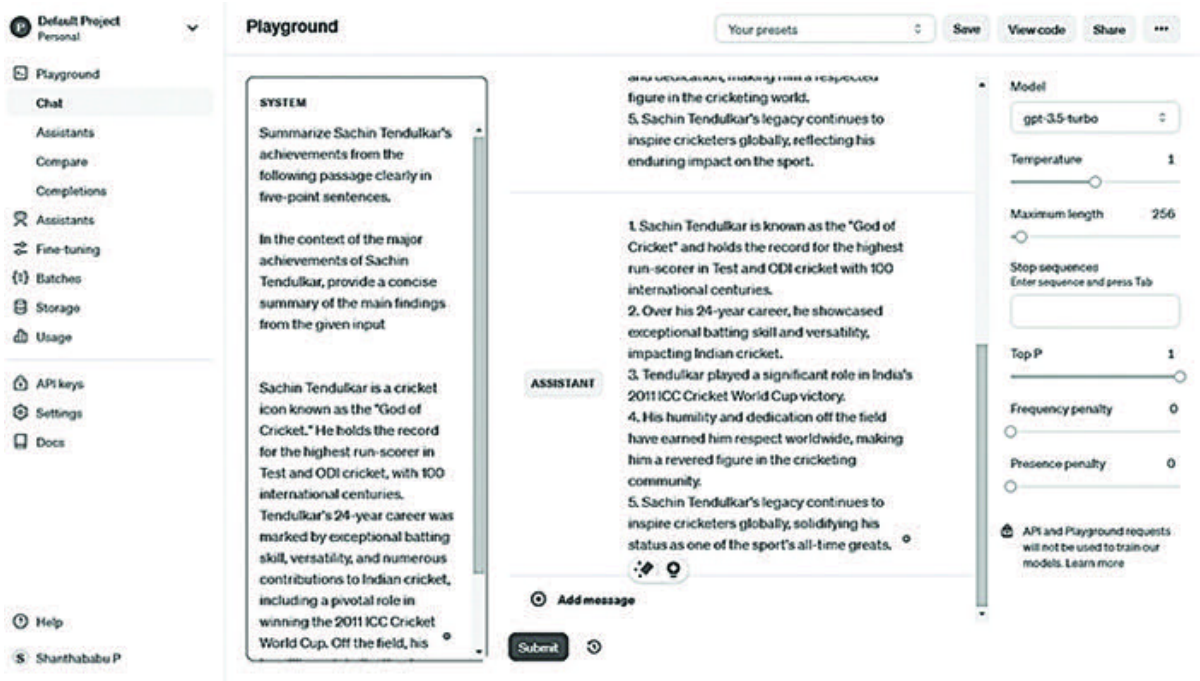

Figure 4.2: *Prompt Elements in the OpenAI Playground*

Output Indicator: This represents the format and type of the output with a specific indicator. This means that the sentiment is an indicator of expected output, and based on the input text, the output would be neutral, negative, or positive.

Instructions: Classify the text as neutral, negative, or positive.

Here is an example:

System Input

Input: The food in the restaurant is good.

Output indicator: Sentiment:

Figure 4.3 shows the output through an assistant as

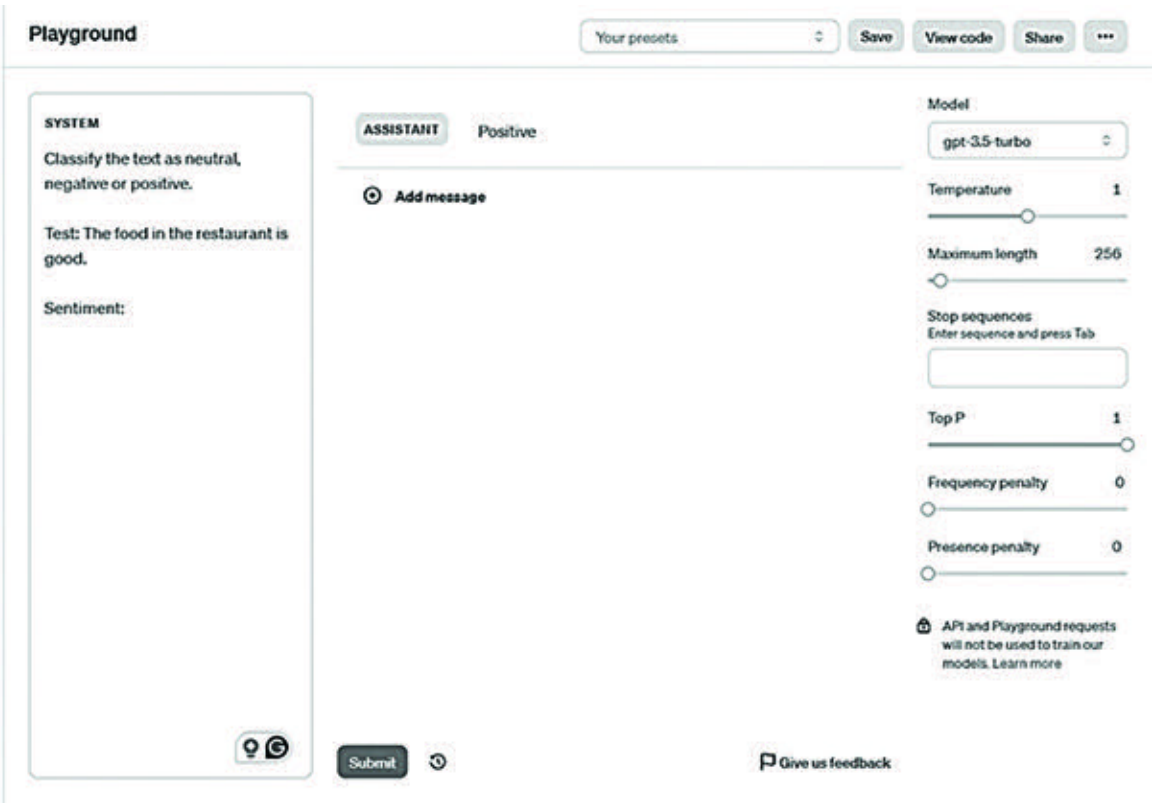

Figure 4.3: *Prompt Elements with Output Indicator (Sentiment)*

Placeholder Tokens: The placeholder token element is a powerful way to design your prompt engineering, and it helps us manage the output format and structure it properly. It supports multiple output generations. We use placeholder tokens as variable inputs for the prompt input and to indicate that the LLM should insert its response into the generated text from the dataset residing in the vector database.

Here is an example:

System Input

you are visiting a **[Country]** for a business trip. You have **[Number_of_Days]** days to participate in the tech event. What kind of visas and other papers would you need? Provide a brief description of each and explain. Given the output in point and priority-based.

Placeholder tokens are **[Country]** and **[Number_of_Days]**

Output

To visit New York, USA, for a business trip to participate in a techno event for five days, you would typically need the following visas and documents:

1. Business Visa:
- A Business Visa is necessary to enter the USA for business purposes, including attending meetings, conferences, or trade shows.
- You will need to apply for a Business Visa (B1 Visa) at the U.S. Embassy or Consulate in your home country.
- The visa application may require documentation such as a letter of invitation from the tech event organizers, proof of accommodation, flight itinerary, and proof of sufficient funds.

2. Electronic System for Travel Authorization (ESTA):

- ESTA is required for citizens of countries under the Visa Waiver Program (VWP) who are traveling to the USA for short business trips or tourism.
- You need to apply for an ESTA at least 72 hours before your departure to the USA.
- ESTA grants you authorization to travel to the USA without a visa for stays up to 90 days.

3. Invitation Letter:

- An invitation letter from the tech event organizers in New York can be helpful in supporting your visa application and explaining the purpose of your visit.
- The invitation letter should detail the event dates.

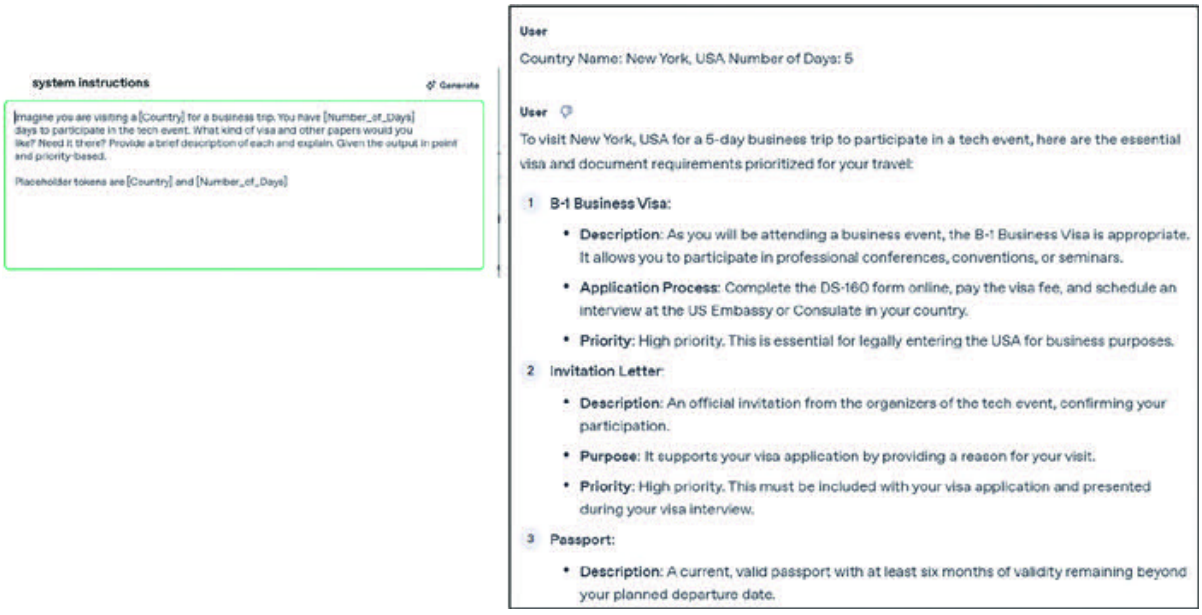

Figure 4.4: *Prompt Elements with Placeholder Tokens*

Sample Prompt elements with placeholder tokens such as country (for instance, France), city (for instance, Paris), and number of days (for instance, 10).

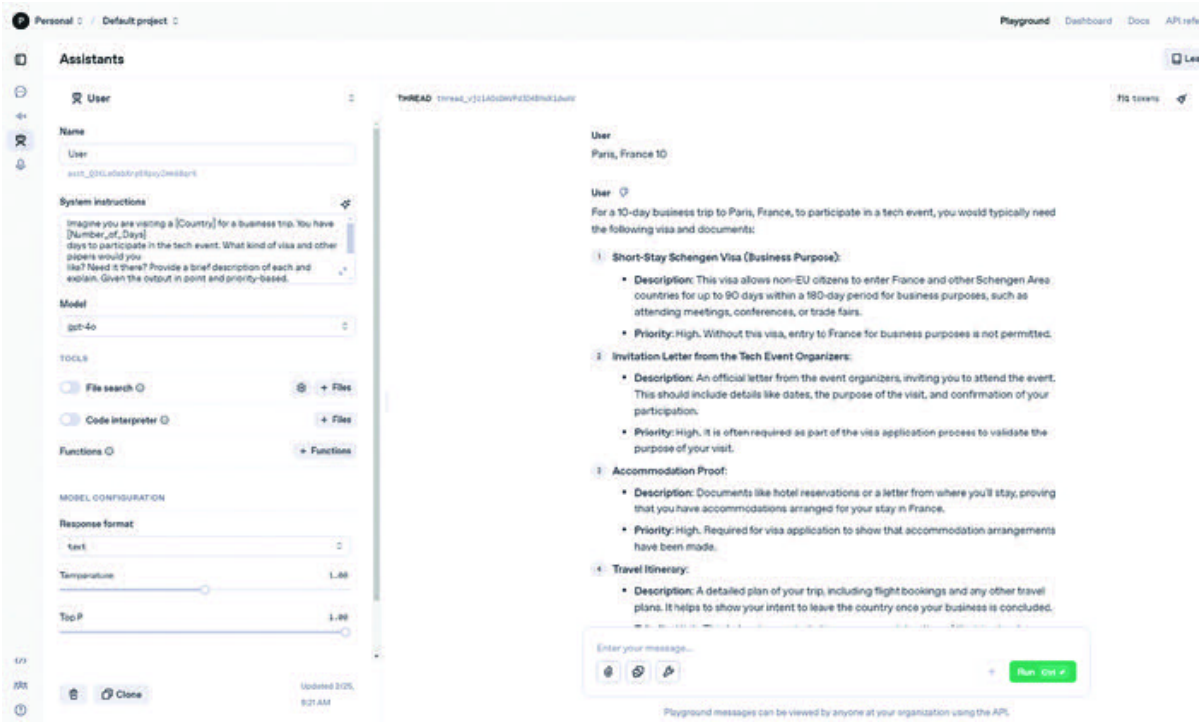

Figure 4.5: *Prompt Elements with Placeholder Tokens (country Paris, France, and 10 days)*

Constraints: This element imposes specific constraints on the LLM response to ensure it meets specific requirements and guidelines,

and prevents awkward content to maintain ethics. The required constraints must be explicitly or implicitly expressed within the prompt by carefully creating the description and context.

Here is an example,

System input

Compose a **short 8-line** poem about Gen AI and its impact on industry with an AXAA rhyme scheme.

Constraints: **short 8-line**

Output

Gen AI, a force in the industry's fray,
Revolutionizing with each passing day,
Tech advancements lead the way,
Bringing efficiency, come what may.

Automation and machine learning in play,
Boosting productivity with a seamless array,
Future of work, brighter display,
Gen AI, here to stay.

System input

Compose a **short 8-line** poem about the city of London with an AABB rhyme scheme.

Constraints: **short 8-line**

In London town, the red buses race
Past palaces and landmarks of grace
The Thames flows steady, a silent guide
Through streets where history can't hide

From Big Ben's chime to the Tower's keep
In its heart, old stories sleep
London, a city alive and grand
Where the past and present hand in hand

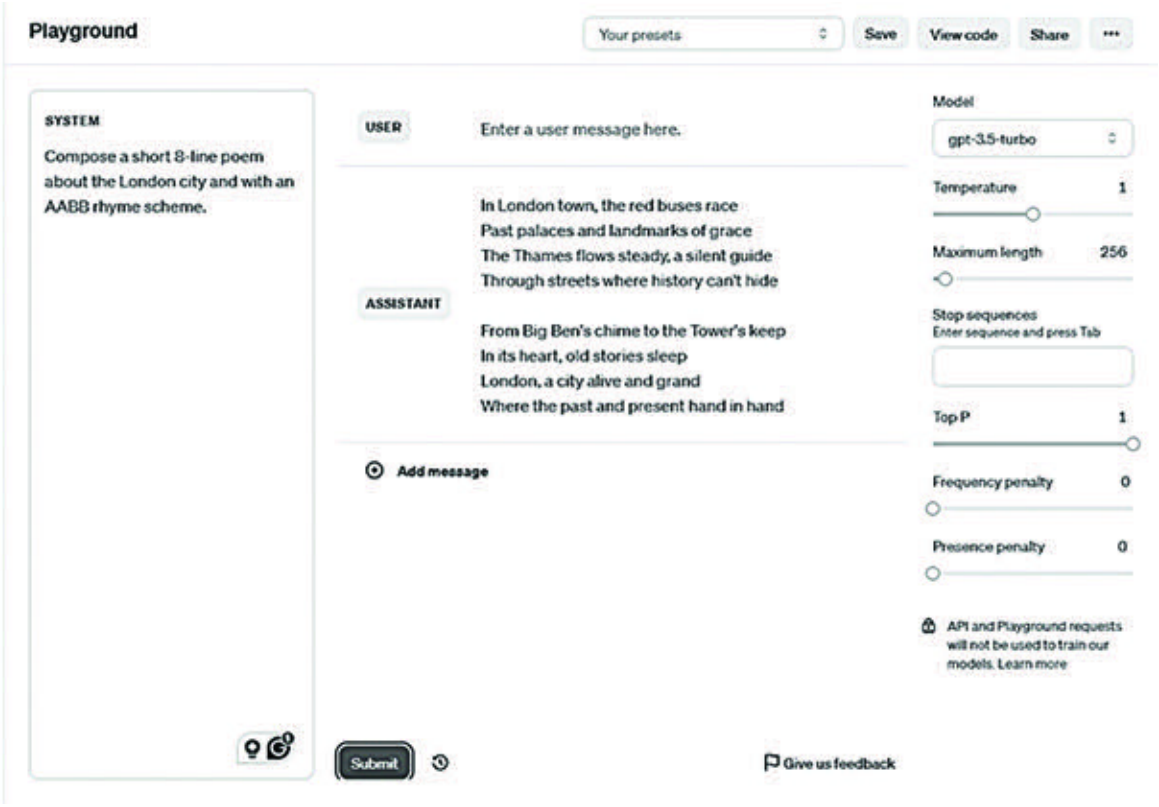

Figure 4.6: *Prompt Elements with Constraints*

Tone and Style: Provide a certain tone and style to the input prompt to modulate the model response based on the business requirements. Since the prompt is in simple English language, we can provide the desired tone specific to the business requirements, such as conversational, formal, assertive, authoritative, optimistic, casual, animated, influential, fun tone, or more.

Based on everyone's understanding, the preceding tone is a parameter of the prompt, and it helps us generate outputs that

align with the intended purpose and audience.

System input

System input with /